Finanziato dall'Unione europea NextGenerationEU

Ministero dell'Università e della Ricerca

Italiadomani PIANO NAZIONALE DI RIPRESA E RESILIENZA

UNIVERSITÀ DELLA CALABRIA

UNIVERSITA' DELLA CALABRIA

Dipartimento di Ingegneria Informatica, Modellistica, Elettronica e Sistemistica

Dottorato di Ricerca in

Information and Communication Technologies

CICLO

XXXVIII

AI METHODS FOR PHYSIOLOGICAL TIME SERIES IN HEALTHCARE: CGM ANALYTICS AND SPEECH-BASED NEUROLOGICAL ASSESSMENT

Settore Scientifico Disciplinare IINF-05/A

Coordinatore: Ch.mo Prof. (Giancarlo Fortino)

Firma _____

Supervisore: Ch.mo Prof. (Sergio Flesca)

Firma Sergio Flesca

Co-Supervisore: Ch.mo Prof. (Pierangelo Veltri)

Firma _____

Dottorando: Dott./ssa (Muhammad Kashif)

Firma Muhammad Kashif

AI Methods for Physiological Time Series in Healthcare: CGM Analytics and Speech-Based Neurological Assessment

Muhammad KASHIF

Examination committee:
Prof. dr. Pietro Hiram Guzzi, chair
Prof. dr. Domenico Ursino
Prof. dr. Elio Masciari
Prof. dr. Sergio Flesca, Supervisor
Prof. dr. Pierangelo Veltri, Co-supervisor

Dissertation presented in partial fulfillment of the requirements for the degree of Doctor of Philosophy (PhD)

June 2026

Preface

This dissertation marks the culmination of a research journey that began at the University of Calabria and was significantly enriched by a visiting period as an International Scholar at KU Leuven. Along the way, I was fortunate to meet supervisors, colleagues, collaborators, and friends who shaped both my research direction and my personal growth. This preface is a sincere acknowledgement of the people and experiences that made this work possible.

First and foremost, I would like to express my deepest gratitude to my supervisors at the University of Calabria, Prof. Sergio Flesca and Prof. Pierangelo Veltri. Thank you for giving me the opportunity to pursue my PhD and for your continued trust and guidance throughout these years. I still remember my first days at UNICAL, the excitement of starting a new chapter, together with the uncertainty of whether I could meet the expectations of doctoral research. Your support helped me transform that uncertainty into steady progress. You encouraged me to think critically, to build strong methodological foundations, and to pursue research questions with scientific rigor. I am particularly grateful for your feedback on writing and presentation: your comments repeatedly pushed me to improve clarity, structure, and reproducibility, and they shaped the way the results in this thesis are communicated.

I would also like to acknowledge my collaborators at the University of Calabria who contributed to my work on speech-based neurological assessment. In particular, I thank Prof. Patrizia Vizza and Prof. Giuseppe Tradigo for their collaboration and support in developing the studies on multiple sclerosis and Parkinson's disease, leading to the work presented for 23rd International Conference on Artificial Intelligence in Medicine (AIME 2025) and the 17th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB 2026) Posters. Your insights, feedback, and encouragement were invaluable in shaping these projects.

I also want to warmly thank my supervisors and hosts at KU Leuven, Prof. Bart Vanrumste and Dr. Chunzhuo Wang, for welcoming me as a visiting PhD student at the e-Media Research Lab (GroepT). My first day at the GroupT campus remains a vivid memory: a new environment, new people, and a strong sense that I was entering a place where research is carried out with both ambition and discipline. I am grateful for the stimulating discussions, for the emphasis on strong experimental design, and for the constructive feedback that helped me mature as a researcher.

I am especially thankful to Dr. Chunzhuo Wang for his close mentorship during my time in Leuven. Beyond general supervision, he provided detailed guidance on experimental design, helped me refine the technical narrative, and carefully reviewed key manuscripts including the Biomedical Signal Processing & Control (BSPC) review paper and the European Signal Processing Conference

(EUSIPCO 2026) conference submission. His feedback significantly improved the clarity of the contributions, strengthened the evaluation protocol, and helped me present the work in a more rigorous and reproducible manner.

Throughout my PhD, I benefited greatly from the support of colleagues at the University of Calabria. I would like to thank my lab mates and collaborators: Tariq Mehmood, Asaf Raza, Jamsher, Farwa Ikram, Khushal Das, Bharat Lal, Irfan Kareem, Reza, Majid, Rafiq, Babul Islam, Ihtisham, Asaf, AAmir, Liliana, Francesco, Silvia, Tommaso, Raffaele, Lucio La Cava, Andrea Tagarelli for discussions, feedback, and the shared experience of navigating research challenges. The informal conversations in offices and corridors, the advice exchanged during experiments, and the encouragement during difficult phases were all important to sustaining progress. I also appreciate the support I received during local academic activities, including workshops and seminars at UNICAL, which provided valuable opportunities to practice presenting research and to refine ideas through discussion.

A special milestone in this journey was presenting our work at the 20th International Conference on Personalized Health (pHealth 2024). Preparing for the conference, finalizing experiments, and communicating results to an international audience was an important step for me. It helped me see my work through a broader research lens and strengthened my motivation to pursue clinically meaningful evaluation and reproducible pipelines. I am also grateful for opportunities such as the IEEE-related academic activities and workshops associated with UNICAL, which contributed to my research culture by emphasizing both scientific communication and community engagement.

My visiting period in Belgium allowed me to experience a vibrant international research environment. I would like to thank my colleagues at the e-Media Research Lab and ESAT-STADIUS, Campus Group T Leuven: Weiyi Xu, Juha Carlon, Meixing Liao, Amsalu Tomas Chuma, Po-Kai Yang, Diwas Lamsal, Martina Montenegro, Vayalet Stefanova, Max Yudayev, Parya Khankeh, Jit Chatterjee, Hannelore Strauven, Melkamu Hunegnaw Asmare, Alexandre Personnic, Kathleen MacWilliam. Their scientific and informal interactions created a welcoming atmosphere and made daily research life enjoyable. The lab culture at GroepT encouraged collaboration and open exchange of ideas, and I learned a great deal from observing and participating in that environment.

My time in Leuven was also personally meaningful. During Ramadan, I had the chance to meet many kind people at the masjid, including Tauseef, Waseem, Ishaq Awan, Haroon, Raees, Rayan, Abdul Razzak and Ahmad, whose companionship and warmth made the month special while being away from home. I am also grateful to Sher Ali Khan for showing me around and for the memorable walks in the Provinciedomein Kessel-Lo park. I also enjoyed experiencing Leuven across different seasons and local events. Arriving in December, I was welcomed by the Christmas atmosphere and the beautifully decorated Leuven city center. In March, the carnival celebrations added another memorable dimension to daily life in the city. In April 2025, Leuven hosted the European Running Championships, and it was inspiring to witness the energy of the event and the city-wide excitement around it. I would like to thank my flatmates and friends in Leuven for making daily life easier and more enjoyable, including: Joppe, Chau Dinh, Mattes, and Swagata in the Leuven center. I especially appreciate the farewell party we arranged together filled with music, cookies, and games, which became a memorable moment during my time in Leuven.

Beyond formal research, some experiences became memorable highlights of my PhD life. One of these was attending Hertie School Data Science Summer School (DSSS 2023 and DSSS 2025), Germany and the ELLIS Unit Amsterdam Winter School on Foundation Models (FoMo 2026). The school provided exposure to an exciting range of topics, including foundation models and multimodal learning, and it offered the chance to meet researchers from different backgrounds. I still remember the unusual weather during the week hail, cold winds, and sudden snow, which made Amsterdam feel even more dramatic, especially for someone coming from sunny Calabria. Traveling each day by ferry to the venue at Tolhuistuin and sharing coffee breaks, discussions, and evening conversations created a unique atmosphere. The social activities and informal interactions during the winter school were as valuable as the lectures themselves: they reminded me that research is not only about technical progress, but also about community, exchange, and shared curiosity.

I would also like to extend my sincere thanks to all participants and contributors whose data made this research possible. Whether in CGM-related studies or in speech-based assessment settings, data collection requires time, effort, and trust. I am grateful for their willingness to contribute, which enabled the analyses and evaluations presented in this dissertation.

I am grateful for the financial and institutional support that enabled my PhD research. In particular, I would like to thank the University of Calabria and the FAIR (*Future Artificial Intelligence Research*) initiative for supporting my doctoral work and related research activities. This support provided the essential foundation for conducting the experiments, attending scientific events, and completing the research presented in this dissertation.

I would also like to express my sincere gratitude to my Master's supervisors, Prof. Ayaz Hussain, whose guidance and encouragement played an important role in shaping my research mindset and motivating me to pursue doctoral studies. I am thankful as well to Asim Munir for his support and encouragement during my academic journey.

I would like to thank my friend Shoaib and my cousin Faisal Malak for their companionship and steady support during my time at IIUI Islamabad. I especially value the many conversations we shared during walks, travel, and study sessions in the library. Our discussions, often continuing over meals, provided a welcome space for reflection and helped me clarify ideas during important stages of my academic journey.

Finally, my heartfelt gratitude goes to my family. I am deeply thankful to my parents, brothers and sister for their unconditional support, patience, and encouragement throughout my education. Their sacrifices and belief in me made it possible for me to reach this stage. I also thank my friends for their support during challenging times and for celebrating even small successes along the way.

Muhammad Kashif
June 2026, Cosenza, Italy

Abstract

Continuous physiological sensing enables data-driven monitoring of chronic conditions and neurological disorders. This thesis investigates artificial intelligence (AI) methods for prediction and event detection from health-related time series, with a primary focus on continuous glucose monitoring (CGM) and a complementary focus on speech-based neurological assessment.

For CGM analytics, the thesis addresses two clinically relevant tasks: glucose forecasting and meal episode detection. First, we study personalized glucose prediction using recurrent neural networks and evaluate forecasting performance under standard regression metrics. Second, we examine the problem of identifying meal episodes from CGM in free-living conditions, where labels can be noisy and positive events are rare. Building on a systematic synthesis of prior work, we implement and evaluate a CGM-only meal episode detection pipeline using sliding-window sequence modeling and constraint-based decoding, emphasizing episode-level evaluation and operational metrics such as false positives per day and detection delay.

To demonstrate methodological generalization beyond metabolic sensing, the thesis further investigates speech-based neurological assessment. We present an LSTM-based framework for multiple sclerosis classification from acoustic features and a Parkinson's disease screening approach that combines task-conditioned parameter-efficient fine-tuning of a speech foundation model with prompt-aligned ASR error structure representations.

Across these domains, the thesis highlights the importance of rigorous preprocessing, leakage-aware validation, and task-aligned evaluation. The presented results support the broader conclusion that clinically useful time-series models require not only effective learning architectures but also principled definitions of targets, operating points, and metrics that reflect real-world deployment constraints.

Keywords. Continuous Glucose Monitoring; meal detection; glucose prediction; BiLSTM; constrained decoding; speech biomarkers; Parkinson's disease; multiple sclerosis.

List of Abbreviations

Abbrev.	Meaning
CGM	Continuous Glucose Monitoring
T1D	Type 1 Diabetes
T2D	Type 2 Diabetes
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
FP/day	False Positives per Day
ASR	Automatic Speech Recognition
PEFT	Parameter-Efficient Fine-Tuning
MS	Multiple Sclerosis
PD	Parkinson's Disease

Contents

Abstract v

List of Abbreviations vii

Contents ix

List of Figures xiii

List of Tables xv

1 Introduction 1

1.1 Motivation 1
1.2 Thesis Scope and Definition 2
1.3 Challenges 2
1.3.1 Noisy and missing data 2
1.3.2 Class imbalance and operational constraints 3
1.3.3 Evaluation beyond pointwise metrics 3
1.4 Summary of (sub-)Objectives 3
1.5 Thesis Outline 4

I CGM-Based Glucose Prediction 7

2 Glucose Forecasting from Continuous Glucose Monitoring 9

2.1 Background 9
2.2 Data Collection and Methods 12
2.2.1 Data Collection and Preprocessing 12
2.2.2 Glucose Prediction Model 13
2.3 Experimental Analysis and Discussion 14
2.4 Conclusions 16

II CGM-Based Eating Event Detection and Prediction 17

3 AI-Based Detection and Prediction of Eating Events Using CGM 19

3.1 Introduction . . . 20
3.1.1 Motivation for CGM-Based Eating Event Analysis . . . 20
3.1.2 Emergence of AI in Detection and Prediction . . . 21
3.1.3 Challenges and Open Questions . . . 21
3.1.4 Scope and Contribution . . . 21
3.1.5 Inclusion and Exclusion Criteria . . . 22
3.1.6 Organization of the Paper . . . 23
3.2 Sensor Modalities for Eating Event Detection and Prediction . . . 24
3.2.1 Continuous Glucose Monitoring . . . 24
3.2.2 Strengths and Challenges of Using CGM for Meal Detection . . . 24
3.2.3 Multimodal Sensor Fusion . . . 25
3.3 Modeling Approaches for Eating Event Detection and Prediction . . . 26
3.3.1 Model-Based Estimation Techniques . . . 26
3.3.2 Machine Learning Approaches . . . 30
3.3.3 Deep Learning-Based Models . . . 33
3.3.4 Hybrid and Multimodal Approaches (CGM-Focused) . . . 38
3.4 Datasets, Labeling Strategies, and Evaluation Metrics . . . 42
3.4.1 Datasets and Participant Populations . . . 42
3.4.2 Labeling Strategies for Eating Events . . . 47
3.4.3 Evaluation Metrics and Training Protocols . . . 47
3.5 Cross-Study Comparison and Benchmarking Discussion . . . 48
3.5.1 Variability in Modeling Paradigms . . . 48
3.5.2 Differences in Input Modalities and Fusion Strategies . . . 49
3.5.3 Dataset Sources and Population Diversity . . . 49
3.5.4 Disparities in Evaluation Metrics . . . 49
3.5.5 Benchmarking and Reproducibility Gaps . . . 49
3.6 Challenges and Limitations . . . 50
3.6.1 Sensor-Based Limitations . . . 50
3.6.2 Algorithmic and Modeling Challenges . . . 51
3.6.3 Deployment and Regulatory Barriers . . . 51
3.7 Chapter Summary . . . 52

4 Meal Episode Detection from CGM 53
4.1 Introduction . . . 54
4.2 Related Work . . . 54
4.3 Methods . . . 55
4.3.1 Dataset and Label Construction . . . 55
4.3.2 Feature Engineering . . . 56
4.3.3 Models and Episode Decoder . . . 56
4.4 Experimental Setup and Results . . . 58
4.5 Discussion, Limitations, and Chapter Summary . . . 61

III Speech-Based Neurological Assessment 63

5 Multiple Sclerosis Classification from Vocal Signals 65
5.1 Introduction . . . 65
5.2 Materials and Methods . . . 66
5.2.1 Data collection and processing . . . 66
5.3 Experimental Results . . . 67
5.3.1 MFCC features Analysis . . . 67
5.3.2 Acoustic Features Analysis . . . 68
5.4 Discussion, Limitations, and Chapter Summary . . . 68

6 Parkinson's Speech Assessment using Task-Conditioned PEFT and Prompt-Aligned ASR Error Structure (PAESL) Fusion 69
6.1 Introduction . . . 70
6.2 Related Work . . . 72
6.2.1 Speech-Based Parkinson's Disease Assessment . . . 72
6.2.2 Self-Supervised Speech Models and Parameter-Efficient Adaptation . . . 73
6.2.3 ASR-Based Features and Error Modeling in Pathological Speech . . . 73
6.3 Dataset and Materials . . . 74
6.3.1 Dataset Description . . . 74
6.3.2 Preprocessing and Audio Segmentation . . . 74
6.3.3 Task Conditioning and Label Structure . . . 75
6.3.4 ASR Transcription and Alignment . . . 75
6.4 Proposed Method . . . 75
6.4.1 Overview . . . 75
6.4.2 TC-PEFT Acoustic Branch . . . 76
6.4.3 PAESL Branch: Prompt-Aligned Error Structure Learning . . . 77
6.4.4 Fusion Strategy . . . 78
6.4.5 Training and Evaluation Protocol . . . 79
6.4.6 Implementation Details . . . 79
6.5 Experimental Setup . . . 79
6.5.1 Evaluation Protocol . . . 79
6.5.2 Compared Methods . . . 79
6.5.3 Proposed Models . . . 79
6.5.4 Implementation Details . . . 80
6.5.5 Evaluation Metrics . . . 80
6.6 Results . . . 80
6.6.1 Main Results . . . 80
6.6.2 Baseline Comparison . . . 81
6.6.3 Task-wise Results . . . 81
6.6.4 Permutation Testing . . . 81

6.6.5 Cohort-Restricted Evaluation 81
6.7 Ablation and Additional Analyses 81
6.7.1 Effect of PAESL Fusion 81
6.7.2 ASR Feature Analysis 82
6.7.3 Task Contribution 82
6.7.4 Threshold Stability 82
6.7.5 Robustness to Label Perturbation 82
6.8 Discussion, Limitations, and Chapter Summary 82

IV General Discussion and Conclusions 85

7 Discussion, Limitations, and Future Work 87
7.1 Summary of Contributions 87
7.2 Limitations 88
7.2.1 Dataset scale, cohort diversity, and external validity 88
7.2.2 Label noise and event definition ambiguity 88
7.2.3 Modeling scope and deployment constraints 89
7.2.4 Comparability across tasks and metrics 89
7.3 Future Work 89
7.3.1 CGM: broader validation and stronger operational evaluation 89
7.3.2 CGM: improved labeling and semi-automated ground truth refinement . . 90
7.3.3 Speech: robustness across tasks, conditions, and cohorts 90
7.3.4 Reproducibility: shared evaluation toolkits and reporting standards 90
7.4 Concluding Remarks 90

Bibliography 91

Curriculum Vitae 100

List of Publications 101

List of Figures

2.1 Different features are shown from different sensors and food log for 014 participant such as heart rate, glucose level, sugar, and carb. 13
2.2 Architecture of the proposed model 14
2.3 The comparison between actual and predicted glucose levels for participant 014 is illustrated. (a) displays the first 10-minute interval, while (b) presents the data for all intervals. 15

3.1 PRISMA-style flow diagram illustrating the literature search and study selection process for CGM-based eating event detection and prediction studies. Records were identified from major scientific databases including IEEE Xplore, PubMed, Scopus, Web of Science and Google Scholar, screened using predefined inclusion and exclusion criteria, and filtered through full-text review to obtain the final set of studies analyzed in this review. 23
3.2 A pattern-based machine learning pipeline to detect meals from CGM subsequences by identifying individualized glucose response patterns [1]. 32
3.3 Trade-off between sensitivity and false alarms per day reported by representative CGM-based meal detection methods. Colors indicate the methodological category. 32
3.4 Detection delay across representative studies, grouped by methodological family. [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14] from left to right. 36
3.5 A hierarchical fusion model for eating detection [15]. 39
3.6 An IoT-enabled framework using the Dexcom G6 CGM and cloud-based analytics to infer daily meal routines [16]. 39

4.1 Proposed CGM-based meal detection pipeline. 56
4.2 Episode-level F1 (mean $\pm$ SD) for LR,RF, and the proposed BiLSTM window model on OhioT1DM. 57
4.3 Episode-level recall vs false positives per day for LR, RF, and proposed BiLSTM window model across 11 subjects. 58
4.4 Example day illustrating CGM trace, probability decoding into episodes and the resulting detections 59
4.5 Distribution of window-level eating probability for eating vs non-eating windows, with the indicated calibrated decision threshold. 60

4.6 Threshold-performance trade-off showing episode-level F1, recall, and FP/day, with the selected operating threshold marked. 61

5.1 A comprehensive flow chart of the data collection and research methodology . . . 67

6.1 Overview of the proposed framework. The TC-PEFT branch extracts task-conditioned acoustic embeddings using WavLM with LoRA adaptation, while the PAESL branch encodes prompt-aligned ASR error structures using a Transformer encoder. The two representations are standardized using training-split statistics, fused by concatenation and passed to an MLP classifier for subject-level Parkinson's disease prediction. 76

List of Tables

1.1 Mapping between thesis chapters, associated publications, and sub-objectives (SO). 5

2.1 Participant demographics (age group: 35-65 years). 12
2.2 Hyperparameters 15
2.3 Comparison with state of the art 16

3.1 A comprehensive overview of model-based estimation approaches for eating event detection and prediction using CGM data. Abbreviations: FL = free-living; IS = in-silico; RCT = randomized controlled trial; CHO = carbohydrate; ROC = rate of change; AUC = area under the receiver operating characteristic curve. 29
3.2 A comprehensive overview of machine learning based approaches for eating event detection and prediction using CGM data. 35
3.3 Overview of deep learning based approaches for eating event detection and prediction using CGM data. 37
3.4 A comprehensive overview of hybrid/multimodal approaches for eating event detection and prediction using CGM data. 41
3.5 Overview of publicly available and experimental datasets for meal detection and glucose dynamics analysis. Abbreviations: HR = heart rate; SMBG = self-monitoring of blood glucose; GIM = glucose-insulin model; MUA = Material Use Agreement. 44

4.1 Episode-level performance of baseline models and proposed BiLSTM sliding window model on OhioT1DM. 58

5.1 Experimental results comparing different models and features. 68

6.1 Subject-level 5-fold cross-validation results for the evaluated configurations. Values are reported as mean $\pm$ standard deviation across folds. 80

Chapter 1

Introduction

1.1 Motivation

Continuous physiological monitoring is rapidly transforming chronic disease management and population-level health assessment. Among the available wearable modalities, continuous glucose monitoring provides a dense and clinically actionable time series of interstitial glucose values typically sampled every 5 minutes over multiple days, enabling high-resolution characterization of glycemic variability and supporting data-driven decision support systems [17, 18, 19].

Two problems have emerged as central to CGM analytics: (i) *glucose prediction (forecasting)*, anticipating near-future glucose trajectories to reduce exposure to hypo- and hyperglycemia, improve decision support, and inform automated control systems; and (ii) *meal/food intake episode detection*, identifying when meals occur from sensor data, with the goal of supporting unannounced meal handling in closed-loop insulin delivery and digital health interventions [20, 12].

While CGM is physiologically meaningful, CGM-based modeling is nontrivial. The glucose response to meals is delayed and confounded by insulin, activity, stress, illness, and circadian effects. In free-living data, labels may be incomplete or noisy (e.g., self-reported meal times), and performance cannot be judged solely by point-wise accuracy. In practice, the operational utility of a detection model depends on *alarm burden* (false alarms per day) and *timeliness* (detection delay), not only on standard classification metrics [7, 12].

In parallel to metabolic monitoring, another major direction in health sensing is speech-based neurological assessment, where speech recordings are used to support screening and monitoring for neurological disorders. Speech signals are attractive because they are non-invasive, inexpensive to collect, and can be acquired remotely. However, speech assessment also faces real-world constraints similar to CGM analytics: limited labeled data, variability across tasks/prompts, and the need for robust evaluation. Recent work increasingly leverages self-supervised speech foundation models such as wav2vec 2.0 and WavLM to learn robust representations that transfer well under limited labeled data [21, 22]. However, adapting large pretrained models to small clinical datasets can be challenging; parameter-efficient fine-tuning (PEFT) methods such as LoRA provide an effective alternative by updating only a small subset of parameters [23]. Furthermore, modern ASR systems such as Whisper enable robust transcription across languages and conditions, motivating

ASR-informed representations for pathological speech analysis [24].

This thesis addresses these challenges under a unifying theme: *AI methods for clinically meaningful prediction and event detection from physiological time series*, with CGM as the primary focus and speech as a complementary application domain.

1.2 Thesis Scope and Definition

This thesis is organized as a publication-aligned monograph that integrates the author's published and submitted works into a coherent research narrative. The scope is defined by two levels:

- **Primary focus:** CGM-based modeling for (i) glucose prediction and (ii) eating/meal episode detection, with emphasis on realistic constraints in free-living data and clinically meaningful evaluation.
- **Secondary focus:** time-series speech modeling for neurological assessment, illustrating how modern deep learning pipelines and evaluation practices extend beyond metabolic sensing.

Accordingly, the thesis is structured into four parts:

- **Part I:** CGM-based glucose prediction (foundation and forecasting).
- **Part II:** CGM-based eating event detection and prediction (systematic review and episode detection).
- **Part III:** Speech-based neurological assessment (multiple sclerosis classification and Parkinson's speech assessment).
- **Part IV:** General discussion and conclusions (synthesis, limitations, and future work).

The central methodological principle throughout the thesis is that sensor analytics must be evaluated at the level that matches the real task: forecasting error for prediction, and *episode-level* criteria (e.g., FP/day and detection delay) for event detection, complemented by rigorous validation protocols that minimize information leakage.

1.3 Challenges

1.3.1 Noisy and missing data

Real-world sensing data are imperfect. CGM signals may contain calibration drift, transient dropouts, and sensor noise. Even when devices report regularly sampled readings (e.g., 5-minute CGM), practical datasets include missing intervals due to connectivity issues or sensor replacement. These effects are visible across widely used research datasets (e.g., OhioT1DM and DiaTrend) and must be handled explicitly to avoid artifacts during training and evaluation [25, 26].

Similarly, speech recordings collected under real conditions can vary in duration, channel conditions, background noise, and adherence to prompted protocols. These sources of variability necessitate explicit preprocessing policies (e.g., segmentation, normalization) and careful separation of training vs. evaluation transformations.

1.3.2 Class imbalance and operational constraints

Meal detection from CGM is inherently imbalanced: most time points correspond to non-meal states, and positive windows occupy a small portion of a day. As a result, models can achieve deceptively high accuracy by predicting the majority class. Operationally, meal detectors must achieve a useful trade-off: detecting a large fraction of meals while maintaining an acceptable false alarm burden [7, 12]. A key constraint in practical unannounced meal detection is *false positives per day (FP/day)*, which directly reflects alarm fatigue and system usability.

Speech-based screening problems can also exhibit imbalance or dataset-specific biases (e.g., cohort composition), requiring evaluation strategies that report robust metrics beyond accuracy.

1.3.3 Evaluation beyond pointwise metrics

A recurring limitation in the literature is the mismatch between the evaluation unit and the target task. For meal detection, point-wise metrics (accuracy, AUC, time-point F1) are insufficient because meals are episodes with temporal extent. Two models with similar point-wise performance can yield very different episode-level behavior after post-processing. Therefore, evaluation must explicitly report:

- Episode-level precision/recall/F1,
- FP/day,
- Detection delay (and/or onset error) under clearly defined matching rules.

For forecasting, standard regression metrics (RMSE, MAE, MSE, R^2) remain appropriate but must be reported under consistent protocols (e.g., subject-wise splits, horizons, and feature availability).

1.4 Summary of (sub-)Objectives

The thesis addresses the following sub-objectives (SO), each corresponding to one or more chapters:

1. **SO1 — CGM glucose prediction:** Develop and evaluate deep learning models for CGM-based glucose prediction using real participant data, and analyze the impact of additional physiological and behavioral features on forecasting performance.
2. **SO2 — Evidence synthesis for CGM-based eating event analysis:** Provide a systematic and critical review of AI-based eating event detection and prediction using CGM, summarizing datasets, labeling strategies, modeling paradigms, and evaluation practices, and identifying reproducibility gaps.
3. **SO3 — CGM meal episode detection:** Design and evaluate a CGM-only meal episode detection pipeline based on sliding-window sequence modeling and explicit decoding, with evaluation at clinically meaningful operating points (e.g., FP/day-controlled thresholds).
4. **SO4 — Generalization to neurological speech assessment:** Demonstrate that robust time-series modeling pipelines extend beyond CGM to speech-based neurological assessment, including multiple sclerosis classification and Parkinson's disease screening using modern speech representation learning and structured ASR-derived features.

1.5 Thesis Outline

This thesis is structured into four parts comprising seven chapters. Chapters 2–6 are based on peer-reviewed research outputs, summarized in Table 1.1. The structure is aligned with the sub-objectives defined in Section 1.4: Part I addresses SO1 (CGM glucose prediction), Part II addresses SO2 and SO3 (CGM-based eating event evidence synthesis and CGM meal episode detection), Part III addresses SO4 (speech-based neurological assessment), and Part IV synthesizes findings across SO1–SO4. The remainder of the thesis is organized as follows.

Chapter 2 (Part I): CGM-based glucose prediction. Chapter 2 presents the work on glucose forecasting, *GLSTM: On Using LSTM for Glucose Level Prediction*. The chapter investigates personalized LSTM-based forecasting using CGM and auxiliary signals and reports performance using standard regression metrics (e.g., RMSE, MAE, MSE, and R^2). Beyond the publication, the chapter provides additional implementation and experimental details to clarify the full modeling pipeline.

Chapter 3 (Part II): Review of CGM-based eating event detection and prediction. Chapter 3 expands the work *AI-Based Detection and Prediction of Eating Events Using Continuous Glucose Monitoring: A Comprehensive Review*. It synthesizes model-based, machine learning, deep learning, and hybrid approaches, and compares datasets, labeling strategies, and evaluation metrics. The chapter highlights key reproducibility gaps and motivates the need for episode-level reporting and operational metrics.

Chapter 4 (Part II): CGM meal episode detection. Chapter 4 presents the work *Meal Episode Detection from Continuous Glucose Monitoring Using a BiLSTM Sliding-Window and Constraint-based decoding*, The chapter introduces a CGM-only pipeline that separates probabilistic prediction from episode formation, emphasizing fair comparison against classical baselines under identical features, decoding, and per-subject operating-point selection. Performance is reported at the episode level, including false positives per day (FP/day) and detection delay, to reflect practical deployment constraints.

Chapter 5 (Part III): Multiple sclerosis classification from speech. Chapter 5 presents the work on speech-based multiple sclerosis classification, *Using LSTM-Based Model on Vocal Signal Analysis for the Classification of Multiple Sclerosis*. It evaluates MFCC-based and acoustic-feature-based pipelines using LSTM/BiLSTM and classical baselines, and discusses the role of data augmentation and class balancing in small-cohort pathological speech settings.

Chapter 6 (Part III): Parkinson's speech assessment with foundation models and ASR structure. Chapter 6 describes the work on Parkinson's speech assessment, *Task-Conditioned PEFT with Prompt-Aligned ASR Error Structure Fusion*. The chapter develops a two-branch framework that combines task-conditioned parameter-efficient adaptation of a speech foundation model with

Table 1.1: Mapping between thesis chapters, associated publications, and sub-objectives (SO).

SO(s)	Chapter	Publication
SO1	2	M. Kashif, S. Flesca, and P. Veltri, "GLSTM: On using LSTM for glucose level prediction," *Stud. Health Technol. Inform.*, vol. 314, pp. 103–107, 2024.
SO2	3	M. Kashif, S. Flesca, and P. Veltri, "AI-Based Detection and Prediction of Eating Events Using Continuous Glucose Monitoring: A Comprehensive Review," (*Biomedical Signal Processing and Control*, accepted).
SO3	4	M. Kashif, S. Flesca, and P. Veltri, "Meal Episode Detection from Continuous Glucose Monitoring Using a BiLSTM Sliding-Window and Constraint-Based Decoding," (*EUSIPCO 2026*, accepted).
SO4	5	M. Kashif, P. Vizza, A. Delfino, R. B. Bossio, G. Tradigo, S. Flesca, and P. Veltri, "Using LSTM-based model on vocal signal analysis for the classification of multiple sclerosis," in *Lecture Notes in Computer Science*, Springer Nature Switzerland, 2025, pp. 192–196.
	6	M. Kashif, P. Vizza, G. Tradigo, and S. Flesca, "Task-Conditioned PEFT with Prompt-Aligned ASR Error Structure Fusion for Parkinson's Speech Assessment," (*ACM-BCB 2026 poster abstract*, accepted).

prompt-aligned ASR error-structure representations, and evaluates subject-level Parkinson's disease screening under subject-independent cross-validation.

Chapter 7 (Part IV): Discussion and conclusions. Finally, Chapter 7 synthesizes the main contributions of the thesis, discusses limitations across CGM and speech sensing settings, and outlines future research directions toward clinically reliable, deployable time-series models for prediction and event detection.

Part I

CGM-Based Glucose Prediction

Chapter 2

Glucose Forecasting from Continuous Glucose Monitoring

The work presented in this chapter is based on:

M. Kashif, S. Flesca, and P. Veltri, "GLSTM: On using LSTM for glucose level prediction," *Stud. Health Technol. Inform.*, vol. 314, pp. 103–107, 2024.

Abstract

The Prediabetes impacts one in every three individuals, with a 10% annual probability of transitioning to type 2 diabetes without lifestyle changes or medical interventions. It's crucial to manage glycemic health to deter the progression to type 2 diabetes. In the United States, 13% of individuals (18 years of age and older) have diabetes, while 34.5% meet the criteria for prediabetes. Diabetes mellitus and prediabetes are more common in older persons. Currently, nevertheless, there aren't many noninvasive, commercially accessible methods for tracking glycemic status to help with prediabetes self-management. This study tackles the task of forecasting glucose levels using personalized prediabetes data through the utilization of the Long Short-Term Memory (LSTM) model. Continuous monitoring of interstitial glucose levels, heart rate measurements, and dietary records spanning a week were collected for analysis. The efficacy of the proposed model has been assessed using evaluation metrics including Root Mean Square Error (RMSE), Mean Squared Error (MSE), Mean Absolute Error (MAE), and the coefficient of determination (R2).

2.1 Background

Prediabetes manifests as elevated blood glucose levels, falling below the threshold for type 2 diabetes but exceeding normal levels. It's estimated that 70% of individuals with prediabetes will eventually progress to diabetes [27]. Diabetes stands as one of the foremost global health concerns today, with its prevalence only rising. Regular monitoring of blood glucose levels is crucial for managing glucose levels and establishing effective diabetes treatment plans [28]. Failure

to diagnose this condition can significantly elevate the risk of severe, life-threatening ailments such as cardiac arrest, cerebral hemorrhage (commonly known as strokes), kidney damage, among others. Diabetes comes in two major forms: type 1 and type 2. The autoimmune disease known as type 1 diabetes, also referred to as juvenile diabetes, is characterized by the body's immune system attacking and destroying the cells that make insulin, a hormone that controls blood sugar levels. However, type 2 diabetes is a metabolic disease brought on by an interaction of lifestyle factors including weight and inactivity, as well as genetics. With 90-95% of cases being of this kind, it is the most prevalent kind of diabetes. Individuals who have type 2 diabetes either have insufficient insulin production or insulin resistance [29]. Current clinical definitions of blood glucose fluctuations are based on population averages and are primarily used to characterize glucose variations in people diagnosed with Type 1 Diabetes (T1D) and Type 2 Diabetes (T2D) [30]. These definitions include hyperglycemia (elevated glucose levels) and hypoglycemia (lowered glucose levels). Achieving adequate glycemic management stands as the main and continuous aim for those living with diabetes. Practically, this entails addressing numerous factors that influence blood glucose levels, including dietary choices, medication, physical activity, and various other considerations.

In the United States, approximately 37.3 million adults are affected by diabetes, with 95% of them diagnosed with Type 2 Diabetes (T2D). Additionally, another 96 million adults in the U.S. are estimated to have prediabetes. Within this prediabetic population, 5-10% transition to diabetes annually, and it's projected that 70% of individuals with prediabetes will develop T2D over their lifetime. The annual economic burden of diabetes in the USA alone amounts to $327 billion. The annual cost of health care for those with diabetes is $16,752, which is 2.3 times higher than the cost for people without diabetes [31]. Diabetes is one of the four main non-communicable diseases (NCD) that raise the risk of early mortality in adults aged 30-69 each year, according to the World Health Organization (WHO) [6]. Projections indicate that by 2030, approximately 578 million people globally will have diabetes, with this number expected to rise by 51% to reach 700 million by 2045 unless immediate and substantial action is taken [32].

This study introduces a Long Short-Term Memory (LSTM) deep learning approach for predicting blood glucose levels in a personalized prediabetes participant dataset [30, 33]. Pre-processing of sensor data is conducted, which involves ensuring sampling consistency and interpolating to address missing samples. Furthermore, the effectiveness of the proposed model is assessed using metrics such as RMSE, MSE, MAE, and R2, and its performance is compared with state-of-the-art methods.

Given the physiological lag that occurs between blood and Insulin Sensitivity Factor (ISF) glucose levels, it is imperative to anticipate or predict glucose levels to avoid hypoglycemic situations caused due to insulin dosage delays. An extensive LSTM approach is presented to provide accurate glucose prediction using different features, which resulted in Root Mean Squared Error (MSE) of 5.04 and Mean Absolute Relative Difference (MARD) of 2.61 [34]. Deep learning architecture based on LSTM neural network is compared for blood glucose prediction Auto Regressive Integrated Moving Average (ARIMA). The demonstrated results showed that MSE and RMSE of 108.140 and 10.399 for ARIMA while MSE and RMSE of 24.075 and 4.906 for LSTM [35].

Customized LSTM models have been presented for the 100 patients simulated by the UVA/Padova simulator. These models forecast blood glucose (BG) levels over a prediction horizon of 40 minutes,

utilizing inputs such as meal intake, insulin administration, and past BG readings. The proposed LSTMs achieved outcomes with FIT of 75.86% and RMSE of 7.67 [36]. A multi-layer (LSTM) based on recurrent neural network is proposed to predict blood glucose levels in patients with type 1 diabetes by transforming event-based data into discriminative continuous features. The proposed approach is applied to forecast blood glucose levels with prediction horizons of 30 and 60 minutes by evaluating three patients from the Ohio T1DM dataset. The proposed method describes the performance for 30 minutes prediction horizons is 14.76 mg/dL RMSE while 25.48 mg/dL 60 minutes prediction horizons, respectively [37]. Introducing a dual-band fusion approach and an enhanced long short-term memory (LSTM-R) method leveraging UWB microwave technology aims to counteract the challenges posed by antenna miniaturization. A mean absolute relative difference (MARD) of 2.79% and a root-mean-squared error (RMSE) of 4.2367 mg/dL are the outcomes of testing the medium under tests [38]. Multisensory data was gathered by integrating millimeter-wave (mm-wave), near-infrared (NIR), and other auxiliary sensors. Correlations between mm-wave, NIR, skin temperature, pressure data, and blood glucose levels were analyzed. A random forest model was assessed using data from 5 healthy individuals who were undergoing an intravenous glucose tolerance test. This evaluation yielded an RMSE of 21.06 mg/dL and a MAE of 7.31% for glucose prediction [39]. The comparison of two models LSTM and ARMA (autoregressive moving average) was performed by evaluating the Type 1 diabetes mellitus of actual 29 patients for different prediction horizon. The output result denoted that LSTM has achieved better than ARMA dL for 5, 15 and 30-minutes prediction horizon of 3.13, 6.41 and 8.81 mg/dL RMSE and MAE of 1.98, 5.06 and 6.47 mg/dL [40]. The study [41] uses the OhioT1DM dataset and deep learning methods, such as Long Short-Term Memory (LSTM) and Convolutional Recurrent Neural Network (CRNN), to forecast blood glucose levels. The outcomes show that both models perform well in terms of accuracy; the LSTM model achieved an RMSE of 4.208 ± 0.92 mg/dL and an MAE of 3.22 ± 0.58 mg/dL, while the CRNN model achieved an RMSE of 4.0846 ± 0.9 mg/dL and an MAE of 3.0845 ± 0.52 mg/dL.

A multilayer deep neural network is proposed, incorporating a combination of LSTM and gated recurrent unit (GRU), to forecast future glucose levels based on historical data including glucose levels and meal information. This methodology underwent testing on a simulated insilico dataset comprising of 10 Adult T1D sce-narios generated using the UVA/Padova simulator. the findings demonstrate signifi-cant enhancements in the RMSE, MAE and MARD of 5.27 mg/dL, 4.04 mg/dL and 2.911%, respectively for 15-minute prediction horizon while RMSE, MAE and MARD of 14.85 mg/dL, 11.04 mg/dL and 8.42%, respectively for 30-minute prediction horizon. [42]. The implications of psychological stress and regular physical activity on variations in blood glucose (BG) were evaluated by including behavior-related biomarkers into two cutting-edge deep learning-based glucose prediction models: an LSTM and a CNN-LSTM network. Prediction horizons (PHs) of 30, 60, and 90 minutes are assessed for these models. It shows that the BG prediction models work better when estimated behavior-related biomarkers are included, using an ablation study. The LSTM model achieves mean absolute error (MAE) of 9.13 ± 0.95, 17.75 ± 1.93, and 31.85 ± 2.88 mg/dL, root mean square error (RMSE) of 12.35 ± 1.06, 24.71 ± 2.31, and 41.64 ± 4.12 mg/dL, and coefficient of determination (R2) of 95.34 ± 3.34%, 78.87 ± 4.35%, and 60.11 ± 4.76% respectively.

Table 2.1: Participant demographics (age group: 35-65 years).

Male (n=7)		Female (n=9)	
ID	**HbA1c**	**ID**	**HbA1c**
002	5.6	001	5.5
009	6.1	003	5.9
011	6.0	004	6.4
012	5.6	005	5.7
013	5.7	006	5.8
014	5.5	007	5.3
016	5.5	008	5.6
		010	6.0
		015	5.5

Similarly, the CNN-LSTM model achieves MAE of 9.37 ± 0.88, 17.87 ± 1.67, and 29.47 ± 2.13 mg/dL, RMSE of 12.51 ± 1.40, 24.37 ± 2.49, and 39.52 ± 3.89 mg/dL, and R2 of 94.65 ± 3.90%, 78.37 ± 4.11%, and 61.12 ± 4.30% respectively, across all PHs [43]. The paper is structured as follows: Section 2.2 provides a description of the Data Collection and Methods, Section 2.3 presents the experimental analysis, and Section 2.4 concludes the study while also outlining directions for future research.

2.2 Data Collection and Methods

Following the background to prediabetes and diabetes state-of-the-art techniques, this section elaborates on the dataset, data preprocessing, and proposed methodology in detail.

2.2.1 Data Collection and Preprocessing

Data was collected from 16 participants through the Duke University Health Side system under the Biomedical Engineering department. The preeminent blood glucose in the normal range were monitored with the Dexcom G6 continuous glucose monitors and Empatica E4 wrist-worn wearable devices for 8-10 days for each participant resulting in 7 features [33]. Participants with a medical history of chronic obstructive pulmonary disease (COPD), cardiovascular disease, cancer, or chronic kidney disease were ineligible for inclusion. Additionally, participants without access to a laptop or computer, necessary for daily data upload from the wearable device, were also excluded. Institutional Review Board (IRB) approved phone script was for communication with patients by study coordinator. Participants received total compensation of $150 for their visits and the dataset is described in Table 2.1.

The Dexcom G6 measures interstitial glucose concentration (mg/dL) every 5min using CGM sensor. Continuous glucose monitors (CGMs) provide high-frequency information regarding daily glucose variation and are recognized as effective for improving glycaemic control in individuals living with diabetes [44]. The Empatica E4 records the data through four sensors such as PPG (photoplethysmography), EDA (electrodermal activity), skin temperature (TEMP), and tri-axial

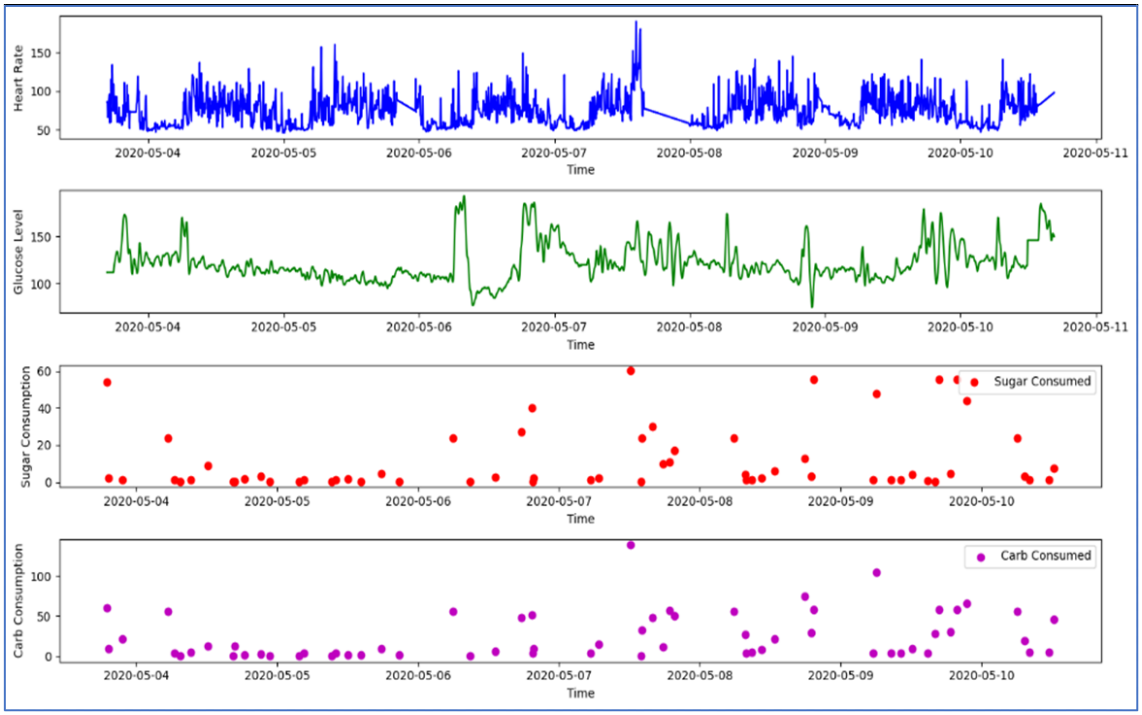


Figure 2.1: Different features are shown from different sensors and food log for 014 participant such as heart rate, glucose level, sugar, and carb.

accelerometer (ACC). PPG was sampled at 64 Hz by providing heart rate which was recorded once per second along with a signal of BVP (blood volume pulse) from which the IBI (interbeat interval) data was computed. EDA and skin temperature were sampled at 4 Hz and accelerometer was sampled at 32 Hz. Several additional diet habit variables were calculated in food log. All data for each participant is contained in a folder numbered 001 through 016, with a separate CSV file for each feature with corresponding timestamp and values. The seven features are: ACC, BVP, Dexcom, EDA, HR, IBI, TEMP and Food Log which refers to food items consumed by that participant throughout the study containing date, time, calorie, total carb, dietary fiber, sugar, protein and total fat. Gen-der and HbA1c information for each participant is also recorded and can be found in Demographics file [30].

The prediabetes dataset is publicly available [33]. The dataset underwent rigorous preprocessing, excluding entries from participants 003, 007, 013, 015, and 016 due to irregularities such as data recorded in different months, which could impact performance. After this pruning phase, the dataset contains data of 11 prediabetics patients. This data is characterized by the following features: glucose levels, heart rate measurements, sugar intake, and carbohydrate consumption, measured over one week for each patient, as reported in Figure 2.1. A linear approximation method synchronized these features to ensure dataset coherence, facilitating meaningful analysis.

2.2.2 Glucose Prediction Model

LSTM, a type of recurrent neural network (RNN), excels at processing diverse data types like speech and images. It's widely used in deep learning for tasks such as speech and handwriting recognition.

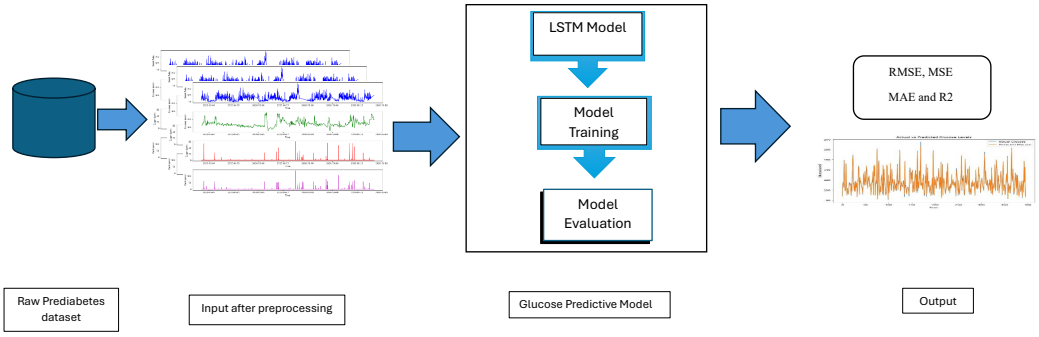


Figure 2.2: Architecture of the proposed model

LSTM's unique design includes gates to handle long-term dependencies effectively. This makes it suitable for complex tasks like glucose prediction with extended observation windows [35]. The TensorFlow LSTM Sequential model is designed for sequence prediction tasks, comprising an input layer, LSTM layer, and dense layer. The input layer receives input sequence features, while the LSTM layer learns temporal dependencies with its memory cells and gates. The final prediction is generated by the fully connected Dense layer, as illustrated in the LSTM flow Figure 2.2.

2.3 Experimental Analysis and Discussion

We utilized the Python programming language along with various libraries to preprocess the data and develop predictive models. First necessary libraries including pandas, NumPy, scikit-learn, and TensorFlow were imported for data manipulation, numerical computations, and deep learning tasks, respectively. Followed by outline steps to our approach sequences of glucose values along with related features patient were created. standard scaler is used to scale the data followed by splitting it into training and testing sets. As input sequences to the predictive model 36 lookback values means 3-hour horizon were iteratively used for over 7 days for each participant. LSTM-based deep learning model using TensorFlow's Keras API is deployed by training the prediabetes data. The performance of trained model is evaluated based on various metrics such as MSE, RMSE, MAE, and R2. Hyperparameters used for the proposed model are presented in Table 2.2.

Extensive analysis has been conducted and the comparison of actual vs predicted glucose

Table 2.2: Hyperparameters

Hyperparameter	Value
Lookback	36
Batch size	32
Activation function	relu
Optimization algorithm	Adam
Loss function	MSE
Number of epochs	150
monitor	val_loss Validation loss for each epoch

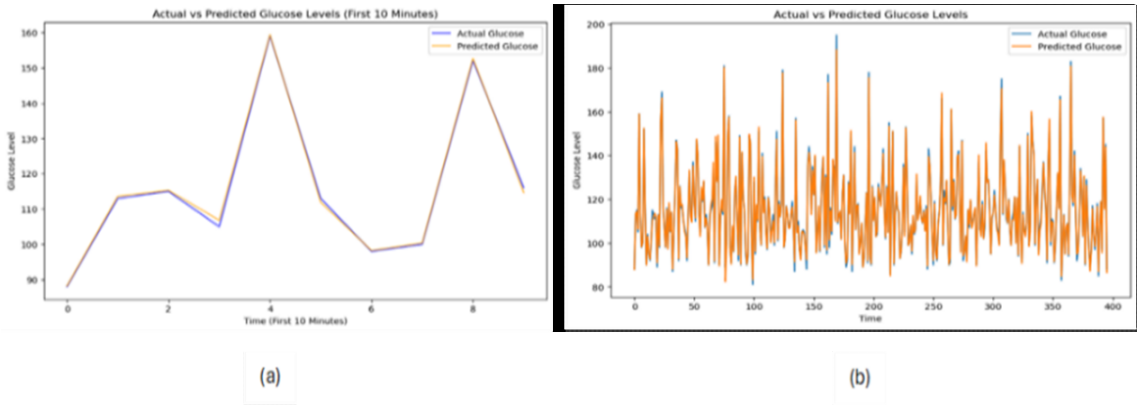


Figure 2.3: The comparison between actual and predicted glucose levels for participant 014 is illustrated. (a) displays the first 10-minute interval, while (b) presents the data for all intervals.

levels has been illustrated in Figure 2.3. Two sets of experiments were undertaken to evaluate the performance of the LSTM model in predicting glucose values. Initially, the LSTM model was assessed using only the historical glucose feature. The proposed model exhibited better performance as evidenced by improved metrics including RMSE, MSE, MAE, and the coefficient of determination (R2), achieving average values of 3.70, 15.08,1.82 and 0.97, respectively.

Subsequently, an expanded feature set comprising additional physiological parameters such as heart rate, along with glucose levels and dietary information including sugar and carbohydrate intake, was incorporated into the LSTM model for glucose prediction. The augmented model demonstrated enhanced predictive capabilities, as indicated by improved RMSE, MSE, MAE, and R2 scores, achieving average values of 1.83, 3.44, 1.24 and 0.99, respectively. The performance comparison of the proposed model is described in Table 2.3. LSTM outperforms ARIMA for blood glucose prediction, with MSE and RMSE of 24.075 and 4.906, respectively, compared to 108.140 and 10.399 for ARIMA [35]. An LSTM approach accurately predicts glucose levels, with MSE of 5.04 and MARD of 2.61, mitigating hypoglycemia risks due to insulin dosage delays [34].

Summarizing, the fact that the model used for experiment II yield a better accuracy compared with the model used for experiment I suggests that the additional features used for yielding predictions in experiments II have a relevant impact on the variations of the glucose levels, opening the way for further analysis of this aspect.

Table 2.3: Comparison with state of the art

Reference	Technique	RMSE
[35]	LSTM	4.906
[36]	LSTM	7.67
[34]	LSTM	5.04
[38]	LSTM-R	4.2367
Proposed Model Experiment I	LSTM	3.70
Proposed Model Experiment II	LSTM	1.83

2.4 Conclusions

This study introduces an efficient approach for precise and continuous blood glucose level prediction in individuals with prediabetes. Data preprocessing, including interpolation techniques are evaluated to enhance data quality. A Long Short-Term Memory (LSTM) model based on recurrent neural networks is trained for glucose prediction, outperforming existing tools. However, the current study is limited by its application at prediabetics patients only. The approach needs to be tested over diabetics patients data in order to prove its utility in practical contexts. Nonetheless, the good performances exhibited by the LSTM model encourage its application in context where additional digital biomarkers are available where higher accuracy levels may be achieved, particularly by integrating features like accelerometry and skin temperature.

Part II

CGM-Based Eating Event Detection and Prediction

Chapter 3

AI-Based Detection and Prediction of Eating Events Using CGM

The work presented in this chapter is based on:

M. Kashif, C. Wang, S. Flesca, and P. Veltri. "AI-Based Detection and Prediction of Eating Events Using Continuous Glucose Monitoring: A Comprehensive Review". Biomed. Signal Process. Control, vol. 127, article no. 111207, 2026"

Abstract

Continuous Glucose Monitoring (CGM) has become a cornerstone in diabetes technology, enabling high-resolution monitoring of interstitial glucose dynamics. Beyond its clinical use in glucose tracking, CGM data provide a unique opportunity to automatically detect and predict eating events, which are critical inputs for closed-loop insulin delivery, digital health interventions, and lifestyle monitoring. Over the past decade, diverse computational paradigms have been developed to address this challenge. Control-theoretic approaches, such as Moving Horizon Estimation, Kalman filtering, and nonlinear observers, exploit physiological models of glucose-insulin dynamics and achieve low-latency detection in simulations, but remain limited by parameter sensitivity and poor adaptability to free-living data. Machine learning (ML) frameworks have leveraged engineered features and postprandial glucose patterns to achieve balanced sensitivity and specificity, though performance is constrained by small cohorts and heterogeneous labeling. More recently, deep learning (DL) architectures, including long short-term memory (LSTMs), Convolutional neural networks (CNNs) and hybrid recurrent convolutional models, have demonstrated superior ability to capture temporal dependencies, yielding high accuracy and F1-scores on both in silico and real-world datasets. Hybrid and multimodal strategies that integrate CGM with insulin logs, physiological wearables, or motion sensors further improve robustness and reduce false positives, though at the expense of user burden and generalizability. Despite promising progress, major barriers remain, including dataset heterogeneity, inconsistent ground-truth labeling, variable evaluation metrics, and limited interpretability. The aim of this review is to systematically synthesize current evidence on

CGM-based eating event detection and prediction, critically compare methodological advances across modeling paradigms, and highlight key challenges and opportunities for achieving clinically reliable, real-world deployment.

3.1 Introduction

Monitoring eating behavior is fundamental to understanding human metabolism, managing chronic diseases such as diabetes, and enabling precision health interventions. Accurate identification of when and how individuals consume meals plays a crucial role in optimizing insulin dosing, preventing glycemic excursions, and informing personalized nutritional strategies. Traditionally, dietary assessment has relied on self-report tools including food diaries, 24-hour recall, and food frequency questionnaires that are inherently limited by recall bias, low adherence, and a lack of real-time capability [45, 46]. These challenges have motivated the development of automated, passive, and continuous methods for detecting and predicting eating events.

Automated food intake monitoring has emerged as a multidisciplinary research domain aimed at objectively capturing eating behavior using wearable and ambient sensors. A wide range of modalities such as wrist-worn motion sensors for hand-to-mouth gesture detection [47, 45, 48, 49, 50], acoustic sensors for chewing and swallowing recognition [51, 52, 53], and radar [54] or camera-based systems [55] for bite counting have been investigated to overcome the limitations of self-report methods. These approaches primarily capture behavioral proxies of eating activity. In contrast, CGM provides a physiological perspective by quantifying the body's metabolic response to food intake. This physiological grounding makes CGM a unique and clinically meaningful signal source within the broader landscape of automated eating event detection.

CGM refers to the use of wearable biosensors that continuously measure interstitial glucose levels every few minutes, offering a dynamic and minimally invasive window into glycemic fluctuations. These devices typically use subcutaneous electrochemical sensors to provide continuous glucose readings, enabling real-time assessment of postprandial metabolic activity. Initially developed for individuals with type 1 diabetes, CGM technology is increasingly used across a wider spectrum of populations including people with type 2 (T2D) diabetes, prediabetes, and even healthy individuals for monitoring glucose variability, studying metabolic responses, and guiding behavior change [11, 56]. The continuous nature of CGM data makes it a promising signal for the inference of eating behavior, with many studies leveraging these data to develop detection or prediction algorithms that are clinically meaningful and scalable to real-world applications [57].

3.1.1 Motivation for CGM-Based Eating Event Analysis

Automating the identification of meals from CGM data is not merely a technical convenience, but also a clinical and behavioral necessity. In diabetes care, failure to recognize meals in real-time can lead to mismatches between insulin delivery and food intake, resulting in hypo- or hyperglycemic events [58]. Beyond clinical populations, understanding eating patterns from glucose signals can

support personalized nutrition systems, digital coaching platforms, and behavior-aware health technologies [59, 60].

Unlike motion-based or audio-based sensing, CGM offers a physiological window into the body's metabolic response to food, capturing the downstream effects of carbohydrate intake. However, CGM data also come with inherent challenges: physiological delays (10-15 minutes), inter-individual variability in glycemic responses, and confounding factors such as physical activity, stress, or illness [45, 46]. To address these issues, AI-based methods have been increasingly explored to extract meaningful patterns from CGM time series.

3.1.2 Emergence of AI in Detection and Prediction

The transition from rule-based approaches to machine and deep learning has fundamentally changed how CGM data are interpreted. Early works applied classical ML algorithms such as Support Vector Machines (SVMs), Random Forests (RF), and XGBoost on engineered features derived from glucose trajectories [46, 60]. While interpretable, these models are often limited in their ability to generalize across individuals or adapt to noisy and free-living data.

In contrast, DL models especially CNNs, LSTMs, Gated Recurrent Units (GRUs), and Temporal Convolutional Networks (TCNs) learn hierarchical representations of glucose patterns directly from raw or smoothed CGM sequences. For example, personalized LSTM models have been shown to accurately predict meal times in insulin pump users by learning temporal dependencies from both CGM and insulin delivery logs [58, 61]. Similarly, CNN-based models have been used to detect local patterns of glucose excursions associated with meals and classify windowed CGM segments accordingly [60].

Importantly, recent efforts are not limited to retrospective meal detection. Predictive modeling which seeks to forecast an imminent eating event before it occurs is emerging as a critical capability. Such systems have the potential to support anticipatory insulin dosing, just-in-time dietary interventions, or real-time behavioral feedback [26, 45]. These applications require models capable of capturing subtle preprandial trends and making decisions under uncertainty, often in the presence of noise and limited labeled data.

3.1.3 Challenges and Open Questions

Despite advances in CGM sensing and AI-based algorithms, several challenges persist, ranging from sensor noise and physiological delay to inconsistent labeling, limited generalizability, and lack of standardized evaluation metrics [46, 11, 45, 60]. These open issues, along with real-world deployment barriers, are analyzed in detail in Section 3.6.

3.1.4 Scope and Contribution

Recent reviews have also examined broader modeling strategies for CGM data, including mechanistic models, machine learning techniques, and hybrid approaches for diabetes management. However, these works primarily focus on glucose prediction, insulin therapy optimization, or artificial pancreas

systems rather than specifically addressing the detection and prediction of eating events from CGM signals [62, 63]. This review paper provides a comprehensive, structured overview of existing literature on the AI-based detection and prediction of eating events using CGM. In contrast to broader reviews that include multimodal wearable sensors (e.g., IMUs, radar, audio), our focus is specifically on CGM as the primary input signal, with optional inclusion of insulin delivery or behavioral data when integrated directly into the CGM modeling process. Key contributions of this review include:

- A comparative analysis of AI models including both classical and deep learning approaches for CGM-based detection and prediction
- Outlines key datasets, labeling protocols, and population characteristics
- A synthesis of evaluation methodologies, including performance metrics and validation protocols
- A discussion of challenges and open problems, such as data variability, annotation quality, and deployment limitations
- A perspective on emerging directions, such as real-time systems, personalized modeling, and integration with digital therapeutics

This review targets researchers, clinicians, and engineers interested in developing robust, scalable systems for interpreting CGM data in the context of eating behavior.

3.1.5 Inclusion and Exclusion Criteria

We conducted a keyword-based search using the unified query (("Continuous Glucose Monitoring" OR CGM) AND ("meal detection" OR "eating event detection" OR "meal prediction" OR "eating event prediction" OR "meal onset prediction" OR "meal size estimation" OR "carbohydrate estimation") AND ("machine learning" OR "deep learning" OR "artificial intelligence" OR "model-based" OR "artificial pancreas" OR "hybrid" OR "multimodal")) across PubMed, IEEE Xplore, Springer, Scopus, and Google Scholar. This query was designed to capture studies linking CGM with both eating event detection and prediction tasks, including meal onset detection, carbohydrate estimation, and anticipatory meal modeling using model-based, machine learning, deep learning, or hybrid frameworks, as illustrated in Fig. 3.1. The screening process focused on studies that explicitly used CGM signals for detecting, estimating, or predicting eating events, ensuring consistency with the scope of the review. The retrieved records were screened based on title and abstract relevance, followed by full-text assessment of potentially eligible studies.

Inclusion criteria:

- Peer-reviewed journal articles or conference papers (2010- Feb 2026).
- Use of CGM data as a primary input for meal detection, meal size estimation, or prediction of eating events.
- Studies applying model-based estimation, machine learning, deep learning, or hybrid/multimodal approaches.

- Both simulated datasets (e.g., UVA/Padova T1DM simulator) and real-world human data (e.g., OhioT1DM, DiaTrend, clinical trials, feasibility pilots).
- Studies reporting quantitative outcomes (e.g., sensitivity, specificity, F1-score, detection delay, carbohydrate error, time-in-range).

Exclusion criteria:

- Studies without CGM data (e.g., only accelerometer, radar, or video-based gesture detection).
- Works focusing solely on glucose forecasting without explicit detection or labeling of meals.
- Papers limited to theoretical modeling without validation.
- Purely behavioral or dietary studies without algorithmic evaluation.

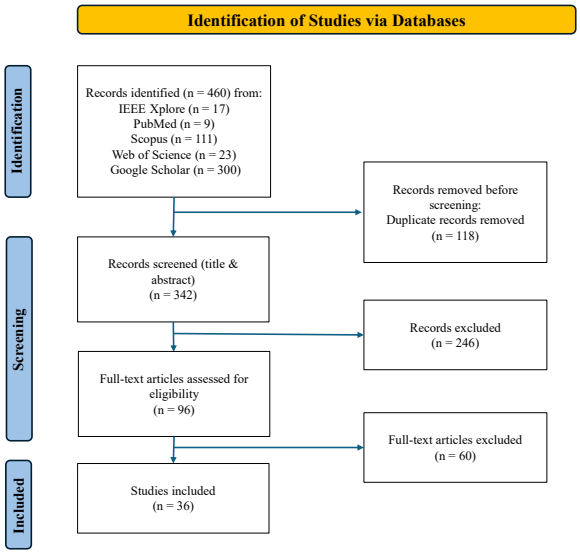


Figure 3.1: PRISMA-style flow diagram illustrating the literature search and study selection process for CGM-based eating event detection and prediction studies. Records were identified from major scientific databases including IEEE Xplore, PubMed, Scopus, Web of Science and Google Scholar, screened using predefined inclusion and exclusion criteria, and filtered through full-text review to obtain the final set of studies analyzed in this review.

3.1.6 Organization of the Paper

The paper is organized as follows. Section 3.2 describes sensor modalities for eating event detection and prediction. Section 3.3 categorizes modeling approaches. Section 3.4 summarizes datasets, labeling strategies, and evaluation metrics. Section 3.5 presents cross-study comparison and benchmarking. Section 3.6 discusses challenges and limitations. Section 3.7 concludes the review and outlines future directions.

3.2 Sensor Modalities for Eating Event Detection and Prediction

The performance of AI models in detecting or predicting eating events strongly depends on the quality and relevance of the input signals. Among various sensing approaches, CGM serves as the primary modality due to its direct measurement of metabolic responses to food intake. While some studies incorporate additional physiological or behavioral data, such as insulin, activity logs, heart rate or motion signals, CGM remains the central and most informative source for modeling meal-related dynamics. This section outlines the sensing strategies reported in recent studies, emphasizing CGM-based systems while briefly acknowledging multimodal fusion approaches where CGM acts as the core signal for eating event detection and prediction.

3.2.1 Continuous Glucose Monitoring

CGM is among the most widely used sensor modalities for meal detection, particularly in diabetes management contexts. These devices provide real-time continuous readings of interstitial glucose concentrations, typically sampled every 1-5 minutes. Because glucose levels rise in response to food intake especially carbohydrate-rich meals, CGM can serve as a valuable proxy for inferring the occurrence of meals.

Modern CGM devices such as the Dexcom G6, Freestyle Libre, and Medtronic Guardian use electrochemical sensors inserted subcutaneously to measure glucose concentrations in the interstitial space. The readings are transmitted wirelessly to receivers or smartphones and are accompanied by timestamped logs of calibration events, alerts, and sensor status. The 10-15 minute lag between blood glucose and interstitial glucose, due to fluid transport delay, is well-known and must be considered when interpreting postprandial glucose rises [45, 11].

In the context of eating event detection, CGM signals serve as physiological proxies for food intake: meals typically cause a rise in glucose values after 10-30 minutes, depending on meal composition and individual metabolic profiles. This makes CGM data a promising input stream for detecting postprandial events and even predicting likely meal onset using preprandial glucose dynamics. However, in free-living scenarios, CGM traces may be affected by signal noise, temporary dropouts, or calibration shifts. These artifacts must be carefully handled during preprocessing, as they can introduce false positives or obscure actual meal-related excursions.

3.2.2 Strengths and Challenges of Using CGM for Meal Detection

CGM has several compelling advantages as a primary sensor for eating event analysis:

- **Unobtrusiveness**: Once applied, CGM devices operate continuously for 7-14 days with minimal user interaction.
- **Physiological Relevance**: Unlike motion sensors, CGM measures the metabolic effect of meals rather than behavioral proxies.
- **Clinical Integration**: CGM is already part of routine diabetes care, making it deployable in real-world clinical settings.

- **Temporal Resolution**: With a high sampling rate (up to 288 samples/day), CGM captures fine-grained changes associated with meals.

  However, using CGM alone for detecting or predicting meals also presents notable challenges. While CGM provides a valuable physiological signal, its performance can be limited compared to multimodal systems that integrate contextual data such as insulin dosing, heart rate, or activity information. The main challenges include:
- **Physiological Lag**: The inherent sensor delay limits real-time responsiveness and complicates precise timestamp alignment for detection models.
- **Inter-individual Variability**: Different individuals exhibit different glycemic responses to the same foods, especially across diabetic and non-diabetic populations.
- **Signal Ambiguity**: Non-meal events such as stress, exercise, or illness can trigger glucose fluctuations similar to meals.
- **Annotation Noise**: Ground truth labels often rely on self-reported food logs or insulin bolus timestamps, which may be incomplete or inaccurate [56, 46].

3.2.3 Multimodal Sensor Fusion

While this review focuses on CGM as the primary signal, it is important to note that some of the most effective models incorporate insulin pump data, bolus timestamps, or carbohydrate entries to improve meal detection precision. For example, Prioleau et al. used personalized LSTM models trained on CGM plus insulin data to identify meals with high sensitivity [58]. Similarly, CGM with motion and insulin data have been integrated to distinguish between meals, snacks, and physical activity [46].

Several studies demonstrate that combining sensor modalities leads to improved detection accuracy and robustness. CGM and hand gestures were fused to automatically trigger insulin boluses without requiring manual meal announcements [60]. Glucose profiles compared under different eating routines (controlled vs. free-living), using fixed mealtimes and wearable monitoring to study variability in glycemic patterns [11]. In a similar multimodal setup, wrist-worn motion sensing with CGM combined to refine the temporal annotation of eating episodes, particularly in prediabetic and obese populations. Their study found that self-reported mealtimes were often misaligned with physiological patterns, which CGM could help correct [59]. Beyond diabetes-focused studies, CGM applied as an objective marker of meal intake in individuals with binge-spectrum eating disorders, showing its applicability in behavioral health contexts as well [56]. In these studies, CGM remained the central physiological source, with auxiliary data serving to enhance robustness or disambiguate non-meal glucose rises.

CGM offers a rich and increasingly accessible physiological data stream for modeling eating behavior. Its continuous, unobtrusive nature and direct link to metabolic outcomes make it an ideal primary modality for AI-based detection and prediction of eating events. At the same time, researchers must contend with physiological delays, inter-individual variability, and weak labels. These factors underscore the need for robust model architectures and careful annotation strategies to realize the full potential of CGM in meal detection systems.

The next section will delve into how AI models process these signals ranging from handcrafted feature classifiers to end-to-end deep learning architectures. Meal detection models also rely heavily on how "meal events" are defined. Studies vary in their labeling windows from a fixed 10-15 minutes before to 60 minutes after recorded eating which complicates cross-study comparison and ground truth consistency. These issues will be explored in the next section.

3.3 Modeling Approaches for Eating Event Detection and Prediction

The modeling landscape for meal detection and prediction using CGM data has evolved substantially over the last decade, driven by the need for automated, personalized, and real-time solutions to support diabetes management and behavior-aware health interventions. Across the literature, a variety of paradigms have emerged, ranging from physiological model-based estimation (e.g., Moving Horizon Estimation and Kalman filters), to supervised ML and DL architectures capable of capturing temporal dependencies in glucose fluctuations. In CGM-based approaches, meal detection typically relies on identifying sharp glucose excursions or characteristic postprandial patterns. Early studies employed rule-based signal processing and peak detection algorithms, while more recent work applied machine learning classifiers trained on glucose dynamics [45]. Some frameworks have integrated auxiliary cues, such as insulin dosing or wearable motion data, to improve accuracy and clinical usability [60]. These advances highlight the practicality of CGM signals for detecting eating episodes in both controlled and free-living settings [46, 64].

This section reviews the main modeling paradigms used for CGM-based eating event detection and prediction, including model-based estimation, machine learning, deep learning, and hybrid/multimodal frameworks. To improve cross-study interpretability, each subsection summarizes methodological characteristics, datasets, and reported performance metrics. Comparative insights are further illustrated using summary figures and tables that highlight trade-offs between sensitivity, detection delay, and false alarm rates across representative studies.

3.3.1 Model-Based Estimation Techniques

Moving Horizon Estimation (MHE) has been widely used for meal detection due to its ability to model the glucose-insulin dynamics in a predictive control framework. Several control-theoretic frameworks have been explored. MHE is a model-based state estimation technique that reconstructs hidden physiological variables such as glucose rate and insulin sensitivity by minimizing the difference between measured and model-predicted glucose over a sliding time window. This allows the estimation of disturbances in glucose trajectories caused by unannounced meals. Committed Moving Horizon Estimation (CMHE) has been introduced for detecting and quantifying unannounced meals in type 1 diabetes. This framework runs multiple MHE instances in parallel and applies a "commitment" rule to stabilize meal detection decisions. Evaluated in the UVA/Padova in silico type 1 diabetes simulator, CMHE achieved daily detection rates up to 88.5%, with 100% detection of main meals and significantly reduced false alarms compared with standard MHE approaches. While promising, the reliance on physiologically realistic models limits adaptability to free-living data

[65]. MHE was combined with Linear Discriminant Analysis (LDA) using estimated glucose rates into binary detection decisions as features. LDA is a supervised classification method that separates "meal" and "non-meal" states based on the temporal patterns of estimated glucose appearance rates. It provides a simple yet effective way to classify the outputs of the MHE framework, enhancing robustness and reducing false detections. The framework tested in the UVA/Padova simulator, the system detected all simulated meals within 35 minutes and showed better sensitivity than threshold methods. While effective in simulation, its performance in real-world CGM data remained untested [2]. In a follow-up study, the MHE + LDA framework applied to CGM data from 12 pediatric T1D patients. The approach achieved ∼88% sensitivity, false alarms/day less than 0.5, and detection delays of 20-30 minutes outperforming threshold-based detectors. However, low-carbohydrate meals were harder to detect due to subtle CGM changes [3].

A study applied Kalman filtering with residual hypothesis testing for CGM-based meal estimation. Simulation results showed reduced detection delays compared to thresholds, but no large-scale clinical validation or detailed sensitivity metrics were reported. The approach remains primarily proof-of-concept with limited real-world evidence [66]. Turksoy et al. integrated a meal detection module into the Multivariable Artificial Pancreas (MAP) system, where CGM signals were processed with a state-augmented minimal model estimated using an Unscented Kalman Filter (UKF). Residual innovations from the filter were subjected to hypothesis testing to detect unannounced meals. In clinical studies involving nine type 1 diabetes subjects, the method successfully identified 61 out of 63 meals/snacks, with only two missed detections and one false alarm. The average glucose rise at detection was 16 ± 9.4 mg/dL, indicating that the algorithm triggered shortly after the onset of postprandial excursions, thus achieving low latency while maintaining high reliability. However, performance declined for low-carbohydrate meals, where the CGM excursions were less pronounced, underscoring the dependence on detectable postprandial patterns [20].

A super-twisting observer (STO) meal detector was refined by combining residual generation with a Kalman-filtered glucose derivative. Tested on 14 days of free-living CGM data from 30 type 1 diabetes patients, the method achieved 70% recall, 73% precision, ∼45 min detection delay, and 1.4 false positives/day. False alarms were linked to rising glucose trends, while missed detections often followed calibrations or small-carbohydrate meals. The approach shows promise for real-time use but still requires validation in larger and more diverse populations [4]. Two offline methods were developed to detect meals from CGM: a heuristic approach (Savitzky-Golay + rate-of-change thresholds) and a pattern-matching method using Dynamic Time Warping. Tested on 37 adults with T1D in a 7-day clinical trial, the heuristic method showed the shortest detection delay (19.5 min, 0.42 false positives/day), while DTW achieved higher breakfast sensitivity of 90% but longer delays of 34.6 min while Dinner remained hardest to detect with ∼60% sensitivity. Neither method estimated carbohydrate size [5]. An Unscented Kalman Filter with cross-covariance analysis was applied to a glucose-insulin model to detect meals from CGM. In in-silico UVA/Padova simulations of 10 T1D adult patients, ten days (30 meals/subject), the method achieved specificity 95%, 90% sensitivity, 94% accuracy, and a mean delay of 28 minutes. While robust to variability, validation was limited to simulation only [6].

Overall, model-based estimation techniques such as MHE, Kalman filtering, and observer-

based frameworks provide interpretable and physiologically grounded solutions for meal detection. These approaches are particularly attractive for real-time control systems due to their relatively low detection latency and explicit representation of glucose-insulin dynamics. However, their performance often depends on individualized parameter tuning and may degrade under free-living variability, limiting scalability across heterogeneous populations. A comprehensive analysis is shown in Table 3.1.

Table 3.1: A comprehensive overview of model-based estimation approaches for eating event detection and prediction using CGM data. Abbreviations: FL = free-living; IS = in-silico; RCT = randomized controlled trial; CHO = carbohydrate; ROC = rate of change; AUC = area under the receiver operating characteristic curve.

Ref	Features	Methodology	Dataset	Performance	Task Type	Challenges / Limitations
[65]	Model residuals; glucose appearance R_a	CMHE with commitment logic	IS UVA/Padova; T1D adults; multi-day scenarios	Daily detection up to 88.5%; 100% main meals; reduced false alarms vs MHE	Real-time	Simulation-only evaluation; parameterized physiology; limited free-living adaptability
[2]	Estimated R_a; trend	MHE + LDA classifier	IS UVA/Padova; T1D, controlled meal scenarios	All meals detected within ≤ 35 min; sensitivity $>$ threshold baseline	Real-time	Simulation-only evaluation; real-world performance unknown
[3]	Estimated R_a features	MHE + LDA (clinical)	12 pediatric participants with T1D; CGM	approximately 88% sensitivity; < 0.5 FP/day; delay 20–30 min	Real-time	Low-CHO meal detection; small pediatric cohort
[66]	Filter innovations/residuals	Kalman filtering + residual hypothesis testing	IS proof-of-concept; limited clinical reporting	Reduced delay vs thresholds; no large-scale metrics	Real-time	Limited clinical validation; parameter/model mismatch risk
[20]	UKF innovations; residual tests	State-augmented minimal model + UKF; hypothesis testing	9 participants with T1D; clinical multivariable AP study; 63 meals/snacks	61/63 detected; 1 FP; mean glucose rise at detection (16 ± 9.4) mg/dL	Real-time	Low-CHO meals less pronounced; dependence on postprandial pattern
[4]	CGM derivative + STO residuals	Super-twisting observer with residual fusion	30 participants with T1D; 14-day free-living study	Recall 70%; Precision 73%; delay 45 min; 1.4 FP/day	Real-time	Rising-glucose FPs; misses after calibration/small meals; scalability
[5]	Smoothed CGM; ROC; DTW distances	Heuristic (Savitzky-Golay + thresholds) and DTW	37 participants with T1D; 7-day controlled study	Heuristic: delay 19.5 min; 0.42 FP/day. DTW: breakfast sensitivity 90%; overall delay 34.6 min; dinner sensitivity 59.3%	Retrospective	Offline processing; no CHO estimation; meal-type variability
[6]	Disturbance parameter D from CGM	UKF + cross-covariance detector	IS UVA/Padova; 10 virtual adults; 30 meals each	Specificity 95%; Sensitivity 90%; Accuracy 94%; delay 28 min	Real-time	Simulation-only evaluation; parameter tuning sensitivity; lack of clinical validation

Task Type indicates whether the method performs real-time meal detection (online detection during CGM monitoring) or retrospective meal detection (offline analysis of completed glucose trajectories).

3.3.2 Machine Learning Approaches

To overcome the limitations of strict physiological assumptions, several studies have explored supervised machine learning for meal detection. The viability of using CGM signals has been explored to objectively detect meal consumption in individuals with binge-spectrum eating disorders (B-EDs). They collected data from 22 adults wearing CGMs over a 2-week period, with self-reported meal episodes used as ground truth. They applied Random Forest classification models on features derived from CGM readings to distinguish "eating" vs "non-eating" windows. Their optimal model achieved 82% accuracy, 71% sensitivity and 94% specificity. While the study demonstrates that CGM may be a promising passive sensor for meal detection in a non-diabetic behavioral context, its limitations include the small sample size, reliance on self-report logs (which may contain errors), and lack of validation in broader free-living populations [56]. An ensemble of Artificial Neural Network (ANN), Random Forest, and Logistic Regression has been proposed to detect unannounced meals from CGM and insulin data. Tested on the UVA/Padova in-silico simulator and the OhioT1DM dataset, the approach achieved F1-scores up to 91% in simulation and 66-72% on real data, with 0.17-0.32 false positives/day and detection delays of $\sim$21-41 minutes. Integration into automated insulin delivery improved time-in-range by 10-16% and reduced postprandial hyperglycemia. While effective, the method was sensitive to ensemble voting rules and needs prospective validation in clinical settings [7]. An AI based novel detection algorithm was trained on CGM data from the CDDIAB study (14 patients, 30 days each) to detect meals in real-time. Using a leave-one-patient-out design, it achieved AUC 91.5%, sensitivity 63%, precision 59%, F1-score 60%, and 25.6 min delay. Performance was limited by small sample size and mislabeled meals, but results support integration into diabetes management platforms [8].

A pattern-based machine learning pipeline is developed to detect meals from CGM subsequences by identifying individualized glucose response patterns. Using the OhioT1DM dataset of 12 patients, they trained classifiers such as Random Forest, AdaBoost, and Gradient Boosting on qualitative CGM features as presented in Fig. 3.2. The best models achieved F1-scores between 67% and 83%, with an average of 1.7 false positives/day and about one missed meal/day. While effective, the approach was limited by the quality of self-reported meal logs, as false detections sometimes corresponded to unreported eating events rather than algorithmic error [1]. In related work, population-level versus individual-level clustering of postprandial CGM subsequences have been analyzed to extract meal response patterns. Applying fuzzy C-means clustering on the OhioT1DM dataset, they reported average detection success ratios of 0.54 (15-min margin) to 0.76 (45-min margin). Individualized models slightly outperformed population models for patients with distinct glucose responses, while population models were useful when individual training data was scarce. Limitations included the sensitivity to time-margin assumptions and the need for sufficient logged meals per subject to achieve stable clustering [57]. A Dynamic Mode Decomposition (DMD) framework was developed to identify meal-induced instabilities in CGM signals without requiring insulin data or patient-specific parameters. Sliding-window eigenmodes were extracted from CGM dynamics and classified using logistic regression, with meal onset linked to transient instability markers. In in-silico tests with 100 UVA/Padova virtual subjects, the method detected over 89% of

meals within 10-15 minutes, with sensitivity ranging from 69% for <40 g meals to >98% for >90 g meals. Clinical validation on two trials (35 participants) achieved recall of 80-93%, AUC up to 0.94, and detection delays of 20-25 minutes, outperforming baseline methods. Limitations included reduced sensitivity for small meals and the lack of long-term free-living evaluation [9].

A meal detection framework has been proposed for healthy adults using CGM and physiological monitoring (heart rate, body temperature, respiration). Data were collected from nine metabolically healthy male volunteers under controlled inpatient conditions, with exact meal times logged. Tree-based models were applied, with a Random Forest in a cold-start scenario achieving AUC-ROC of 0.891 and AUC-PR of 0.803 on 110 minutes data, while a Gradient Boosted Tree in a non-cold start case achieved AUC-ROC over 0.996 and AUC-PR of 0.964. The system demonstrated strong potential for objective validation of participant compliance in dietary protocols. Limitations include the small sample size, tightly controlled environment, and reliance on post-exercise nutritional settings, which may not generalize to free-living conditions [67]. A random forest classifier was trained on CGM two-hour morning segments to identify breakfast responses. Data used from the CTRAL cohort of 15 adults without diabetes, 4 weeks CGM and validated on the Digital Health dataset of 36 Hispanic/Latino adults with or at risk for T2D, 14 days CGM and logs. The model showed good agreement with annotated postprandial responses, with no significant differences in glucose rise or incremental AUC. Limitations included small sample sizes, focus on morning data only, and potential errors from noisy CGM signals [68]. A recent in-silico study explored supervised learning for both meal detection and carbohydrate estimation in T1D using the UVA/Padova simulator. The meal detector used current glucose and its backward derivative as inputs, with label shifting and oversampling to address physiological delay and class imbalance, while regression models were trained for carbohydrate estimation. This study is relevant because it jointly addresses meal detection and size estimation, but its conclusions remain limited by exclusive in-silico validation [69]. Machine learning has also been used to retrospectively identify missing meal records in diabetes datasets. A recent study proposed a logistic-regression-based algorithm that detects unreported meal events using CGM-derived features and residuals between observed glucose values and model-predicted glucose dynamics. This approach can help identify missing meal annotations and improve the quality of datasets used for training automated meal detection systems [70].

A fuzzy logic-based algorithm has been developed that combines wavelet filtering, qualitative trend analysis, and CGM plus insulin data to both detect meals and estimate their carbohydrate (CHO) content. Evaluated in 30 in-silico subjects from the UVA/Padova simulator (10 adults, 10 adolescents, 10 children), the algorithm achieved 91.3% sensitivity and 9.3% false positives, with a mean CHO estimation error of 23.1%. It maintained an average time-in-range (70-180 mg/dL) of 76.8% and mean glucose $\sim$142 mg/dL across simulations. While the method demonstrated strong performance in simulation and the ability to reduce postprandial hyperglycemia without manual meal announcements, its dependence on fuzzy membership tuning and the absence of real-world validation remain limitations [71].

Machine learning approaches improve adaptability by learning data-driven representations of CGM dynamics without relying on fixed physiological assumptions. Ensemble models and

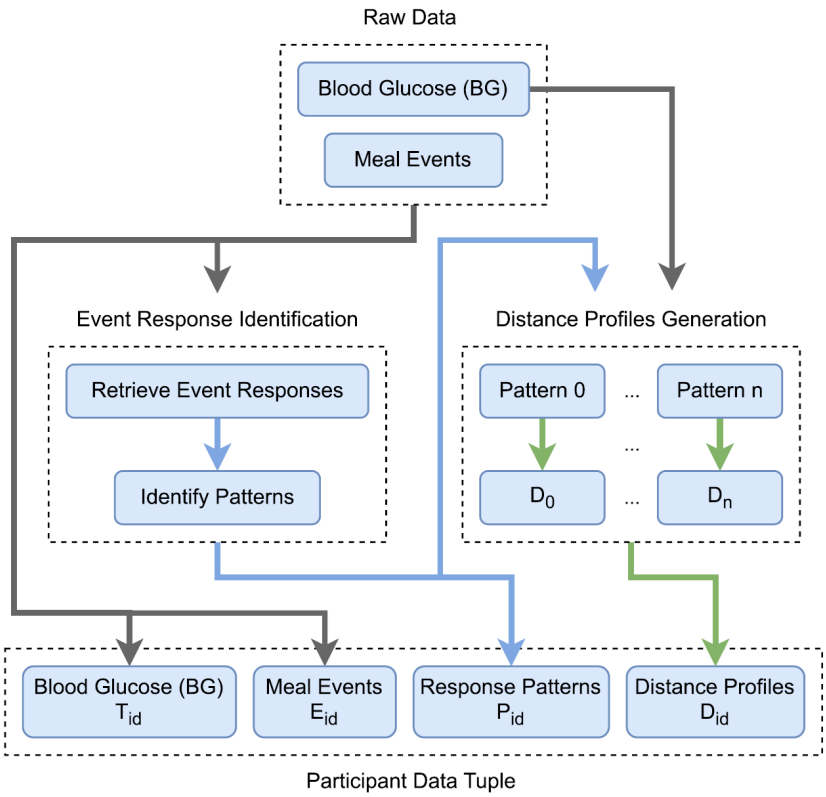


Figure 3.2: A pattern-based machine learning pipeline to detect meals from CGM subsequences by identifying individualized glucose response patterns [1].

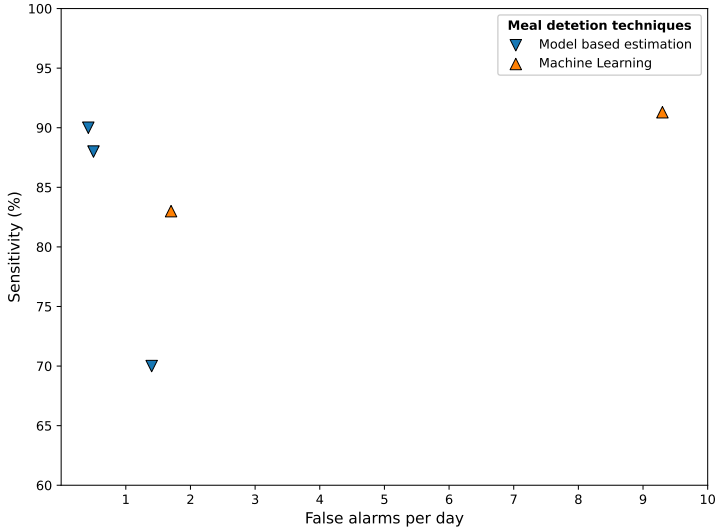


Figure 3.3: Trade-off between sensitivity and false alarms per day reported by representative CGM-based meal detection methods. Colors indicate the methodological category.

pattern-based pipelines have demonstrated competitive sensitivity and F1-scores, particularly when personalized training data are available. However, their performance remains sensitive to label quality, dataset size, and heterogeneous evaluation protocols, making standardized benchmarking across studies challenging as shown in Table 3.2.

To facilitate cross-method comparison across the reviewed literature, Fig. 3.3 summarizes the trade-off between sensitivity and false alarm rates for representative model-based and machine learning approaches. While model-based estimation methods tend to achieve relatively low false alarm rates due to explicit physiological modeling, machine learning approaches can provide competitive sensitivity but may produce higher false alarm frequencies depending on the training data and feature design. Such visualization helps highlight the practical trade-offs faced when designing real-time meal detection systems.

3.3.3 Deep Learning-Based Models

Deep learning models have gained popularity due to their ability to learn temporal dependencies in time-series CGM data. Multiple AI and classical models have been investigated for meal detection from CGM signals in in-silico cohorts of 10 adults, 10 adolescents, and 10 children with type 1 diabetes, each simulated for 30 days with 5 meals/day using the UVA/Padova simulator. They compared Decision Trees, Random Forests, SVM, Naïve Bayes variants, Feedforward NNs, LSTM NNs, and a threshold-based method. Results showed that the Feedforward NN achieved the best performance in adults, Random Forests achieved the highest performance F1-score of 72%, while SVMs performed best in adolescents with 80% F1-score and 76% for children, outperforming the threshold baseline detecting over 81% of meals with relatively few false positives. Threshold methods reached high precision of over 92% but missed many meals results in lower recall approximately 61%. Overall, AI models offered a more balanced trade-off, highlighting their potential for closed-loop insulin systems, though findings remain limited to simulated data [72].

A sequence-to-sequence LSTM model with quantile regression is designed to detect unannounced meals and estimate carbohydrate size in in-silico type 1 diabetes. Using UVA/Padova simulator data of 10 virtual adults for 2 months, the model achieved 93% precision, 76% recall, and 38 min detection delay, while estimating carbohydrate size with a mean error of 18g. Integration improved glycemic outcomes compared to unannounced meals, though detection of small snacks remained less reliable and clinical validation is still lacking [10]. Recurrent neural networks (RNNs), including LSTM models with 1D convolutional layers, have been applied to detect meal and physical activity events from free-living CGM, insulin pump, and self-reported logs in people with type 1 diabetes. Using de-identified long-term datasets from Tidepool (30-700 days per subject, age 15-72 years), they addressed challenges of missing values and noise through outlier reconciliation and imputation. The best model achieved overall accuracy of 95.7%, with meal detection 98.05%, exercise detection 93.4%, and concurrent meal-exercise 55.6%. Performance declined without preprocessing, highlighting the importance of data cleaning. While highly effective for separate meal or exercise detection, the model struggled with overlapping events and may overproduce false alarms in noisy free-living data [46]. CNN, LSTM, and Bi-GRU-attention models have been

evaluated on CGM data from 38 T1D patients (10, 922 days), with total 2483 registered meals from 16 patients while the rest of the patients not having any meals labeled (consumed but not logged or added to the dataset). After the manual correction, 1167 labeled meals were selected. While recall was high (87-91%), precision was poor (21-23%), giving F1-scores less than 36%, improved to 56% with post-processing. The study highlighted data imbalance and label noise as major barriers to clinically useful performance [73]. LSTM, CNN-LSTM, ConvLSTM, and BiLSTM-CNN models have been trained on free-living data from 11 people with T1D, combining CGM, insulin, carbs, and activity logs. After preprocessing with imputation and denoising, the best models (CNN-LSTM and BiLSTM-CNN) achieved 91.42% - 95.86% accuracy and F1-scores of 91.41% - 96.26%. Exercise-only detection was weaker, likely due to missing wearable signals. Limitations included small sample size, self-reported labels, and reliance on personalized training [74]. A personalized LSTM ensemble is implemented that ingests 120-min CGM windows (25 samples at 5-min rate) to classify whether a meal onset occurs within the most recent 30 min of the window. Data collected in in-silico UVA-Padova T1DM simulator (7-day scenarios; 10 adults, 10 adolescents, 10 children). Among four ensemble combination schemes, the method achieved mean c-statistics $\sim$75-80% and mean detection times $\sim$7-13 min, with the simple-averaging ensemble performing best. Key limitations are the exclusive in-silico evaluation, heuristic labeling (23% positives defined by the last-30-min rule), and lack of validation under free-living variability (e.g., exercise, sensor noise drift), which may affect generalization [64].

Table 3.2: A comprehensive overview of machine learning based approaches for eating event detection and prediction using CGM data.

Ref	Features	Methodology	Dataset	Performance	Task Type	Limitations
[56]	CGM windows; statistical features	Random Forest (eating vs non-eating)	22 adults with binge-spectrum ED; 2 weeks; self-logs	Accuracy 82%; Sensitivity 71%; Specificity 94%	Retrospective	Small cohort; self-report bias; non-diabetic cohort
[7]	CGM + insulin + CHO features	Ensemble (ANN+RF+LR) for unannounced meals	Two IS cohorts comprising 20 and 47 participants; OhioT1DM	F1-score up to 91% (IS) and 66–72% (Ohio); FP 0.17–0.32/day; delay approximately 21–41 min; TIR improved by 10–16%	Real-time	Sensitive to voting rules; prospective clinical validation needed
[8]	Windowed ROC; trend stats	Real-time ML; leave-one-patient-out	CDDIAB; 14 participants; 30 days each	AUC 0.915; Sensitivity 63%; Precision 59%; F1-score 60%; delay 25.6 min	Real-time	Small cohort; mislabeled/missed meals $\rightarrow$ FP, single-center study
[1]	CGM response patterns	Personalized RF/AdaBoost/GBM	OhioT1DM; 12 T1D adults; multi-day	F1-score 67–83%; approximately 1.7 FP/day; approximately 1 missed meal/day	Retrospective	Label quality limits; individualized models outperform generalized models
[57]	Postprandial subsequences; clustering	Fuzzy C-means (population vs individual patterns)	OhioT1DM	Detection success 0.54 (15 min margin) to 0.76 (45 min)	Retrospective	Sensitive to time-margin; needs enough logged meals
[9]	DMD eigenmodes; instability markers	DMD + Logistic Regression (online)	100 IS subjects (90 days); 2 clinical studies (35 participants; 2 days)	IS: >89% within 10–15 min; Clinical: recall 80–93%; AUC $\leq$0.94; delay 20–25 min	Real-time	Lower sensitivity for small meals; long-term FL needed
[67]	CGM + HR/Temp/Resp features	RF/GBT; cold-start vs. non-cold-start	9 healthy males; controlled inpatient	AUC-ROC 0.891 (cold-start); 0.996 (non-cold); AUC-PR 0.803/0.964	Retrospective	Small, controlled cohort; limited generalization
[75]	Steps/HR + CGM features	SVM/RF/XGB (wearables with/without CGM)	10 participants; 2 weeks; FL	10–23% F1/MCC with CGM; 15–40% Specificity; 17–35% Precision	Retrospective	Small/short study; app logs; Sensitivity drop for RF
[68]	2-h morning CGM; PPGR features	Random Forest (breakfast detection)	CTRAL: 15 participants (4 weeks); Digital Health: 36 participants (14 days)	Good agreement with PPGR annotations; no significant differences in rise or iAUC	Retrospective	Morning-only focus; small sample size; CGM noise
[64]	120-min windows; 5-min sampling	Personalized LSTM ensemble (seq classification)	IS UVA/Padova; 10 adults, 10 adolescents, 10 children; 7-day scenarios	c-stat 75.12–79.52%; detection time 7.08–12.84 min	Real-time	Heuristic labels based on the last 30-min window; no real-world validation
[71]	CGM + insulin trends; wavelet/qualitative	Fuzzy logic meal + CHO estimation	IS UVA/Padova; 30 participants (10 adults, 10 adolescents and 10 children)	Sensitivity 91.3%; FP rate 9.3%; mean CHO estimation error 23.1%; TIR 76.8%	Real-time	Membership tuning; simulation-only

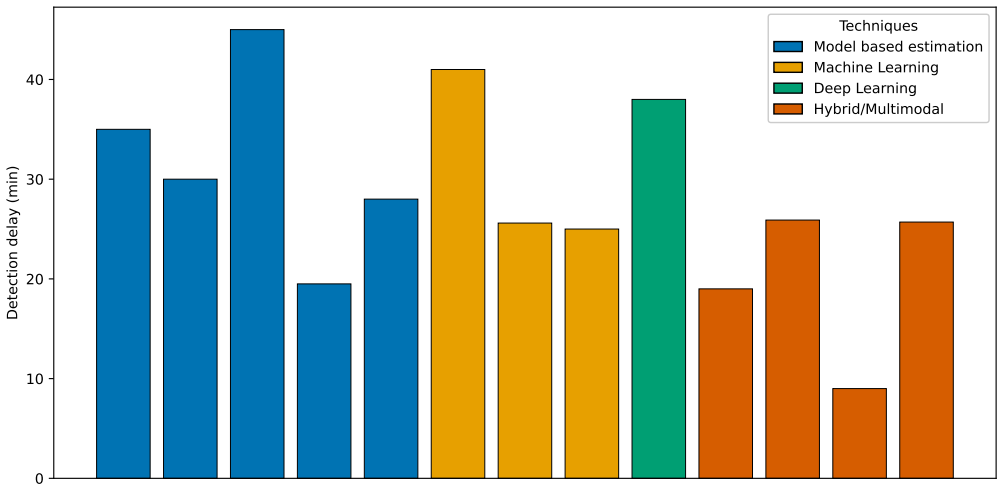


Figure 3.4: Detection delay across representative studies, grouped by methodological family. [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14] from left to right.

Deep learning architectures unify feature extraction and classification, enabling the modeling of complex nonlinear temporal dependencies in CGM time-series data. Compared with traditional ML, DL approaches consistently yield higher detection accuracy and robustness to signal noise. Nevertheless, they typically require large annotated datasets and remain difficult to interpret, while variations in datasets and evaluation protocols complicate direct comparison across studies. Despite these challenges, DL methods show strong potential for future real-time, personalized meal detection systems as described in Table 3.3.

Detection latency is another critical factor for practical deployment of meal detection algorithms, particularly in closed-loop insulin delivery systems. Fig. 3.4 compares reported detection delays across representative studies grouped by methodological paradigm. Model-based techniques typically achieve moderate delays due to explicit glucose-insulin modeling, while machine learning and hybrid approaches may exhibit wider variability depending on training data and detection criteria. These differences highlight the importance of balancing sensitivity with clinically acceptable detection latency.

Table 3.3: Overview of deep learning based approaches for eating event detection and prediction using CGM data.

Ref	Features	Methodology	Dataset	Performance	Task Type	Limitations
[72]	Simulated CGM windows by age group	Classical ML/DL vs. threshold baseline	IS UVA/Padova; 10 adults, 10 adolescents, and 10 children; 30 days; 5 meals/day	Adults: FFNN F1-score 72.2%; adolescents: SVM F1-score 80.1%; children: SVM F1-score 76%; thresholds: precision >92%, recall 61.2–61.8%	Retrospective	Simulation only, age-dependent tuning
[10]	Simulator sequences; multi-task outputs	Seq2Seq/CNN + quantile regression	IS UVA/Padova; 10 virtual adults; approximately 2 months	Precision 93%; Recall 76%; delay 38 min; CHO MAE 18 ± 15 g	Real-time	No real CGM input; clinical transfer unproven
[46]	CGM + insulin + wearable subset	Multimodal RNN/GBM ensembles	Tidepool T1D; 30–700 days per subject	Accuracy 95.7%; meal detection 98.05%; exercise detection 93.4%; concurrent meal-exercise detection 55.6%	Real-time	Overlapping events hard to detect; false alarms if data are not cleaned
[73]	CGM + insulin windows; engineered/DL features	CNN/LSTM/GRU + post-processing	38 participants with T1D; 10,922 days; 2,483 registered meals; 1,167 selected meals; meals labeled for 16 patients	Recall approximately 87–91%; Precision approximately 21–23%; F1-score approximately 34–59.7% after post-processing	Retrospective	Label noise; class imbalance; precision bottleneck
[74]	CGM statistics/derivatives; insulin/CHO	LSTM, CNN-LSTM, ConvLSTM, BiLSTM–CNN	11 participants with T1D; multi-week FL	Accuracy 91.4–95.9%; F1-score 91–96% with best CNN-LSTM/BiLSTM-CNN models	Real-time	Small sample size; self-reported labels; weak exercise-only detection

3.3.4 Hybrid and Multimodal Approaches (CGM-Focused)

Recognizing the limitations of CGM only approaches, several studies explored hybrid models incorporating additional wearable or contextual data. Self-reports, wrist motion (ActiGraph), and CGM (Abbott Libre Pro) were compared for identifying eating occasions in adults with prediabetes or moderately controlled T2D. In a 10-day study of 39 participants, overlap between CGM and self-reports was 55-80% (±30-120 min tolerance), between wrist motion and self-reports 52-65%, and between CGM and wrist motion only 23%. CGM detected earlier first meals and later last meals, while wrist motion captured fewer late-night events. These results suggest that CGM and wrist motion offer complementary information and can reduce dependence on potentially biased self-reports [59]. A hierarchical fusion model combined CGM (Abbott Libre Pro, 15-min sampling) with hand motion (Apple Watch, 10 Hz), electrodermal activity, skin temperature, and heart rate (Empatica E4) from wearables to detect meals in 6 adults over 10 days. Each modality was first modeled with XGBoost, then fused through logistic regression with temporal smoothing as described in Fig. 3.5. Individually, CGM achieved ROC-AUC 0.64 and hand motion 0.77, while multimodal fusion improved to 0.79 with higher F1-scores. The approach reduced false detections, but performance varied across participants and was limited by the small sample size and reliance on self-reports [15]. A study with 36 healthy individuals monitored by dual FreeStyle Libre sensors examined pre- and postprandial glucose responses under fixed meal times and free-living conditions. During fixed meals, pre-meal glucose was 92.8 mg/dL, rising by 47 mg/dL to a peak of 143.3 ± 23.5 mg/dL about 50 minutes post-meal. Under free-living conditions, median glucose was slightly higher (98.1 mg/dL) but postprandial peaks were less distinct. Participants spent over 90% of time in the 70-180 mg/dL range, confirming that transient excursions up to 160-180 mg/dL are normal in non-diabetic individuals. Limitations include the small, homogeneous cohort, short study duration, use of older CGM sensors, and averaging effects that may smooth individual peaks [11].

A Random Forest based detector was embedded in the Robust Artificial Pancreas (RAP) framework, combining CGM, insulin delivery data, and control algorithms to detect unannounced meals and estimate carbohydrate size. In a randomized crossover trial with 15 adults with T1D, the algorithm achieved 83.3% sensitivity, 16.6% false discovery rate, and a mean detection delay of 25.9 minutes. Compared with a standard MPC controller, the system reduced time-above-range (>180 mg/dL) by 10.8% and showed a 9.1% trend toward improved time-in-range (70-180 mg/dL), without increasing hypoglycemia. Although promising, performance dropped for high-fat meals, and the trial required user confirmation before insulin dosing, limiting full automation [12]. A feedback estimator using CGM deviations from model-predicted glucose was developed to detect meal onset and estimate carbohydrate intake. In in-silico evaluation with 30 UVA/Padova virtual subjects, it achieved 98.78% sensitivity, 1 false positive, and 14% mean CHO error and mean onset bias 4.6 min. Retrospective testing on 11 adults with T1D yielded similar results, with 98.91% sensitivity, 8.43% CHO error, and a mean onset bias of 9 minutes. The method was accurate and robust but relied on personalized functional insulin therapy (FIT) parameters, had a small clinical sample, and risked false positives from unmodeled factors such as exercise [13]. A simulation-based explanation (SBE) algorithm that combines CGM, insulin pump, and heart rate data with the

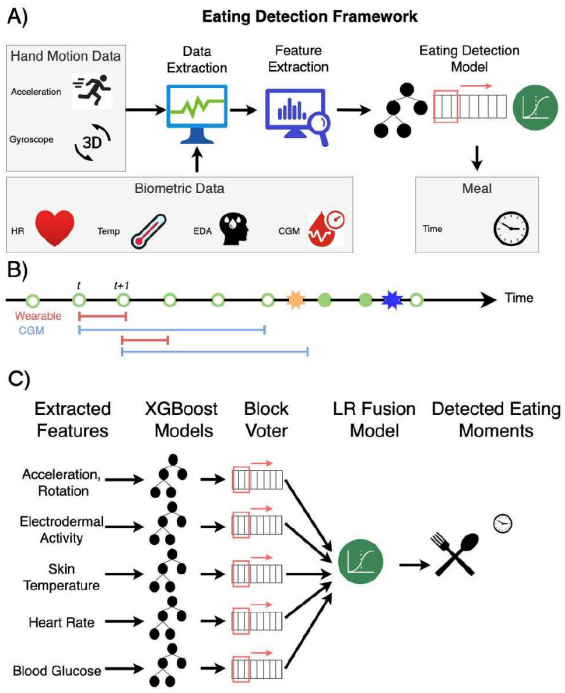


Figure 3.5: A hierarchical fusion model for eating detection [15].



Figure 3.6: An IoT-enabled framework using the Dexcom G6 CGM and cloud-based analytics to infer daily meal routines [16].

FDA-approved Glucose-Insulin Model to infer both meal timing and size. In in-silico experiments with 100 simulated subjects, SBE detected meals with an average delay of 25.7 ± 5 minutes and a meal size error of 1.2 ± 3.6 g, outperforming the Variable State Dimension (VSD) method with 48.3 ± 9 minutes delay and 17.2 ± 8 g error. Applied to the real-world DMITRI dataset consist of 15 individuals with T1D, 1437 hours of CGM, insulin and HR. The algorithm identified 30 plausible meals with 76.7% acceptance versus only 39.4% acceptance for VSD. While robust to noise and exercise, evaluation was limited by the absence of synchronized ground-truth meal logs [14].

The role of CGM is investigated as an additional modality for automatic eating detection alongside consumer wearables (Fitbit, Mi Band). Data from 10 participants over two weeks included steps, heart rate, and interstitial glucose (FreeStyle Libre 2), with eating events logged via a mobile app. Features were extracted over 3,000 without CGM, 3,800 with CGM, and SVM, Random Forest, and XGBoost classifiers were trained. Incorporating CGM improved F1-scores and MCC by 10-23%, reduced false positives, and increased specificity by 15-40% and precision by 17-35%, though sensitivity for Random Forest declined. Results confirmed that CGM data helps balance

performance between eating and non-eating detection, but the small cohort and short monitoring period limit broader conclusions [75]. An IoT-enabled framework using the Dexcom G6 CGM and cloud-based analytics was tested on 7 participants over 10 days to infer daily meal routines shown in Fig 3.6. A sliding-window standard deviation and outlier detection method identified consistent patterns for 4 participants, with meals typically at 8:00, 12:00, 17:00, and 19:00-20:00. However, data loss in 3 participants and reliance on simple statistical thresholds limited robustness, underscoring the need for more adaptive methods and validation against ground-truth logs [16].

A recent probabilistic framework used particle filtering for retrospective acausal meal detection and meal shape estimation in T1D, combining CGM and insulin pump data with population-level priors derived from meal timing distributions, National Health and Nutrition Examination Survey (NHANES) meal sizes, and triple-tracer meal response curves. Preliminary Tidepool-based results suggested that the method was effective for meal detection, while also supporting meal size estimation for closed-loop applications. Its main strengths are flexibility and personalization potential. However, the current study is still preliminary and retrospective, and further validation is required before real-time clinical deployment [76].

Hybrid and multimodal frameworks integrate CGM with complementary physiological or behavioral signals such as insulin dosing, heart rate, or motion to reduce ambiguity and improve robustness and contextual awareness. These approaches demonstrate potential for reducing false positives and improving clinical relevance, particularly in hybrid closed-loop systems. However, their increased complexity, sensor-synchronization requirements, and user burden remain practical challenges for large-scale real-world deployment. An overview is shown in Table 3.4.

Table 3.4: A comprehensive overview of hybrid/multimodal approaches for eating event detection and prediction using CGM data.

Ref	Features	Methodology	Dataset	Performance	Task Type	Challenges / Limitations
[59]	CGM trends; wrist motion; self-reports	Multimodal alignment/analysis	39 participants with prediabetes or moderate T2D; 10-day FL	Overlap: CGM vs. logs 55–80%; wrist vs. logs 52–65%; CGM vs. wrist 23%	Retrospective	Complementarity; modality misalignment; late-night underdetection by motion sensors
[15]	CGM (15-min); accelerometer; gyroscope; EDA; HR; temperature	Per-modality XGBoost classifiers + logistic-regression fusion + temporal smoothing	6 healthy participants; 10-day free-living study	ROC-AUC: CGM 0.64; motion 0.77; multimodal fusion 0.79; F1-score improved	Real-time	Small sample size; self-reports; diet heterogeneity
[11]	Pre/postprandial excursions; TIR	Descriptive normative profiling	36 healthy participants; 14 days (fixed meals + FL)	Fixed meals: peak 50 ± 19 min; rise 47.3 mg/dL; TIR >90%	Retrospective	Healthy cohort; short duration; older sensors
[12]	CGM residuals; CHO/insulin dynamics	RF detector + CHO estimation in RAP (closed-loop)	15 participants with T1D; inpatient crossover RCT	Sensitivity 83.3%; delay 25.9 min; TAR reduced by 10.8%; TIR increased by 9.1% (trend)	Real-time	User confirmation required before dosing; high-fat meals harder; small RCT
[13]	CGM-model residuals; estimator peaks	Feedback estimator (meal + CHO)	IS (30) + retrospective clinical study (11 participants; 5 days)	Sensitivity 98.78–98.9%; low FP; CHO error 8.43–14%; onset bias 4.6–9 min	Real-time	FIT parameter reliance; small clinical sample; exercise confounds
[14]	CGM + insulin + HR; model residuals; SBE	Simulation-Based Explanation (model + signals)	IS (100) + real-world DMITRI data (15 participants; 1437 h)	Delay 25.7 min (IS); plausible meals 76.7% vs. 39.4% baseline	Retrospective	Real data lacked synchronized meal logs; partial ground truth
[16]	Sliding standard deviation/outliers; routine timing	IoT statistical rules (cloud pipeline)	7 adults; 10-day FL; Dexcom G6	Routines detected in 4/7 participants; descriptive accuracy	Retrospective	Data loss in 3/7 participants; simple thresholds; no ground-truth validation

3.4 Datasets, Labeling Strategies, and Evaluation Metrics

The development and validation of AI-based eating event detection or prediction models critically depend on the choice of datasets, the quality of event annotations, and the evaluation methodologies used to assess model performance. This section synthesizes how the reviewed studies handled data collection, event labeling, and model evaluation across a range of sensor modalities and population settings.

3.4.1 Datasets and Participant Populations

This section describes public datasets, institutional studies, and custom experimental protocols, with varying population sizes and health contexts. One of the most frequently used datasets is the OhioT1DM Dataset [25], which is a real-world, free-living dataset collected from 12 adults, age range 20-60 years with Type 1 diabetes, each monitored for approximately 8 weeks while using Medtronic 530G or 630G insulin pumps paired with Medtronic Enlite CGM. The dataset integrates 5-minute CGM readings, insulin delivery data (basal and bolus), self-reported meals, exercise, and lifestyle events, and physiological signals recorded via wearable devices. Physiological signals were collected using the Basis Peak device for the 2018 cohort and Empatica Embrace device for the 2020 cohort. Similarly, the DiaTrend dataset [26] provides multi-day sensor streams from real-world use of CGM and insulin pumps, often used for personalized or population-based modeling. The DiaTrend dataset contains real-world CGM and insulin pump data from 54 adults with Type 1 diabetes, age range 19-74 years (17 male, 37 female) collected across two studies, Digital SMD (2019) [77] at Dartmouth Health and SweetGoals [78], an ongoing randomized trial recruiting participants via social media. Both were approved by the Committee for Protection of Human Subjects at Dartmouth College (STUDY00031632 and STUDY00023559). The dataset includes 27,561 days of CGM and 8,220 days of insulin pump data, with CGM readings every 5 minutes from Dexcom [17], Abbott [18], and Medtronic [19] devices. Pump data include basal/bolus doses, carbohydrate logs, insulin-carb ratios, and related settings.

Several studies implement custom data collection protocols to enable free-living, multimodal sensor analysis. For example, free-living data were collected from individuals with T1D wearing CGM and smart devices over extended periods in their natural environments. The study utilized a large, de-identified Tidepool free-living dataset, comprising up to 700 days of continuous data per subject from more than 40 participants with T1D, age range 15-72 years (23 males, 21 females). Each record contained 5-minute CGM readings (mmol/L), insulin pump logs (basal and bolus doses), and self-reported meal and physical-activity events. Data were collected under unrestricted, real-world conditions using Automated Insulin Delivery (AID) or Sensor-Augmented Pump (SAP) systems equipped with Medtronic Enlite CGM sensors [46]. The dataset comprises of 10 days of multimodal free-living data from 39 adults (62% female; mean age = 52 ± 11 years; BMI = 34.2 ± 4.7 kg/m^2) with prediabetes or moderately controlled type 2 diabetes. Participants wore an ActiGraph GT9X wrist sensor (15 Hz) and an Abbott FreeStyle Libre Pro CGM (15-minute sampling interval) while self-logging eating occasions through a mobile application. The dataset combines CGM, wrist-motion, and self-report modalities to evaluate alignment and timing of eating events across modalities under real-world, free-living conditions [59]. In study [11], the dataset includes 10 days of CGM data from 36 healthy adults (15 men, 21 women; mean age 23.7 ± 5.7 years; BMI 23.8 ± 3.5 kg/m^2) collected at the Institute for Diabetes Technology, University of Ulm, Germany (ClinicalTrials.gov ID: NCT03405415). Participants wore dual Abbott FreeStyle Libre sensors recording glucose every 15 minutes under both fixed-meal condition for 4 days and free-living condition for 7 days. After calibration and cleaning, the dataset comprises over 60,000 glucose readings and 432 standardized meal events. SBE study has been collected in-silico tests with 100 virtual subjects and DMITRI real-world dataset (15 T1D subjects, 1437 hours of data) [14]. Recently released datasets further expand the availability of multimodal CGM data. For example, the BrisT1D dataset provides six months of free-living CGM, insulin pump, and smartwatch signals

collected from 24 young adults with Type 1 diabetes, enabling longitudinal analysis of glucose dynamics and lifestyle behavior [79]. The CGMacros dataset provides CGM recordings paired with detailed meal annotations including macronutrient composition (carbohydrates, fats, and proteins), enabling research on both meal detection and macronutrient estimation from glucose dynamics [80]. Other works included IoT frameworks recorded 7 participants data over 10 days with Dexcom G6 CGM sampled every 5 minutes [16], multimodal wearables study has evaluated 6 healthy adults over 10 days [15], and closed-loop clinical trials with 15 T1D adults [12]. Insulin pump logs used from 82 subjects spanning over 45,000 meal bolus entries, relying on meal announcements as ground truth for personalized LSTM-based prediction models [58]. Recently, efforts have emerged to improve benchmarking by consolidating heterogeneous diabetes datasets. The MetaboNet dataset integrates multiple publicly available type 1 diabetes datasets into a harmonized repository containing CGM measurements, insulin delivery records, and contextual clinical variables. Such unified resources may facilitate more standardized benchmarking of machine learning algorithms for glucose modeling and meal detection [81]. A summary of datasets and participant populations used across studies is provided in Table 3.5.

Table 3.5: Overview of publicly available and experimental datasets for meal detection and glucose dynamics analysis. Abbreviations: HR = heart rate; SMBG = self-monitoring of blood glucose; GIM = glucose-insulin model; MUA = Material Use Agreement.

Ref.	Status	Population	Duration / Records	Devices / Signals	Availability
[25]	OhioT1DM	12 adults; age 20–60 years; 7 female and 5 male	8 weeks; CGM, insulin, meal, and activity records; free-living	Medtronic 530G or 630G insulin pumps paired with Medtronic Enlite CGM; Basis Peak; Empatica Embrace; self-reported meals and activities	Access upon request (MUA required, Ohio University)
[46]	T1D	44 participants; 23 male and 21 female; age 15–72 years	Up to 700 days per subject; CGM, insulin-pump logs, self-reported meal and physical-activity events; long-term de-identified Tidepool data; free-living	Medtronic Enlite CGM; sensor-augmented pump	Public (Tidepool Big Data)
[59]	Prediabetes / T2D	39 participants (62% female); mean age 52 ± 11 years; BMI 34.2 ± 4.7	10 days; multimodal data (CGM, motion, self-report) for meal-timing alignment; 15-min / 15-Hz sampling; free-living	ActiGraph GT9X wrist sensor; Abbott Libre Pro CGM; smartphone meal logs	By request (authors)
[11]	Healthy adults	36 adults; 15 male and 21 female; age 23.7 ± 5.7 years	14 days (10 analyzed); 60,000 CGM readings; 432 standardized meal events; fixed-meal (4 days) and free-living (7 days)	Abbott FreeStyle Libre CGM; capillary glucose meter (Contour Next One)	Restricted (clinical)
[14]	DMITRI real-world T1D	15 adults; 8 male and 7 female	3 days per subject (1,437 h total); 30 meals (23 valid detections); 60 h multimodal data per subject; 15-min sampling; free-living	Dexcom 7+ CGM; insulin pump; Polar HR strap	By request (DMITRI)
[14]	Simulated T1D	100 virtual subjects	100 days per subject (10,000 total); controlled synthetic data	Simulated CGM; insulin (basal/bolus); HR; exercise (GIM-based)	Generated using the GIM model

Continued on the next page

Table 3.5 continued from the previous page

Ref.	Status	Population	Duration / Records	Devices / Signals	Availability
[26]	DiaTrend T1D	54 adults; age 19–74 years; 17 male and 37 female	27,561 days of CGM and 8,220 days of insulin-pump data; basal/bolus doses, carbohydrate logs, and insulin–carbohydrate ratios; free-living	Dexcom, Abbott, and Medtronic	Partially public (Synapse-certified access required)
[15]	Healthy	8 adults (6 selected); age 20–35 years; BMI < 35; normal blood pressure and resting HR < 120 bpm	10 days of CGM, accelerometer, gyroscope, electrodermal activity, skin temperature, and HR data; 21–57 meals per participant; free-living	Abbott FreeStyle Libre Pro CGM; Apple Watch Series 6; Empatica E4 wristband; custom iOS logging app	By request; code on GitHub
[12]	T1D	15 adults (9 female and 6 male); mean age 37.6 ± 10.4 years; diabetes duration 24.8 ± 9.0 years; HbA1c $= 6.8 \pm 0.5\%$	Two single-day inpatient visits per participant; unannounced meals (45–120 g CHO); 199 simulated subjects $\times$ 14 days; 150 Tidepool users ($> 115{,}000$ meals)	Dexcom G6, Medtronic, or Abbott Libre CGMs; Omnipod insulin pump; Polar M600 smartwatch; iPancreas app	By request (OHSU)
[58]	T1D	82 participants from two cohorts: C1, 54 participants, age 33.6 ± 15.6 years; C2, 34 participants, age 42.3 ± 8.6 years	10,497 days of insulin-pump data and 43,596 meal logs; mean 3–4 meals/day; meal size 38 ± 25 g CHO; free-living retrospective study	No direct CGM or additional sensors included	Partially public (Tidepool) and partially upon request
[68]	At risk / T2D and non-diabetic	CTRAL: 15 adults, age 65–80 years; Digital Health: 36 adults, mean age 52 ± 12 years, 78% female	CTRAL: 4 weeks of CGM; Digital Health: 14 days per participant; CGM, activity, and meal logs; free-living observational studies	Abbott FreeStyle Libre Pro CGM; ActiGraph wGT3X-BT; MyFitnessPal app logs	Available upon reasonable request (Texas A&M / Rice University)
[80]	CGMacros (2025)	45 participants: 15 healthy adults, 16 with prediabetes, and 14 with T2D; age 18–69 years; BMI 21–46 kg/m^2	Multi-day CGM recordings with meal annotations and macronutrient information	Abbott FreeStyle Libre Pro CGM; Dexcom G6 Pro CGM; Fitbit Sense; MyFitnessPal app; detailed carbohydrate, fat, and protein records	Public dataset

Continued on the next page

Table 3.5 continued from the previous page

Ref.	Status	Population	Duration / Records	Devices / Signals	Availability
[79]	BrisT1D Dataset (2025)	24 young adults with diabetes; age 18–26 years	Six-month longitudinal study; 3,903 days; 52,105 h of free-living data	FreeStyle Libre 2; Dexcom G6; Guardian 4; Fitbit devices; CGM, insulin, meals, and smartwatch signals	Public dataset

This diversity reflects both diabetic and non-diabetic cohorts, with sample sizes ranging from small feasibility pilots (6-12 participants) to large in-silico populations (30-100 virtual subjects in UVA/Padova simulator studies). Such variability directly impacts model robustness and external validity. A recurring limitation across datasets is the lack of public availability and benchmark splits. While OhioT1DM and DiaTrend are accessible, most multimodal or clinical datasets [12, 59, 15] remain private, limiting reproducibility.

3.4.2 Labeling Strategies for Eating Events

A major challenge across studies is the lack of standardized labeling methods for eating events. The reviewed works use a variety of annotation sources:

- **Self-reported food logs:** Participants record meal times and food intake. Such logs, used in [59, 15], are common but prone to recall bias and timestamp misalignment (10-30 min).
- **Bolus insulin logs:** In diabetes datasets, mealtime boluses serve as proxies for meal times. This is useful in T1D populations but excludes non-diabetic behavior.
- **CGM signal heuristics:** Some methods label eating events based on glucose excursion patterns, e.g., a rise of over 30 mg/dL within 30-60 minutes. However, these methods are prone to false positives due to exercise, stress, or hormonal fluctuations [16].
- **Model-based reconstructions:** As in [13], inferred meals by comparing measured vs. model-predicted glucose, bypassing explicit logs.
- **Hybrid approaches:** Combining CGM with other modalities were used in [14, 12], improving robustness by aligning physiological and contextual cues.

Overall, hybrid and physiology-informed labeling reduces reliance on imperfect self-reports, but scalability remains limited by manual correction and the absence of consensus definitions for "meal events."

3.4.3 Evaluation Metrics and Training Protocols

The evaluation of AI models in eating event detection and prediction is challenging, due to temporal dynamics, class imbalance, and uncertainty in meal-event annotations. The research studies reported a range of performance metrics. Accuracy is still common but often misleading under class imbalance. Most studies report precision, recall, sensitivity and F1-score [13, 12, 46, 73, 10]. For temporal evaluations, detection delay remains one of the most commonly reported metrics (e.g., 25.9 min in [12], 25.7 min in [14]), as illustrated in Fig. 3.4. Area Under the ROC Curve (AUC-ROC) is reported in some CGM-based studies for threshold-based classifiers [15]. Some works include carbohydrate estimation error (e.g., 8.43%-14% in [13], 18 ± 15 g in [10]) as secondary outcomes.

Training protocols also differ. Subject-independent splits (e.g., leave-one-out in [67], cross-validation in [46]) provide stronger generalization but are more difficult to train, as inter-individual variability complicates optimization and model convergence. In contrast, subject-dependent splits may inflate performance due to data leakage across individuals. Only a few studies addressed class imbalance explicitly through sampling or weighted losses.

In summary, the reviewed studies demonstrate substantial diversity in datasets, labeling strategies, and evaluation protocols for CGM-based eating event detection and prediction. This heterogeneity highlights both innovation and fragmentation across the field. Differences in data quality, annotation accuracy, task setting, and metric selection make cross-study comparisons difficult and limit reproducibility. To improve benchmarking and methodological transparency, future studies should report a consistent minimum set of descriptors, including: (i) task setting (e.g., real-time or retrospective), (ii) dataset characteristics and population type, (iii) label source and annotation protocol, (iv) validation strategy such as subject-dependent or subject-independent splitting, and (v) core evaluation metrics such as sensitivity/recall, precision, F1-score, detection delay, and false positives per day. Establishing shared benchmark datasets, standardized ground-truth definitions, and unified reporting practices will be critical for enabling transparent, comparable, and clinically meaningful assessment of AI models for meal detection and prediction.

3.5 Cross-Study Comparison and Benchmarking Discussion

To improve the comparability of existing approaches, the reviewed studies are further distinguished according to their task setting, namely real-time meal detection and retrospective meal detection. Real-time detection refers to algorithms designed to identify eating events during ongoing CGM monitoring, typically for applications such as artificial pancreas systems or real-time decision support. In contrast, retrospective detection analyzes complete CGM trajectories after data collection and is primarily used for behavioral analysis, dataset annotation, or retrospective clinical studies. As highlighted by Brummer et al. [45], these two settings impose different constraints on detection latency, computational complexity, and clinical deployment, making direct comparison across them potentially misleading. Therefore each method is explicitly labeled with its task type in the summary tables to facilitate clearer benchmarking. To facilitate fair comparison across studies, each method is explicitly labeled according to its task setting (real-time or retrospective) in the summary tables (Tables 3.1, 3.2, and 3.3, 3.4).

While earlier sections have highlighted methodological limitations, this section focuses on cross-study variability and benchmarking challenges observed across the literature. Rather than representing additional limitations, these comparisons synthesize methodological diversity showing how different modeling paradigms, input modalities, datasets, and evaluation criteria shape reported outcomes and influence reproducibility.

3.5.1 Variability in Modeling Paradigms

Studies span a wide range of modeling paradigms. Model-based approaches such as feedback estimators and Kalman filters have shown high sensitivity but remain simulation-heavy [6, 13]. Machine learning methods (Random Forest, Gradient Boosting) have been applied to CGM features and clinical datasets with moderate success [1, 68, 8]. Deep learning models dominate recent work, including CNN-LSTM hybrids and RNNs [74, 46, 10, 73]. While each paradigm shows promise, their heterogeneous architectures and assumptions make direct comparison difficult.

3.5.2 Differences in Input Modalities and Fusion Strategies

Some models rely solely on CGM excursions or rate-of-change features [5, 9], while others integrate insulin and carbohydrate logs [74, 12]. Hybrid frameworks fuse CGM with wearable biomarkers or IMU signals, such as wrist motion, heart rate, and EDA [15, 59, 82]. While multimodal approaches improve precision and reduce false positives, they also introduce higher user burden and technical integration challenges compared to CGM-only models.

3.5.3 Dataset Sources and Population Diversity

Dataset origin strongly shapes outcomes. Public datasets like OhioT1DM and DiaTrend provide broad coverage of T1D populations, while Freckmann et al. contributed normative CGM profiles from healthy adults under fixed and free-living conditions [11]. Another study validated across two cohorts, showing how population differences influence postprandial patterns [68]. Controlled studies [5] benefit from precise annotations, whereas free-living observational datasets [8, 74] expose the challenge of mislabeled or missing meals. Importantly, T1D-focused datasets leverage insulin boluses as meal proxies, while healthy-population studies must rely on manual logs or CGM heuristics, limiting comparability.

3.5.4 Disparities in Evaluation Metrics

There is no universal standard for evaluation in this domain. Reported metrics vary widely. Binary metrics such as accuracy and F1-score are common [74, 1], but less informative under class imbalance. Time-aware metrics such as detection delay 19 min in [5], 25 min in [8], 20–25 min in [9] better capture real-world usability. Sensitivity versus false alarms per day is visualized in Fig. 3.3, whereas detection delay across studies is shown in Fig. 3.4. Some studies also report carbohydrate estimation error [13, 10]. Without unified definitions of meal onset, class balancing, or evaluation windows, cross-study comparisons remain inconsistent.

3.5.5 Benchmarking and Reproducibility Gaps

Only a few works provide access to reproducible datasets or code. Most CGM-based studies rely on proprietary or institution-specific datasets, such as the CDDIAB observational study used by [8], or small closed clinical cohorts like those in [11, 68]. This lack of publicly available benchmarks prevents fair comparison across algorithms and limits reproducibility. There is a pressing need for shared CGM repositories, standardized data splits, consistent annotation guidelines, and unified evaluation toolkits that can support both detection and prediction tasks.

When comparing methods within the same task setting, several patterns emerge. Real-time detection approaches are predominantly represented by model-based estimators and hybrid physiological models, which leverage dynamic glucose-insulin models or observer-based frameworks to detect disturbances in glucose dynamics shortly after meal onset. These methods typically report detection delays ranging from approximately 20 to 45 minutes but aim to minimize false alarms in order to support safe automated insulin delivery.

In contrast, retrospective approaches are more frequently associated with machine learning and deep learning classifiers applied to fixed CGM windows or complete glucose trajectories. Because these methods operate offline and have access to the full postprandial glucose excursion, they can exploit richer temporal features and often achieve higher classification accuracy. However, the delayed nature of retrospective detection limits their applicability for real-time therapeutic interventions.

These differences also reflect distinct technical bottlenecks. Real-time systems must operate with streaming CGM data and limited contextual information, requiring robust algorithms that tolerate sensor noise, calibration errors, and inter-individual variability while maintaining low computational overhead. Retrospective methods, on the other hand, can incorporate longer temporal contexts and more complex models but remain limited by labeling inaccuracies, self-reported meal annotations, and dataset heterogeneity. Recognizing these constraints is essential when interpreting reported performance metrics and when designing algorithms intended for clinical deployment.

In summary, this section does not introduce additional methodological limitations but rather highlights the extent of cross-study heterogeneity in CGM-based eating event analysis. Recent meta-analyses of diabetes technologies have similarly highlighted the heterogeneity in datasets, pre-processing pipelines, and evaluation protocols used across CGM-based studies. Such variability complicates direct comparison between algorithms and reinforces the need for standardized benchmarking frameworks and shared datasets for evaluating meal detection systems [83].

Although AI-based meal detection has advanced considerably, inconsistencies in datasets, annotation strategies, task settings (real-time versus retrospective detection), feature engineering, and evaluation criteria continue to hinder comparability and reproducibility. More explicit benchmarking standards are therefore needed. In particular, future work would benefit from harmonized reporting of task type, cohort characteristics, label definitions, validation design, and a core set of performance measures including sensitivity/recall, precision, F1-score, detection delay, and false positives per day. Such reporting practices would support fairer comparison across models and accelerate translation toward clinically meaningful and reproducible CGM-based meal detection systems.

3.6 Challenges and Limitations

Despite substantial progress, real-world deployment of CGM-based eating event detection remains hindered by practical and methodological barriers. This section categorizes the major limitations into three groups: (1) sensor-based, (2) algorithmic and modeling, and (3) deployment and regulatory challenges.

3.6.1 Sensor-Based Limitations

A persistent issue is the 10-15 minute lag between food intake and detectable changes in interstitial glucose, which limits real-time responsiveness. Even under standardized meal conditions, postprandial peaks vary widely in timing and amplitude across individuals [11]. Similarly, high

inter-individual variability is highlighted in glucose trajectories driven by meal composition and metabolic differences, reinforcing the need for personalization rather than population-wide models [84]. CGM traces also suffer from calibration drift, transient dropouts, and noise, particularly in free-living use. Studies have shown that preprocessing and imputation are essential to stabilize such signals [46]. IoT-based frameworks further revealed that transmission losses can exclude entire participants from analysis [16], ambiguous postprandial signals can also result from exercise or stress, complicating labeling and model predictions [15]. In multimodal studies, systematic misalignments of 10-30 minutes between self-reports, wrist motion, and CGM were observed [59]. These factors collectively introduce uncertainty in both labeling and detection performance.

3.6.2 Algorithmic and Modeling Challenges

Labeling quality directly affects model reliability. Many studies rely on self-reports or bolus insulin logs, which are prone to recall errors, under-reporting, or circular dependence on CGM excursions [46, 13]. Algorithmic constraints also limit timeliness: even advanced artificial pancreas frameworks report mean detection delays of 25-26 minutes in clinical and in-silico evaluations [12, 14]. Deep learning models often achieve high offline accuracy [46] but face difficulties in real-time inference due to window aggregation and latency trade-offs.

Dataset heterogeneity ranging from small pilot cohorts to in-silico populations further restricts generalization, as models trained on one demographic often under-perform on others. For example, datasets differ widely in population, protocol, and size from public datasets like OhioT1DM [25], to small pilot study [15], or controlled clinical trials with 15 adults [12]. Healthy cohort studies provide normative postprandial ranges [11], while multimodal frameworks include mixed populations [59]. Models trained on one cohort often under-perform on others, limiting generalizability. A lack of shared benchmarks or public multimodal datasets further hampers reproducibility.

Finally, interpretability remains limited, efforts to integrate explainability, such as SHAP-based feature attribution for nutritional influences, demonstrated how carbohydrate and glycemic index shape forecasts [84]. Few studies assess usability directly, one trial of a gesture-based smartwatch bolus reminder included participant surveys, showing improved HbA1c and time-in-range but raising concerns about false positives and usability [82]. Without transparent decision-making and user-centered validation, clinical acceptance remains limited.

3.6.3 Deployment and Regulatory Barriers

Regulatory and deployment considerations are rarely addressed. Even clinically validated systems such as the robust artificial pancreas (RAP) required manual confirmation before insulin dosing in trials [12]. Integration into commercial CGM and insulin-pump platforms demands compliance with FDA/CE medical device standards, validated data pipelines, and user privacy safeguards. Without such infrastructure, most high-performing prototypes remain confined to research settings.

3.7 Chapter Summary

AI-based modeling of eating events using CGM data has advanced rapidly, offering new possibilities for passive, real-time detection and prediction. This research study revealed a wide range of approaches from physiological estimators like Moving Horizon Estimation to machine learning classifiers and deep learning models (e.g., CNNs, LSTMs, GRUs). While CGM remains the core signal in most studies, fusion with insulin, motion, and self-reported data improves performance. Reported F1-scores range from 0.65 to 0.90, with deep learning models generally outperforming simpler approaches in free-living settings.

Despite this progress, several challenges continue to limit the clinical translation of CGM-based eating event analysis. Datasets vary widely in size, quality, population, and recording conditions, while labeling strategies remain inconsistent and often rely on self-reports, bolus logs, or heuristic rules. These differences complicate cross-study comparison and can inflate reported performance when evaluation protocols are not directly comparable. In addition, subject-dependent training strategies and limited cohort diversity continue to constrain model generalizability.

Future research should prioritize three complementary directions. First, interpretability and explainable AI approaches should be strengthened to improve transparency, trust, and clinical usability of AI-based CGM models. This includes not only identifying important features but also understanding how model decisions relate to glucose dynamics, meal timing, and uncertainty under noisy or weakly labeled conditions. Second, personalization is likely to remain essential, as glycemic responses to meals vary substantially across individuals, meal composition, metabolic status, and daily routines. Subject-adaptive or hybrid population-personalized frameworks may therefore be necessary for robust deployment in free-living environments. Third, stronger clinical validation is needed. Many published methods remain limited to simulation settings, retrospective analyses, or small observational studies. More prospective and free-living evaluations are required to assess not only algorithmic accuracy, but also robustness, alarm burden, usability, and integration with clinical decision-support or closed-loop systems.

In addition, future work should continue toward standardized datasets and labels, improved episode-level evaluation metrics such as detection delay and boundary accuracy, and the development of predictive models capable of anticipating eating events before clear postprandial excursions emerge. Prediction, rather than only retrospective or delayed detection, may offer particularly strong clinical value by enabling anticipatory interventions in diabetes care and beyond.

In summary, CGM-based eating event detection and prediction has evolved into a promising and clinically relevant research area. Continued progress in benchmarking, interpretability, personalization, and prospective validation will be essential for bridging the gap between methodological innovation and real-world healthcare impact.

Chapter 4

Meal Episode Detection from CGM

The work presented in this chapter is based on:

M. Kashif, C. Wang, S. Flesca, and P. Veltri. "Meal Episode Detection from Continuous Glucose Monitoring Using a BiLSTM Sliding-Window and Constraint-Based Decoding". 34th European Signal Processing Conference (EUSIPCO), 2026, pp. 856-860.

Abstract

Detecting eating events from continuous glucose monitoring (CGM) is crucial for understanding daily glucose dynamics and enabling fully automated insulin delivery. Many CGM-based methods perform short-window time-point classification followed by heuristic post-processing to form episodes, often emphasizing timepoint-level metrics while not consistently reporting false positives per day (FP/day) at clinically acceptable alarm burdens. In this work, we propose a CGM-only meal episode detection pipeline based on a sliding-window Bidirectional Long Short-Term Memory (BiLSTM) and explicit episode decoding. Using the OhioT1DM dataset, we construct 2-hour sequences sampled every 5 minutes and derive 5-minute labels from self-reported meal start times using a fixed postprandial labeling window. We extract low-complexity features by detrending CGM with a rolling-median baseline, computing short-term first and second differences, and adding time-of-day context. A compact BiLSTM is trained with class-balanced binary cross-entropy to produce per-time-point probabilities, which are converted into meal episodes using a constraint-based decoder enforcing minimum duration and post-meal cooldown. For each subject, the decision threshold is calibrated on the training split to match a target FP/day, and performance is evaluated on held-out test days against Logistic Regression (LR) and Random Forest (RF) baselines using the same features and decoding. Across eleven subjects, the proposed method improves episode-level F1 score of (0.51 ± 0.2) over LR and RF at the calibrated threshold, providing a transparent CGM-only baseline with explicit FP/day control suitable for use as a first-stage module in more advanced detection and decision support pipelines.

4.1 Introduction

Continuous glucose monitoring is widely used in type 1 diabetes (T1D) to support daily decision-making and hybrid closed-loop therapy. Despite this progress, fully automated insulin delivery still requires reliable detection of meal events and ideally an estimate of meal size to reduce postprandial excursions when meals are unannounced [12, 7]. Detecting meal from continuous glucose monitoring alone is challenging due to sensor noise, delayed glucose response, confounding behaviors (e.g., exercise or stress) and imperfect or missing annotations.

A broad range of meal detection methods has been proposed in recent years, including continuous glucose monitoring (CGM)-only approaches and systems that incorporate CGM with insulin delivery history, carbohydrate announcements or wearable sensors data [12, 7, 75, 46, 68, 48]. While these studies demonstrate that meal timing can be inferred from data, two practical gaps remain for CGM-only episode detection. First, many methods focus on time-point or window-level classification and primarily report point-wise metrics, even though meals are naturally episodic events. In practical deployment, false positives per day (FP/day) directly reflects alarm burden and is often a primary operating constraint for unannounced meal detection, which is often not explicitly controlled or consistently reported [7, 46, 56]. Second, converting model scores into discrete meal episodes often relies on heuristic post-processing, which complicates comparison across thresholds and subjects.

In this work, we revisit CGM-only meal episode detection on OhioT1DM dataset [25]. We proposed a CGM-only pipeline that separates probabilistic prediction from episode formation. A sliding-window BiLSTM produces time-point probabilities on fixed-length 2-hour windows, leveraging its sequence modeling capability [85]. Constraint-based decoder plus per-subject threshold calibration yields episode detections at a prescribed FP/day [12, 7]. We report episode-level F1, precision, recall, FP/day, detection delay, and compare against LR and RF baselines using the identical features, decoder, and FP/day calibration, ensuring fair comparison [7, 86]. An overview of proposed CGM-based meal detection pipeline is shown in Fig. 4.1.

4.2 Related Work

Several recent studies have explored CGM-derived patterns and machine learning (ML) to infer eating events. Data-driven response-pattern approaches detect meals by matching postprandial response shapes via sliding-window similarity within CGM time series data [86]. Proof-of-concept studies have also examined CGM as an objective proxy for eating behavior in non-diabetic cohorts, demonstrating feasibility while highlighting limitations in generalization and label quality (e.g., small cohorts and weak/subjective ground truth) [56]. More recently, ensemble machine learning approaches have been proposed for unannounced meal detection and evaluation on OhioT1DM, using engineered CGM features over recent history with ensemble classifiers, and reporting performance under clinically relevant constraints such as false alarms and detection delay [7]. These CGM-only studies typically operate on fixed windows (or individual time points) and then convert model scores into meal events via post-processing, motivating standardized episode decoding and episode-level reporting.

Beyond CGM-only methods, several studies incorporate CGM with additional information such as physical activity, insulin delivery history, wearable signals or contextual information. For example, joint detection of meals and physical activity events from free-living data uses multiclass sequence models over CGM plus insulin/pump context and derived features to separate meals from competing disturbances [46]. Other work infers specific meal-related characteristics in targeted settings (e.g., breakfast-related attributes) via machine learning using feature sets extracted from pre/post-meal CGM segments [68]. While multimodal or context-enriched approaches can improve robustness, they require additional data streams and integration effort that may not be feasible in CGM-only deployment.

A key application area is closed-loop control, where timely meal detection and meal size estimation can enable proactive insulin action for unannounced meals. System-level automated insulin delivery framework integrating meal detection and size estimation via ML classifiers combined with control logic/decision thresholds, emphasize operational trade-offs between sensitivity, false alarms, and detection delay [12]. Similarly, targeting unannounced meals to improve postprandial control and report performance with false alarms and delay considerations [7]. These requirements motivate pipelines that explicitly control FP/day and report episode-level performance at a defined operating point, rather than only time-point metrics.

In contrast to prior work that often emphasizes point-wise or window level outcomes, our goal is to provide a CGM-only pipeline that: (i) produces calibrated probabilities from a compact BiLSTM model [85], (ii) converts probabilities into meal episodes using a clearly specified constraint-based decoder, and (iii) reports episode-level performance with explicit FP/day.

4.3 Methods

4.3.1 Dataset and Label Construction

We use the OhioT1DM dataset, which contains CGM, insulin and self-reported meal logs from 12 adults with T1D over multiple days in free-living conditions. In this work we focus exclusively on CGM and meal events to keep the setting comparable to CGM-only studies such as [8, 86, 56]. For each subject, we follow the dataset's predefined per-subject train/test partition. CGM values are recorded at 5 min intervals. We standardize timestamps to a continuous 5 min grid and mark samples as valid or invalid using the dataset quality flags `is_valid` and detected gaps `big_gap`. Only valid CGM samples are used for feature extraction and model training, any sequence window touching invalid points is discarded.

Meal annotations consist of start times only. Following prior unannounced-meal detection work that defines a postprandial horizon around meal start, we derive time-point labels on the 5 min grid: a point is labeled positive if it lies in the interval $t_{\text{meal}} - 5\,\text{min},\ t_{\text{meal}}\ 60\,\text{min}$ and negative otherwise. Overlapping meal windows are merged into a single positive region, yielding ground-truth episodes with nominal duration 65 min per meal (longer when meals are close). Across the eleven subjects, 2136 annotated meals and 607 days data were recorded and the resulting positive class prevalence ranged from about 7% to 23%, motivating class-weighted loss during training, test data are evaluated

under the original class distribution.

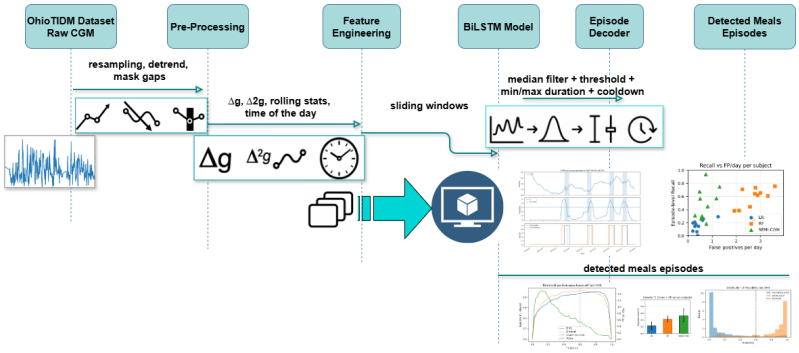


Figure 4.1: Proposed CGM-based meal detection pipeline.

4.3.2 Feature Engineering

At each valid CGM timestamp, we compute a feature vector leveraging short-term trends and daily context. Let g_t denote the raw CGM value at time t (mg/dL). To reduce slow drift while preserving postprandial excursions, We apply a rolling median computed over the previous 75 min, yielding a detrended signal $\tilde{g}_t = g_t - \text{median}\{g_\tau : \tau \in t - 75\,\text{min}, t\}$.

From $\tilde{g}_t$ we derive: (i) the detrended level $\tilde{g}_t$, (ii) the first difference $\Delta g_t = \tilde{g}_t - \tilde{g}_{t-1}$ (mg/dL per 5 min) and (iii) the second difference $\Delta^2 g_t = \Delta g_t - \Delta g_{t-1}$, which reflects local curvature. Additionally we compute rolling summary statistics (mean and standard deviation) of $\tilde{g}_t$ over 30 min and 60 min windows. To incorporate circadian structure, we encode time-of-day context using sine/cosine features with a 24 hour period, and include day-of-week as a categorical indicator.

We explicitly exclude any features derived from insulin or carbohydrate announcements to maintain a CGM-only detector. For each subject and split, we store the per-timestamp feature table containing timestamp, raw glucose, engineered features, binary label and an `is_valid` flag. These same inputs are used by all models to ensure a fair comparison between baselines and the proposed BiLSTM.

4.3.3 Models and Episode Decoder

Point-wise baselines

We use LR and RF as standard, widely used classical baselines for imbalanced biomedical time-series classification, providing interpretable points of comparison under the same features and decoding and trained on single timestamps (D-dimensional feature vectors). For each subject we load the training and testing files, restrict to rows with `is_valid =1`, and drop non-feature columns (timestamp, label, quality flags). LR is trained with an ℓ_2 penalty and class-weight balancing, while RF uses 300 trees and balanced subsampling. Each baseline outputs a probability $p_t \in 0, 1$ that the current time point belongs to a meal labeled interval.

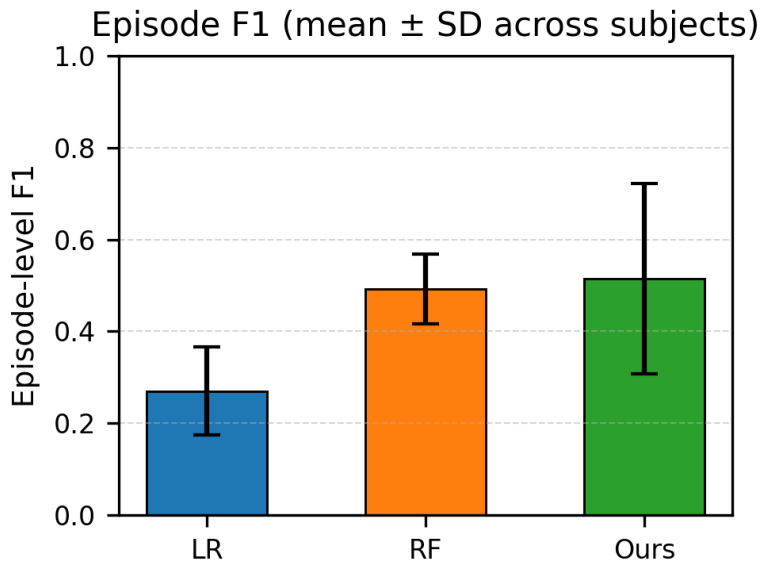


Figure 4.2: Episode-level F1 (mean ± SD) for LR,RF, and the proposed BiLSTM window model on OhioT1DM.

Window-based BiLSTM

Our main model is a compact bidirectional LSTM (BiLSTM) [85] that operates on fixed-length CGM feature sequences. From each subject's time-indexed table, overlapping windows of length $L = 24$ samples (2 h at 5 min) using a sliding window with stride 1 are formed. Any window that includes invalid CGM samples (`is_valid =0`) or missing values is discarded. The resulting input tensor has shape N, L, D, where N is the number of windows and D is the feature dimension. The supervision for each window is the corresponding length L of binary labels. We also tested shorter and longer contexts (1 h and 3 h windows) and selected the 2 h window as a practical trade-off between detection performance and onset-delay behavior.

The network comprises two stacked bidirectional LSTM layers (32 units each) with dropout, followed by a time-distributed dense layer with a sigmoid output layer. Training minimizes time-step binary cross-entropy with class-balanced sample weights computed from the training labels and early stopping on validation loss is used to prevent overfitting. At inference, we extract a single probability per window from the midpoint time step ($L2$), assign it to the corresponding timestamp. Because windows are generated with stride 1 on the 5-minute grid, midpoint assignment yields an almost dense probability trace p_t, except near the boundaries of each contiguous valid segment (the first and last ($L2$) samples).

Constraint-based episode decoder and FP/day calibration

To convert noisy probability traces into discrete meal episodes, we apply a constraint-based decoder to p_t: (i) smooth p_t with a 1D median filter (window $k = 5$), (ii) binarize using a threshold θ: $\hat{y}_t = \mathbb{I}p_t \geq \theta$, (iii) retain only contiguous positive runs whose duration exceeds a minimum length (20 min) and cap any retained runs to a maximum length (80 min) and (iv) enforce a post-episode

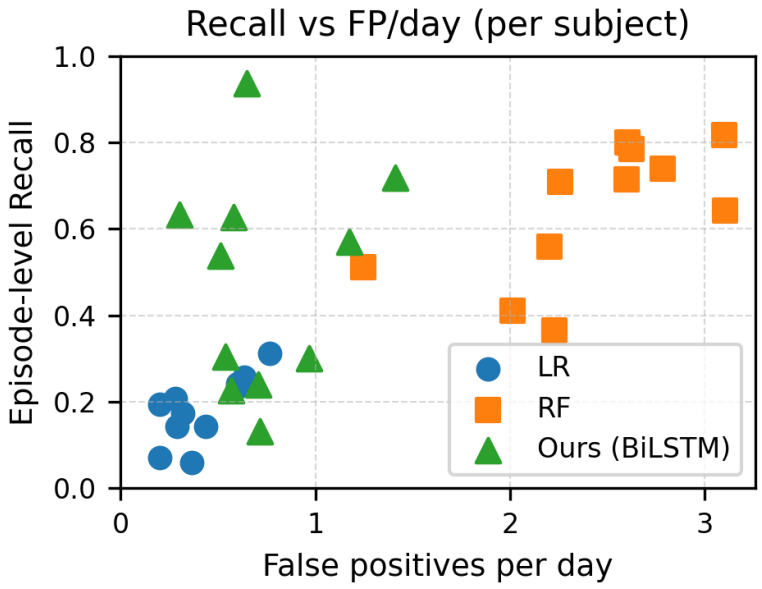


Figure 4.3: Episode-level recall vs false positives per day for LR, RF, and proposed BiLSTM window model across 11 subjects.

cooldown by suppressing new episode onsets for 60 min after an episode ends.

For fair comparison, the same decoder and hyperparameters are used for LR, RF and the BiLSTM; only the probability source changes. Hyperparameters were fixed across subjects based on preliminary runs on the training split and kept constant to avoid per-subject tuning. For each model and subject, we sweep a fixed threshold θ grid on the training split and select an operating point that matches a target FP/day as closely as possible, resulting in improved episode-level F1 score. This follows the common practice of selecting an operating point based on false-alarm/delay trade-offs in unannounced meal detection [87, 7].

Table 4.1: Episode-level performance of baseline models and proposed BiLSTM sliding window model on OhioT1DM.

Model	***F1***	***Precision***	***Recall***	***FP/day***
LR	0.27 ± 0.10	0.54 ± 0.15	0.19 ± 0.08	0.43 ± 0.19
RF	0.49 ± 0.08	0.42 ± 0.10	0.64 ± 0.16	2.43 ± 0.53
Proposed	0.51 ± 0.22	0.60 ± 0.17	0.48 ± 0.25	0.74 ± 0.32

4.4 Experimental Setup and Results

We evaluate on eleven OhioT1DM subjects for whom CGM and meal logs are available and contain at least one positive label in both training and testing splits. For each subject we train LR, RF and the proposed BiLSTM using only the training split and evaluate on the held-out test days. LR uses class-weighted training with ℓ_2 regularization, RF used 300 trees with balanced sub sampling. The

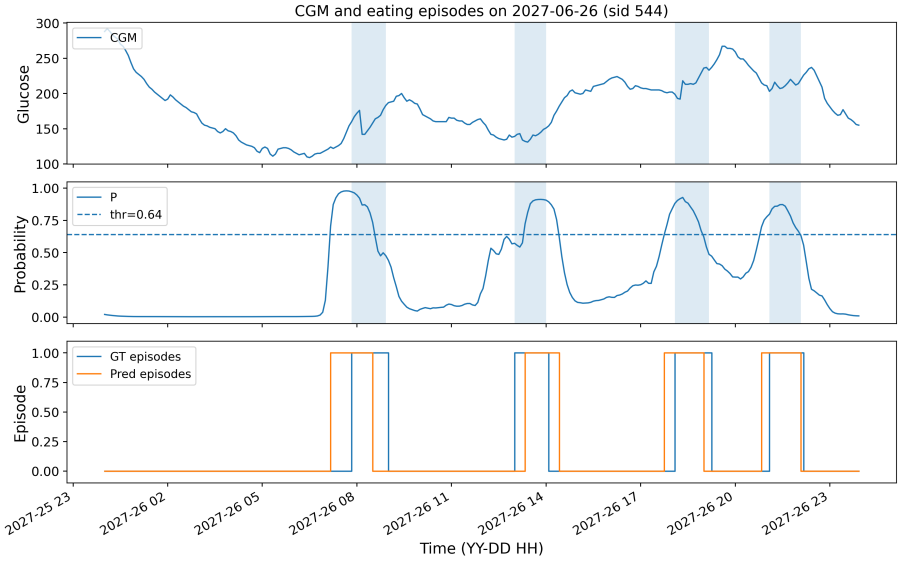


Figure 4.4: Example day illustrating CGM trace, probability decoding into episodes and the resulting detections

BiLSTM uses sequence length $L = 24$ (2 h), hidden size 32, dropout 0.2, Adam optimizer with learning rate 10^{-3}, batch size 128 and up to 10 epochs with early stopping on validation loss (10% of training windows).

We report episode-level metrics by decoding model probabilities into episodes using the same constraint-based decoder for all methods. A predicted episode is counted as a true positive if it overlaps any ground-truth episode, otherwise it is a false positive. Unmatched ground-truth episodes are false negatives.

Table 4.1 summarizes episode-level performance (mean ± std across 11 subjects) at the calibrated FP/day operating point. The proposed BiLSTM achieves the highest overall episode-level F1 score with improved precision compared to RF while maintaining higher recall than LR. RF attains higher recall but at a substantially higher false-alarm burden compared with the calibrated target, indicating reduced specificity under the shared decoding constraints. We additionally report detection delay in minutes (min) as a non-negative onset delay relative to the annotated meal start time. Across subjects, LR exhibits the largest delay (23.41 ± 16.67 min), while the BiLSTM reduces delay substantially (4.55 ± 7.89 min). RF shows near-zero delay (0.68 ± 2.26 min); however, this is consistent with RF often triggering early (negative signed onset errors are clipped to 0 by the non-negative delay definition), and should therefore be interpreted together with its elevated FP/day.

We also compared sequence lengths of 1 h, 2 h, and 3 h (stride 1 on the 5-min grid) only for our proposed pipeline. The 2 h setting provided a strong trade-off (episode-level F1 score, improving over 1 h (0.49 ± 0.20) with FP/day (0.77 ± 0.24) and detection delay of 11.36 ± 10.57 min. A 3 h context yielded a small additional gain in F1 score (0.53 ± 0.18) with FP/day (0.70 ± 0.25) and detection delay of 4.55 ± 7.89 min, but we fix 2 h in the main paper for simplicity and alignment with the postprandial labeling horizon. An ablation is also performed by removing time-of-day features resulted in F1 score of 0.47 ± 0.21, FP/day 0.69 ± 0.25 with detection delay of 10.9 ± 11.6

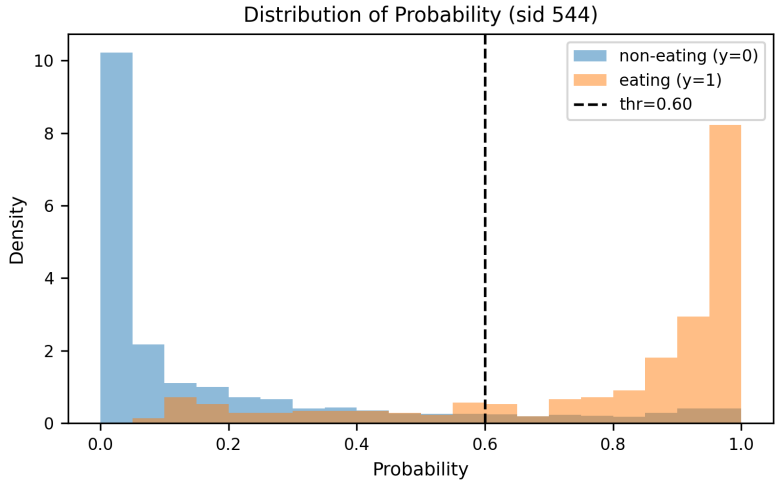


Figure 4.5: Distribution of window-level eating probability for eating vs non-eating windows, with the indicated calibrated decision threshold.

min.

Fig. 4.3 shows recall versus FP/day relationship of all methods observed on the test split. LR tends to cluster at low FP/day but with limited recall, reflecting the difficulty of linearly separating meal vs. non-meal points from CGM statistics alone. RF improves recall at the cost of substantially larger FP/day and noisier episode boundaries. The proposed BiLSTM window model consistently lies on a more favorable part of the trade-off, reaching recall above 0.6-0.9 for several subjects while keeping FP/day close to the calibration target.

Fig. 4.2 reports episode-level F1 score across subjects (mean±SD). The BiLSTM not only improves mean F1 but also reduces subject-to-subject variability. Fig. 4.4 shows one full day for a representative subject, including the CGM trace, the model's eating probabilities, and the ground-truth vs detected eating episodes. Fig. 4.5 shows the distribution of the model probabilities for eating vs non-eating windows together with the calibrated decision threshold. Fig. 4.6 shows the threshold performance trade-off (episode-level F1, recall, and FP/day) used to select the operating point.

We report results at a fixed operating point by calibrating the decision threshold on the training split to a target FP/day (set to 0.5 in our experiments), consistent with prior unannounced-meal studies that report performance under explicit false-alarm/delay trade-offs [7, 87].

An important observation is the substantial inter-subject variability, even at a fixed FP/day target. Subjects with clearer postprandial excursions achieve higher F1 score, whereas noisier CGM profiles, overlapping disturbances, or sparse annotations reduce detection harder. Because acceptable alarm burden is application-dependent, we calibrate and can sweep the FP/day target to trade off missed meals against false alarms.

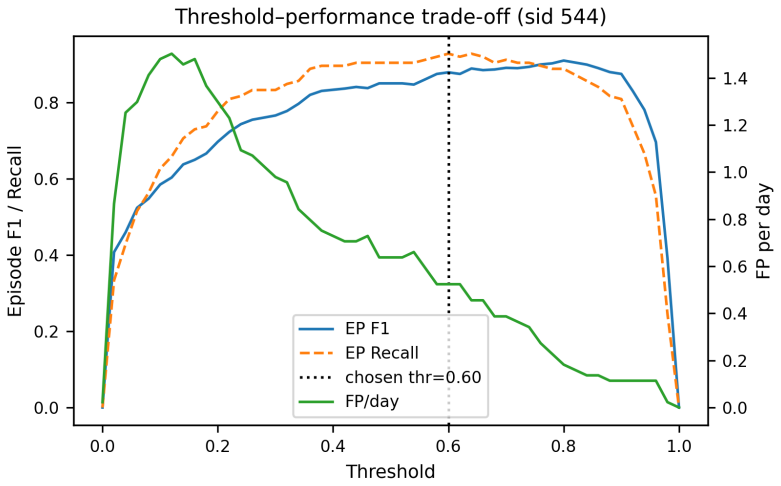


Figure 4.6: Threshold-performance trade-off showing episode-level F1, recall, and FP/day, with the selected operating threshold marked.

4.5 Discussion, Limitations, and Chapter Summary

Our results indicate that a compact window-based BiLSTM trained on low-complexity CGM-derived features can improve episode-level detection performance compared with point-wise LR and RF baselines when evaluated at an explicitly controlled FP/day operating point. A key aspect of the pipeline is the explicit separation between probabilistic prediction p_t and a fixed, clearly specified decoding procedure, enabling fair comparisons across models under identical post-processing and per-subject threshold calibration [8, 86, 56].

This work has several limitations. First, because OhioT1DM meal annotations provide start times only, ground-truth episodes are constructed using a fixed postprandial labeling window, which introduces uncertainty in episode duration and can affect delay-related metrics. Second, the decoder hyperparameters (median smoothing, minimum/maximum duration, cooldown) are fixed rather than optimized on an independent validation cohort, and different clinical applications may prefer different constraints. Third, all models are personalized per subject [7, 86], a subject-independent generalization and few-shot adaptation remain important directions for future work. Finally, training optimizes time-point cross-entropy, while evaluation uses episode-level metrics after decoding; incorporating learning objectives that more directly reflect episode-level performance is a promising next step.

In the longer term, the present proposed BiLSTM window model pipeline is intended to act as a first-stage CGM module. Its episode candidates and probability traces could be passed to mechanistic models (e.g., glucose-insulin simulators or Kalman filters), semi-Markov duration models, or multimodal fusion layers that incorporate wearables. Such second-stage components can refine episode shapes, enforce physiological consistency and support downstream tasks such as meal-size estimation or adaptive insulin dosing.

A data-driven pipeline is presented for detecting eating episodes from CGM alone on the OhioT1DM dataset. The approach combines low-complexity CGM-derived features, a compact

window-based BiLSTM, a constraint-based episode decoder, and per-subject threshold calibration to operate at a prescribed FP/day. Compared with logistic regression and random forest baselines trained on the same features and evaluated under the same decoding and calibration procedure, the BiLSTM consistently improves episode-level performance at comparable alarm burden, supporting its use as a strong CGM-only baseline.

Part III

Speech-Based Neurological Assessment

Chapter 5

Multiple Sclerosis Classification from Vocal Signals

The work presented in this chapter is based on:

Kashif, M., Vizza, P., Delfino, A., Bossio, R. B., Tradigo, G., Flesca, S., & Veltri, P. (2025). Using LSTM-based model on vocal signal analysis for the classification of multiple sclerosis. In Lecture Notes in Computer Science (pp. 192–196). Springer Nature Switzerland.

Abstract

Multiple Sclerosis (MS) is a chronic, neurodegenerative, and autoimmune disease of the central nervous system (CNS) that affects the brain and spinal cord. It impacts motor, sensory, cerebellar, cognitive, and language functions, with more than 2.5 million people affected worldwide. In such a scenario, early detection is crucial for effective management and treatment of MS. Quantitative analysis of MS-related speech disorders could significantly aid physicians in diagnosis and patient monitoring. Neurological voice disorders, including those seen in MS, arise from disruptions in the nervous system's interaction with the larynx. This study investigates the use of classification techniques on speech signal analysis for MS identification by considering acoustic and mel-frequency cepstral coefficient (MFCC) features to differentiate MS patients from healthy controls within a dedicated dataset. The performance of Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), and Support Vector Machine (SVM) models was evaluated across different classification experiments, including those performed on augmented datasets, to explore pathological speech characteristics. These findings provide preliminary information that may support physicians in characterizing the vocal features of MS patients.

5.1 Introduction

Multiple Sclerosis (MS) is a chronic, neurodegenerative, and autoimmune disorder of the central nervous system (CNS) that manifests through a wide array of motor, sensory, cerebellar, cognitive,

and language dysfunctions [88]. With over 2.5 million individuals affected globally, MS poses a significant load on healthcare systems, hence the need for innovative approaches for early detection and effective management of MS [89]. Early diagnosis is particularly critical, as timely intervention can mitigate disease progression and improve patient outcomes. However, the heterogeneity of MS symptoms, particularly those related to speech and voice, presents a diagnostic challenge. Neurological voice disorders in MS arise from disruptions in the neural pathways determining laryngeal function, leading to distinct acoustic alterations that can serve as potential biomarkers for the disease [90]. Advancements in machine learning (ML) and signal processing opened new opportunities for quantitative analysis of speech disorders [91]. Acoustic features and Mel-frequency cepstral coefficients (MFCCs) have been widely employed to characterize pathological speech patterns in neurodegenerative diseases [92, 93]. These features, when analyzed with advanced ML models, enable the differentiation of pathological speech from healthy controls with high accuracy. However, the application of such techniques to MS-specific speech disorders is still limited.

In this study, we use state-of-the-art ML models, including Long Short-Term Memory (LSTM) [94], Bidirectional LSTM (BiLSTM) [95], and Support Vector Machines (SVM) [96], to analyze speech signals from patients affected by MS and healthy individuals. By employing both raw and augmented datasets, we aim to identify distinctive acoustic and MFCC features that can effectively distinguish between MS patients and healthy controls. Experimental results show preliminary evidence that the use of speech signal analysis combined with LSTM-based approach can significantly improve the efficiency of using vocal signal analysis on non-invasive diagnosis and monitoring of MS in automated large-scale scenarios.

5.2 Materials and Methods

This section demonstrates the data collection process and outlines the research methodologies implemented throughout our study. We describe the steps taken to gather the dataset, the features selected for analysis, and the methods used for model development and evaluation also demonstrated in Fig 5.1.

5.2.1 Data collection and processing

The vocal signal dataset was collected from 53 MS patients enrolled at the Neurological Operative Unit, Center of Multiple Sclerosis in Cosenza, Italy, with ages ranging from 25 to 74 years, including 18 men and 35 women. Additionally, voice recordings were acquired from healthy individuals (3 men and 4 women) aged between 24 and 68 years. The study followed an acquisition protocol developed with clinicians, requiring informed consent from each participant, hence the low number of healthy voice signals acquired. To overcome the limitations of such a small number of healthy subjects, synthetic data have been generated. Speech recordings included continuous pronunciation of the five vowels (/a/, /e/, /i/, /o/, /u/) for 5 seconds in a controlled clinical setting. An omnidirectional microphone recorded the signals near the subject's mouth at a 22.05 kHz sampling rate with 16-bit resolution. All recordings were saved in .wav format with the PRAAT software. Participants data

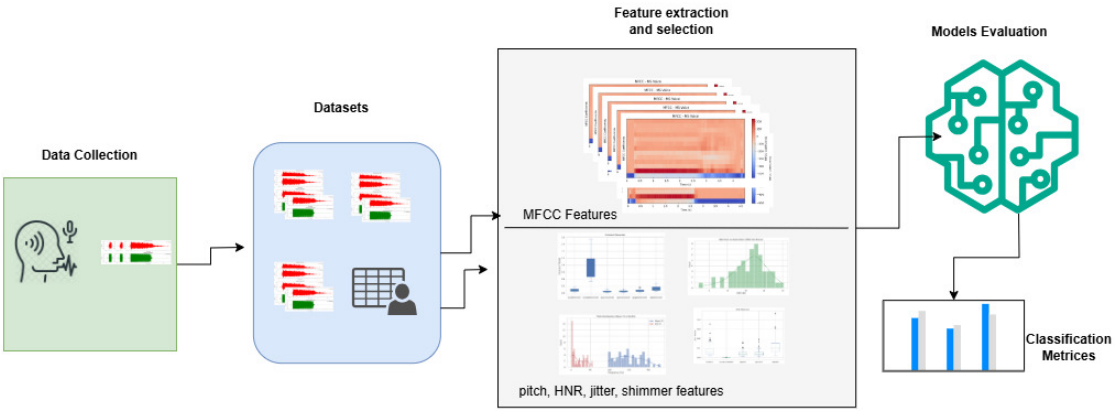


Figure 5.1: A comprehensive flow chart of the data collection and research methodology

was recorded anonymously to ensure privacy and confidentiality [97].

Data augmentation techniques, such as pitch and time shifting, time stretching, and adding noise, have been used to balance the normophonic voice records (HS), enhancing model's generalization and robustness. Synthetic data simulate natural variations in speech (e.g., pitch, background noise, timing shifts), enriching the diversity of healthy voice samples without altering their characteristics and resulting in a more balanced dataset.

5.3 Experimental Results

This study aims to analyze vocal signals from MS and HS to identify relevant patterns and distinguish between healthy and pathological voices. Results presented in this section are useful for characterizing patients' vocal signals and are summarized in Table 5.1 for LSTM, BiLSTM, and SVM classifiers. The hold-out 80/20 validation strategy has been used while performing experiments.

5.3.1 MFCC features Analysis

In the experiment EXP #1, MFCC features were extracted from the augmented dataset, containing speech recordings of 3 to 6 seconds. For each 3-second record, 13 MFCC coefficients were extracted. The MFCC extraction function computes these features from an audio file using a sampling rate of 22,050 Hz. To ensure uniform input dimensions for machine learning models, the extracted features are fixed at 130 frames through zero-padding or truncation. In the experiment EXP #2, acoustic features were extracted using a method adapted from a GitHub repository[1]. The script extracts various acoustic features from voice recordings. Using the Parselmouth library, it computes a set of pitch-related metrics, including the mean pitch (F0), standard deviation, harmonicity, and jitter and shimmer parameters, resulting in a total of 14 features that characterize voice quality. These features are crucial for analyzing the acoustic properties of voices, particularly for applications in speech analysis or classification tasks. In the EXP #3 experiment, Synthetic Minority Oversampling

[1]Feinberg, D.R.: Parselmouth Praat Scripts in Python, osf.io/6dwr3.

Table 5.1: Experimental results comparing different models and features.

Experiments	Data	Features	Model	Accuracy (%)	F1 Score
EXP #1	**Augmented**	MFCC	LSTM	77.27	0.80
			BiLSTM	86.36	0.59
			SVM	72.73	0.67
EXP #2		Pitch, HNR, Jitter, Shimmer	LSTM	86.36	0.84
			BiLSTM	72.73	0.67
			SVM	95.45	0.95
EXP #3			LSTM	86.00	0.84

Technique (SMOTE) was applied to acoustic features, enhancing LSTM performance as shown in Table 5.1. To avoid data leakage, we applied SMOTE only on the training data, after train/test split.

5.3.2 Acoustic Features Analysis

The reported performance metrics include accuracy and F1-score. Although the dataset is imbalanced, we included accuracy for comparison purposes, where F1-score was considered as the primary metric. Notably, the low F1-score observed for BiLSTM in EXP #1 reflects its higher data dependency and limited benefit due to the small training set. Conversely, SVM and LSTM, being less sensitive to training data volume, performed more reliably under the same conditions.

5.4 Discussion, Limitations, and Chapter Summary

The findings from our research provide preliminary data for identifying the voice characteristics of MS patients. In this study, we focused on MFCC and acoustic features. The experiments revealed that MFCC features yielded the best performance with an F1 score of 0.80 when using LSTM, whereas acoustic features achieved an F1 score of 0.95 with SVM. As a result, this work represents a step toward developing an assistive tool for clinicians based on acoustic features.

Funding

PaVi is partially funded by the Next Generation EU - Italian NRRP, project Tech4You - Technologies for climate change adaptation and quality of life improvement (ECS0000009). PiV was partially funded by projects: SERICS (PE00000014) - MUR National Recovery, and SINNephro RNA (CN00000041) - PNRR Next Generation. GT is partially funded by the Italian Ministry of Health by the NRRP ACCELERATE AdvanCed robotiC bEds for the earLy rEhabilitation of cardiac suRgery pATiEnts project (PNRR-MCNT2-2023-12378283).

Chapter 6

Parkinson's Speech Assessment using Task-Conditioned PEFT and Prompt-Aligned ASR Error Structure (PAESL) Fusion

The work presented in this chapter is based on:

M. Kashif, P. Vizza, G. Tradigo, and S. Flesca, "Task-conditioned PEFT with prompt- aligned ASR error structure fusion for Parkinson's speech assessment", in Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics, 2026, pp. 1-2.

Abstract

Speech-based assessment of Parkinson's disease (PD) is appealing because it is non-invasive, scalable, and can be collected remotely. Recent work increasingly leverages self-supervised speech foundation models (e.g., wav2vec 2.0, WavLM) to learn robust representations that transfer well under limited labeled data. However, PD screening is challenged by prompt/task variability and the limited use of structured, prompt-aligned automatic speech recognition (ASR) error patterns beyond coarse word error rate (WER) and character error rate (CER) summaries from modern ASR systems (e.g., Whisper). To adapt these backbones efficiently, we use parameter-efficient fine-tuning (PEFT), specifically low-rank adaptation (LoRA), to reduce trainable parameters while preserving performance. Most PD screening pipelines either pool self-supervised learning (SSL) embeddings per recording and handle elicitation tasks implicitly, or rely on coarse ASR summary scores such as WER/CER. We instead implement task-conditioned PEFT (TC-PEFT) and fuse it with prompt-aligned ASR error learning (PAESL), providing complementary and interpretable cues beyond WER/CER. We propose a two-branch framework for subject-level PD classification using the Italian Parkinson's Voice and Speech (IPVS) dataset, which includes multiple prompted speech

tasks per subject (e.g., phonemically balanced text and words). The proposed framework integrates the following primary stages: (i) the TC-PEFT acoustic branch, where a pretrained WavLM-base backbone is adapted via LoRA and a task embedding is injected to generate task-conditioned pooled acoustic embeddings for each recording; (ii) the PAESL branch, which utilizes a lightweight Transformer encoder to transform prompt-aligned ASR error/structure matrices (computed from ASR transcripts) into compact representations and (iii) the Fusion stage, where these standardized embeddings are concatenated and processed by a small MLP head to predict subject-level PD probability. Performance is evaluated using a 5-fold subject-independent cross-validation scheme by reporting binary classification (e.g., PD vs. Control) using accuracy, macro-F1, sensitivity, and specificity. On the complete core set, the TC-PEFT-only model achieves macro-F1 = 0.9596 ± 0.0553 and accuracy = 0.9633 ± 0.0506. The fusion configuration (TC-PEFT+PAESL) reaches macro-F1 = 0.9798 ± 0.0451 and accuracy = 0.9833 ± 0.0373. Fusion improves over the task-conditioned acoustic-only (TC-PEFT) ablation and over ASR-derived baselines, while remaining competitive with a strong SSL-embedding baseline. Overall, fusing TC-PEFT acoustics with PAESL error structure improves PD screening over acoustic-only and ASR-only baselines; permutation testing yields near-chance performance under label shuffling, supporting that the observed accuracy is not explained by random structure. Future work will focus on longitudinal modeling and testing the system's ability to generalize across different tasks and conditions.

6.1 Introduction

Parkinson's disease is a progressive neuro-degenerative disorder that affects motor control as well as several speech production mechanisms, including phonation, articulation, and prosody. Speech impairment is highly prevalent in PD, with reported rates approaching 89% in affected individuals, which makes speech a clinically meaningful and non-invasive source of biomarkers for screening and monitoring [98]. Because speech can be collected remotely, repeatedly, and at relatively low cost, automatic speech-based assessment has emerged as an attractive direction for computer-aided PD analysis. Traditional speech-based PD assessment has largely relied on handcrafted acoustic descriptors such as jitter, shimmer, harmonicity, prosodic cues, and articulation-related measurements. While such features remain clinically interpretable, their robustness can degrade across datasets, languages, and speaking styles. More recent studies have therefore investigated deep neural and self-supervised speech representations, which often provide stronger predictive performance than manually designed feature sets, particularly in multilingual or cross-lingual settings [99]. At the same time, cross-language and cross-task variability remain major challenges for practical deployment, since speech content, phonetic structure, and elicitation protocol can substantially affect the behavior of automatic assessment systems [100, 99].

The rapid progress of self-supervised learning (SSL) in speech processing has made foundation models especially relevant for pathological speech analysis. Models such as wav2vec 2.0 learn powerful representations from raw audio through large-scale pretraining and subsequent task adaptation [21]. WavLM further extends this direction by jointly modeling masked speech

prediction and denoising, aiming to learn representations that are useful across a broad range of downstream speech tasks rather than only automatic speech recognition [22]. These developments are particularly valuable in clinical settings, where labeled pathological speech data are often limited and heterogeneous. However, directly adapting large pretrained speech models to small clinical datasets raises both statistical and computational challenges. Parameter-efficient fine-tuning methods have become a practical alternative to full model updating by modifying only a small subset of trainable parameters while keeping most pretrained weights frozen. Among these methods, Low-Rank Adaptation introduces trainable low-rank updates into Transformer layers and can substantially reduce the number of trainable parameters and memory demands compared with conventional full fine-tuning [23]. This makes PEFT especially attractive for research settings where computational resources are limited but robust task adaptation is still required. In parallel, automatic speech recognition systems have become increasingly robust and multilingual. Whisper was trained on approximately 680,000 hours of weakly supervised multilingual and multitask audio-text data, demonstrated strong zero-shot transfer and robustness across diverse transcription settings [24]. In speech-based PD research, ASR outputs are often summarized using coarse scalar indicators such as word error rate or character error rate. Although these measures can be informative, they compress recognition behavior into a small number of aggregate values and may discard structured prompt-dependent information, such as error localization, substitution patterns, omissions, and token-level deviations relative to an expected utterance.

This limitation is particularly important in prompted clinical speech assessment. In many PD speech tasks, the participant is asked to produce a known prompt or constrained verbal response. Under this setting, transcription errors are not merely generic ASR failures; they may also reflect disease-relevant deviations in articulation, fluency, timing, or intelligibility. Therefore, instead of relying only on global ASR summary measures, it is beneficial to model the internal structure of prompt-aligned recognition errors and combine that information with acoustic representations extracted directly from the waveform. Motivated by these considerations, this chapter presents a two-branch framework for Parkinson's speech assessment that combines (i) a task-conditioned parameter-efficient acoustic branch built on WavLM and (ii) a prompt-aligned ASR error structure branch derived from Whisper transcriptions. The first branch, referred to as *Task-Conditioned PEFT* (TC-PEFT), adapts a pretrained speech foundation model to heterogeneous prompted tasks using lightweight LoRA-based tuning together with explicit task conditioning. The second branch, referred to as *Prompt-Aligned ASR Error Structure Learning* (PAESL), converts ASR outputs into structured prompt-referenced error representations that capture richer information than global WER/CER summaries alone. The two branches are then fused to produce the final PD classification output.

The main hypothesis of this work is that acoustic SSL embeddings and prompt-aligned ASR error structures provide complementary information. The acoustic branch captures low-level and mid-level speech cues related to dysphonia, articulation, and prosody, whereas the ASR-derived branch captures higher-level manifestations of prompt-dependent intelligibility and production errors. Their fusion is expected to improve robustness over single-branch systems, especially when the dataset contains multiple elicitation tasks with different linguistic and phonetic characteristics.

The contributions of this chapter are fourfold. First, a task-conditioned PEFT strategy is introduced for adapting WavLM to prompted Parkinson's speech assessment under limited data. Second, a structured ASR-error representation is proposed to move beyond coarse transcription summary metrics and explicitly encode prompt-aligned recognition deviations. Third, a fusion architecture is developed to integrate acoustic and ASR-structure information in a unified classification framework. Fourth, the proposed method is evaluated using subject-independent cross-validation with comparisons against acoustic-only, ASR-scalar, and fusion-based baselines.

The remainder of this chapter is organized as follows. Section 6.2 reviews prior work on speech-based PD assessment, self-supervised speech modeling, and ASR-informed pathological speech analysis. Section 6.3 presents the dataset and preprocessing pipeline. Section 6.4 details the proposed TC-PEFT and PAESL branches and the fusion mechanism. Section 6.5 describes the evaluation protocol and implementation details. Section 6.6 reports the main results, followed by ablation and discussion in subsequent sections.

6.2 Related Work

6.2.1 Speech-Based Parkinson's Disease Assessment

Speech analysis has long been explored as a non-invasive tool for Parkinson's disease assessment, due to the high prevalence of motor speech disorders such as hypokinetic dysarthria. Early computational approaches primarily relied on handcrafted acoustic features, including measures of frequency perturbation (jitter), amplitude perturbation (shimmer), harmonic-to-noise ratio (HNR), and prosodic descriptors. These features have been widely used in classical machine learning pipelines and remain clinically interpretable [101, 102].

Little et al. [101] demonstrated that sustained phonation recordings can be used to discriminate PD from healthy controls using nonlinear speech features and support vector machines. Similarly, Tsanas et al. [102] proposed a dysphonia-based regression framework for tracking PD symptom severity, showing that carefully designed acoustic features can correlate with clinical scores. While these studies established the feasibility of speech-based PD detection, their reliance on specific recording conditions and feature engineering limits generalization across datasets and elicitation tasks. To address these limitations, more recent work has explored data-driven and deep learning approaches. Neural models can learn hierarchical representations directly from raw or minimally processed speech signals, often improving robustness to variability in recording conditions and speaker characteristics. The research study [99], highlight that deep neural representations frequently outperform handcrafted features in multilingual and cross-dataset scenarios, although interpretability remains an ongoing concern.

Another important challenge identified in recent literature is the variability introduced by language and task differences. Ríos-Urrego et al. [100] showed that speech-based models trained on one language or task may not generalize reliably to others, emphasizing the need for representations that are robust to phonetic and linguistic variation. This issue is particularly relevant in prompted clinical settings, where different tasks (e.g., reading text vs. isolated words) may elicit distinct

speech characteristics.

6.2.2 Self-Supervised Speech Models and Parameter-Efficient Adaptation

Self-supervised learning (SSL) has become a dominant paradigm in speech representation learning, enabling models to leverage large amounts of unlabeled audio data. wav2vec 2.0 [21] introduced a framework in which latent speech representations are learned through a contrastive objective applied to masked audio segments. This approach significantly improved performance across a range of downstream speech tasks, particularly in low-resource settings.

Building on this line of work, WavLM [22] extended SSL pretraining by incorporating both masked prediction and denoising objectives, and by training on a large and diverse corpus of speech data. The resulting representations have been shown to generalize well across tasks including speech recognition, speaker verification, and paralinguistic analysis. These properties make SSL-based models particularly attractive for pathological speech applications, where labeled datasets are typically limited.

However, adapting large pretrained models to domain-specific tasks such as PD speech assessment introduces computational and statistical challenges. Full fine-tuning requires updating all model parameters and may lead to overfitting when training data are scarce. Parameter-efficient fine-tuning (PEFT) methods address this issue by updating only a small subset of parameters while keeping the majority of the pretrained model fixed. Low-Rank Adaptation (LoRA) [23] is a widely used PEFT technique that injects trainable low-rank matrices into Transformer layers. By constraining updates to a low-dimensional subspace, LoRA reduces memory usage and training cost while maintaining competitive performance. This approach has been successfully applied in both natural language processing and speech tasks, making it a suitable choice for adapting large speech foundation models in clinical applications.

6.2.3 ASR-Based Features and Error Modeling in Pathological Speech

Automatic speech recognition (ASR) systems have also been used in PD assessment, primarily to quantify intelligibility and transcription accuracy. Metrics such as word error rate and character error rate are commonly used as proxies for speech impairment, under the assumption that degraded articulation leads to increased recognition errors. With the emergence of large-scale models such as Whisper [24], ASR systems have become more robust across languages and acoustic conditions, further motivating their use in clinical speech analysis.

However, most existing approaches rely on aggregate ASR metrics, which summarize transcription performance into a single scalar value. While informative, these measures do not capture the structure of recognition errors, such as the location, type, or temporal distribution of deviations from the expected utterance. This limitation is particularly relevant in prompted speech tasks, where the target transcription is known and can be aligned with the ASR output. Some recent studies have begun to explore richer representations derived from ASR outputs, including token-level alignments and error patterns. These approaches suggest that structured ASR information can provide complementary insights beyond acoustic features alone, particularly for capturing intelligi-

bility and linguistic deviations associated with neurological disorders. Nevertheless, systematic integration of prompt-aligned ASR error structures with acoustic representations remains relatively underexplored.

In this context, the present work aims to bridge this gap by explicitly modeling prompt-aligned ASR error structures and combining them with task-conditioned acoustic embeddings within a unified framework.

6.3 Dataset and Materials

6.3.1 Dataset Description

The proposed framework is evaluated on the *Italian Parkinson's Voice and Speech* dataset, which consists of speech recordings collected from subjects diagnosed with Parkinson's disease and healthy control participants. The dataset is designed to support speech-based clinical assessment under controlled prompted conditions, where each subject performs multiple elicitation tasks.

A key characteristic of the IPVS dataset is the presence of *multiple prompted speech tasks per subject*. These tasks are designed to probe different aspects of speech production, including phonation, articulation, and linguistic processing. In this work, we focus on three primary task categories, denoted as B1, B2, and PR, where B1 and B2 correspond to prompted text-reading tasks and PR corresponds to a prompted word-list reading task; PR is equivalent to the FB1 task label used in the original dataset files. This multi-task structure introduces variability in phonetic content, lexical complexity, and speaking style, which makes the dataset suitable for evaluating task-robust speech representations.

Each recording is associated with a subject-level label indicating whether the speaker belongs to the PD group or the control group. The classification problem considered in this chapter is therefore a *binary subject-level classification task* (PD vs. Control), where predictions from multiple recordings of the same subject are aggregated to produce a final decision.

6.3.2 Preprocessing and Audio Segmentation

To ensure compatibility with the pretrained acoustic backbone and to handle variable-length recordings, all audio signals are processed using a standardized segmentation pipeline. Long recordings are divided into short overlapping segments using a sliding window approach. Specifically, each recording is split into fixed-length segments of 5 seconds with an overlap of 2.5 seconds between consecutive segments. This strategy provides a balance between temporal resolution and contextual coverage, while also enabling efficient batch processing during training.

Segmenting the audio into shorter chunks serves multiple purposes. First, it reduces memory requirements when processing recordings with large speech foundation models. Second, it increases the number of training samples, which is beneficial in scenarios with limited subject-level data. Third, it allows the model to capture local temporal variations in speech that may be indicative of PD-related impairments.

All segments inherit the label of the corresponding subject during training. During inference, segment-level embeddings are first aggregated to obtain a recording-level representation, and recording-level representations are then aggregated across available tasks to obtain a subject-level representation.

6.3.3 Task Conditioning and Label Structure

Given the multi-task nature of the IPVS dataset, each audio segment is associated not only with a subject label but also with a task identifier e.g., B1, B2, or PR. This task information is explicitly incorporated into the proposed framework through a task-conditioning mechanism in the acoustic branch (Section 6.4).

The inclusion of task labels is motivated by the observation that different elicitation tasks may emphasize different speech characteristics. For example, reading tasks may highlight prosodic and fluency-related features, while isolated word tasks may emphasize articulation precision. By conditioning the model on task identity, the framework can learn task-aware representations that better capture the variability inherent in prompted clinical speech data.

6.3.4 ASR Transcription and Alignment

For the ASR-based branch, each recording is transcribed using a pretrained automatic speech recognition system. The resulting transcripts are aligned with the expected prompts associated with each task. Based on this alignment, structured representations are constructed to capture differences between the predicted and reference sequences.

Instead of relying solely on aggregate measures such as word error rate or character error rate, the proposed approach extracts *prompt-aligned error structures*, which encode token-level substitutions, insertions, deletions, and alignment patterns. These structured representations are later processed by the PAESL branch (Section 6.4) to learn compact embeddings that reflect intelligibility and linguistic deviations.

6.4 Proposed Method

In this work, we propose a two-branch framework for Parkinson's disease classification from speech, integrating *task-conditioned acoustic representation learning* with *prompt-aligned ASR error structure modeling*. The overall architecture consists of: (i) a Task-Conditioned Parameter-Efficient Fine-Tuning (TC-PEFT) acoustic branch, (ii) a Prompt-Aligned Error Structure Learning (PAESL) branch, and (iii) a late fusion module for joint decision making as shown in Fig. 6.1.

6.4.1 Overview

Let each subject s provide up to three prompted speech recordings corresponding to tasks $\mathcal{T} = \{B1, B2, PR\}$. Each subject is represented as:

$$\mathcal{X}_s = \{x_s^t \mid t \in \mathcal{T},\ x_s^t \text{ available}\} \tag{6.1}$$

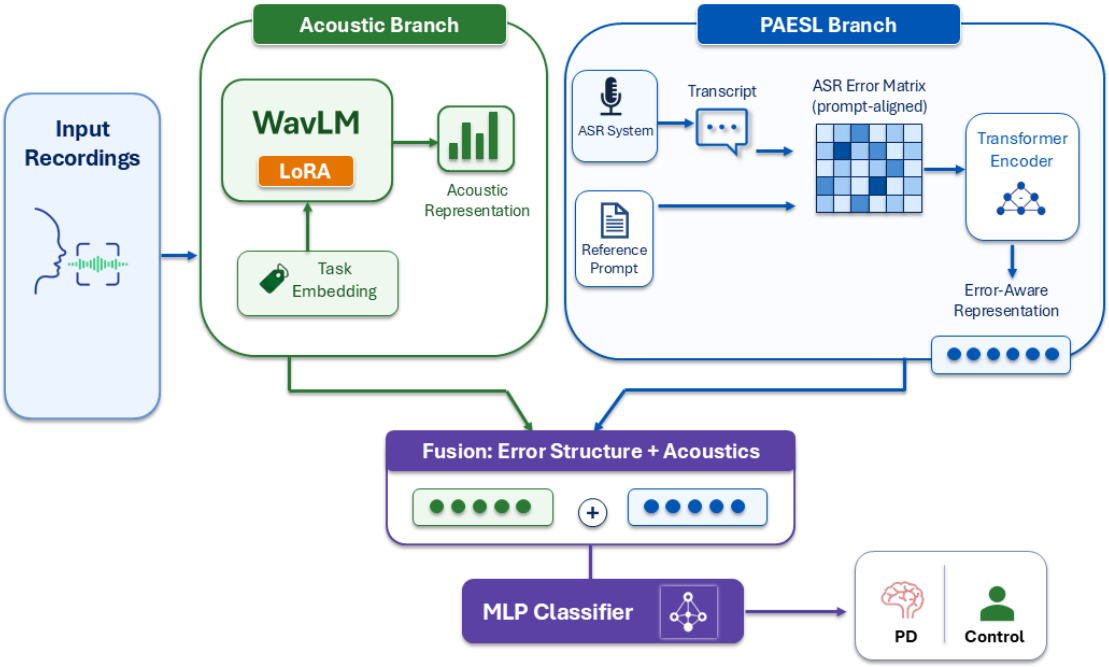


Figure 6.1: Overview of the proposed framework. The TC-PEFT branch extracts task-conditioned acoustic embeddings using WavLM with LoRA adaptation, while the PAESL branch encodes prompt-aligned ASR error structures using a Transformer encoder. The two representations are standardized using training-split statistics, fused by concatenation and passed to an MLP classifier for subject-level Parkinson's disease prediction.

The proposed framework maps $\mathscr{X}_s$ into two complementary representations:

- Acoustic embedding $\mathbf{h}_s^{\text{TC-PEFT}}$
- Structured error embedding $\mathbf{h}_s^{\text{PAESL}}$

These are combined via fusion:

$$\mathbf{z}_s = \text{Fusion}\mathbf{h}_s^{\text{TC-PEFT}}, \mathbf{h}_s^{\text{PAESL}} \tag{6.2}$$

A final classifier produces the probability of PD:

$$\hat{y}_s = \sigma \mathbf{w}^\top \mathbf{z}_s \, b \tag{6.3}$$

6.4.2 TC-PEFT Acoustic Branch

Acoustic Backbone

We adopt a pretrained self-supervised speech model (WavLM-base) as the acoustic backbone, which has demonstrated strong performance in speech representation learning. Given an input waveform x_s^t, the model produces frame-level embeddings:

$$\mathbf{H}_s^t = f_{\text{SSL}} x_s^t \tag{6.4}$$

Chunk-Based Processing

To handle long recordings efficiently, each audio signal is segmented into overlapping chunks of fixed duration (5 seconds) with a hop size of 2.5 seconds. Let:

$$\{c_1, c_2, \ldots, c_K\} = \text{Chunk}x_s^t \tag{6.5}$$

Each chunk is independently processed:

$$\mathbf{h}_k^t = f_{\text{SSL}} c_k \tag{6.6}$$

The chunk-level embeddings are aggregated via mean pooling:

$$\tilde{\mathbf{h}}_s^t = \frac{1}{K} \sum_{k=1}^{K} \mathbf{h}_k^t \tag{6.7}$$

Task-Conditioned PEFT (LoRA)

To adapt the pretrained WavLM backbone efficiently, we employ Low-Rank Adaptation (LoRA). LoRA introduces trainable low-rank updates into selected Transformer attention projection layers while keeping the majority of the pretrained backbone parameters frozen.

For each task $t \in \{B1, B2, PR\}$, a learnable task embedding $\mathbf{e}_t$ is introduced. After chunk-level acoustic embeddings are extracted and pooled, the task embedding is concatenated with the acoustic representation:

$$\mathbf{u}_s^t = \tilde{\mathbf{h}}_s^t ; \mathbf{e}_t . \tag{6.8}$$

The task-conditioned representation is then obtained through a projection layer:

$$\mathbf{h}_s^{t,\text{TC}} = \phi \mathbf{W} \mathbf{u}_s^t \ \mathbf{b}, \tag{6.9}$$

where $\phi \cdot$ denotes a nonlinear activation function.

Subject-Level Aggregation

The final acoustic representation is obtained by aggregating across available tasks:

$$\mathbf{h}_s^{\text{TC-PEFT}} = \frac{1}{|\mathcal{T}_s|} \sum_{t \in \mathcal{T}_s} \mathbf{h}_s^{t,\text{TC}} . \tag{6.10}$$

where $\mathcal{T}_s$ is the set of available tasks for subject s.

6.4.3 PAESL Branch: Prompt-Aligned Error Structure Learning

ASR and Word-Level Alignment

Each speech recording is transcribed using a pretrained ASR model (Whisper). The resulting hypothesis sequence is aligned with the known reference prompt using dynamic programming.

For each reference word w_i, we define an alignment event:

$$e_i \in \{\text{match}, \text{substitution}, \text{deletion}\} \tag{6.11}$$

Insertions are captured as counts before each reference position.

Feature Encoding

Each aligned word is encoded into a feature vector:

$$\mathbf{f}_i = m_i, s_i, d_i, \text{ins}_i, \text{conf}_i, \text{dur}_i, \text{pause}_i, \text{drift}_i \tag{6.12}$$

where:

- m_i, s_i, d_i: match, substitution, deletion indicators
- ins_i: insertion count
- conf_i: ASR confidence
- dur_i: word duration
- pause_i: inter-word pause
- drift_i: timing deviation

This produces a sequence:

$$\mathbf{X}_s^t \in \mathbb{R}^{L \times F} \tag{6.13}$$

Transformer Encoder

For each subject, the task-wise PAESL matrices are arranged into a tensor across the three prompted tasks. In the final implementation, this tensor is reshaped into a sequence representation in which the task-specific feature channels are concatenated:

$$\mathbf{X}_s^{\text{PAESL}} \in \mathbb{R}^{L \times |\mathcal{T}| F}. \tag{6.14}$$

A lightweight Transformer encoder then maps this prompt-aligned sequence into a subject-level PAESL embedding:

$$\mathbf{h}_s^{\text{PAESL}} = g_{\text{PAESL}} \mathbf{X}_s^{\text{PAESL}}. \tag{6.15}$$

6.4.4 Fusion Strategy

Before fusion, TC-PEFT and PAESL embeddings are standardized separately using statistics estimated from the training split only:

$$\bar{\mathbf{h}}_s^{\text{TC-PEFT}} = \text{Standardize}\mathbf{h}_s^{\text{TC-PEFT}}, \quad \bar{\mathbf{h}}_s^{\text{PAESL}} = \text{Standardize}\mathbf{h}_s^{\text{PAESL}}. \tag{6.16}$$

The final fused representation is obtained by concatenation:

$$\mathbf{z}_s = \bar{\mathbf{h}}_s^{\text{TC-PEFT}}; \bar{\mathbf{h}}_s^{\text{PAESL}}, \tag{6.17}$$

which is passed to an MLP classifier to estimate the subject-level PD probability.

6.4.5 Training and Evaluation Protocol

Optimization

The model is trained using binary cross-entropy loss:

$$\mathscr{L} = -\left(y \log \hat{y}\ 1 - y \log 1 - \hat{y}\right) \tag{6.18}$$

Thresholding and evaluation procedures are described in Section 6.5.

6.4.6 Implementation Details

The acoustic backbone uses WavLM-base with LoRA-based fine-tuning. Audio is processed using chunk-based inference for memory efficiency. The PAESL branch uses a lightweight Transformer encoder. All preprocessing steps, including normalization and threshold selection, are performed using training data only to prevent leakage.

6.5 Experimental Setup

6.5.1 Evaluation Protocol

All experiments are conducted using a *subject-independent 5-fold cross-validation* scheme. Subjects are partitioned into five disjoint folds such that all recordings from a given subject appear exclusively in either the training or test set within a fold. This protocol prevents subject-level leakage and provides a realistic estimate of generalization to unseen individuals.

6.5.2 Compared Methods

The proposed framework is evaluated against multiple baselines:

- **SSL Baseline:** WavLM embeddings aggregated at subject level and classified using logistic regression.
- **Task-wise SSL Models:** Separate models using B1-only, B2-only, and PR-only recordings, as well as their concatenation.
- **WER/CER Baseline:** Scalar ASR-derived features (e.g., WER, CER, insertion, deletion counts) used with logistic regression.
- **PAESL-only:** Transformer-based encoding of prompt-aligned ASR error structures.
- **SSL + PAESL Fusion:** Concatenation of SSL embeddings with PAESL features.

6.5.3 Proposed Models

The following configurations are evaluated:

- **TC-PEFT only:** Task-conditioned acoustic model using WavLM with LoRA adaptation.

- **TC-PEFT + PAESL (Fusion):** Joint model combining acoustic and structured ASR representations.

6.5.4 Implementation Details

Audio recordings are processed using a sliding-window segmentation strategy with 5-second chunks and a 2.5-second hop. For each recording, chunk-level embeddings are aggregated to obtain a recording-level representation, followed by subject-level aggregation across tasks.

The acoustic backbone uses WavLM-base with parameter-efficient fine-tuning via LoRA. The PAESL branch uses a lightweight Transformer encoder. All feature normalization is performed using training data only.

Binary classification is optimized using binary cross-entropy loss. Threshold calibration was explored using training data only, including macro-F1 maximization and ROC-based Youden selection. However, because several configurations produced nearly separable scores, calibrated thresholds could become unstable. Therefore, the final reported TC-PEFT and TC-PEFT+PAESL results use the fixed probability threshold of 0.5, while calibrated-threshold results are treated as auxiliary analyses.

6.5.5 Evaluation Metrics

Performance is measured using Accuracy, Macro-F1, Sensitivity (true positive rate), Specificity (true negative rate).

Because several SSL-based configurations showed very strong separability on the current dataset, task-wise, cohort-restricted, and threshold-calibrated analyses are reported as auxiliary checks rather than primary evidence.

6.6 Results

6.6.1 Main Results

The fusion model consistently outperforms the acoustic-only TC-PEFT configuration, particularly in terms of macro-F1 and specificity, indicating improved discrimination of control subjects. Table 6.1 summarizes the performance of the evaluated configurations.

Table 6.1: Subject-level 5-fold cross-validation results for the evaluated configurations. Values are reported as mean ± standard deviation across folds.

Model	Accuracy	Macro-F1	Sensitivity	Specificity
WER/CER scalar baseline	0.638 ± 0.107	0.624 ± 0.111	0.617 ± 0.219	0.677 ± 0.160
PAESL only	0.736 ± 0.195	0.723 ± 0.190	0.681 ± 0.170	0.767 ± 0.268
SSL baseline	0.985 ± 0.034	0.983 ± 0.038	0.960 ± 0.089	1.000 ± 0.000
TC-PEFT only	0.963 ± 0.051	0.960 ± 0.055	0.950 ± 0.112	0.967 ± 0.075
TC-PEFT+PAESL	$\mathbf{0.983 \pm 0.037}$	$\mathbf{0.980 \pm 0.045}$	0.950 ± 0.112	1.000 ± 0.000

The results show that while SSL embeddings already provide strong performance, the proposed TC-PEFT + PAESL framework achieves competitive performance while offering a structured

and interpretable modeling approach. The WER/CER baseline performs substantially worse, highlighting the limitations of coarse ASR-derived features.

6.6.2 Baseline Comparison

The SSL baseline achieves strong performance (accuracy ≈ 0.9846, macro-F1 ≈ 0.9830), highlighting the effectiveness of pretrained representations. However, the proposed fusion model achieves comparable or improved performance while providing a more structured and interpretable framework.

In contrast, the WER/CER baseline performs significantly worse (accuracy ≈ 0.64, macro-F1 ≈ 0.62), demonstrating that coarse ASR summary metrics are insufficient for reliable PD classification.

The PAESL-only model achieves moderate performance (accuracy ≈ 0.74), confirming that structured ASR error information is informative but insufficient when used alone.

6.6.3 Task-wise Results

Task-wise analysis is reported only as an auxiliary check. The PR word-list task was highly discriminative within the current IPVS split protocol, while B1 and B2 also showed strong performance. Because this separability may reflect dataset-specific task characteristics, these results are not used as the primary evidence for the proposed framework.

6.6.4 Permutation Testing

Permutation testing is conducted by randomly shuffling subject labels within each fold. Under this setting, performance drops to near chance (accuracy ≈ 0.52, macro-F1 ≈ 0.48), indicating that the observed results are not due to random structure.

6.6.5 Cohort-Restricted Evaluation

Cohort-restricted analyses were conducted to examine whether the results were driven mainly by the mixture of young and elderly controls. Performance remained high in these restricted comparisons, suggesting that PD/control discrimination is not explained solely by cohort composition. However, because the dataset is small and some task-specific baselines are highly separable, external validation remains necessary.

6.7 Ablation and Additional Analyses

6.7.1 Effect of PAESL Fusion

Comparing TC-PEFT with TC-PEFT + PAESL demonstrates that incorporating structured ASR error information improves overall performance. The improvement is most evident in macro-F1 and specificity, suggesting that PAESL contributes complementary information not captured by acoustic features alone.

6.7.2 ASR Feature Analysis

The WER/CER baseline shows that scalar ASR metrics are insufficient for accurate classification. In contrast, PAESL leverages structured, prompt-aligned error patterns, which provide richer information about articulation and intelligibility.

6.7.3 Task Contribution

Task-wise analysis indicates that the PR task is highly discriminative within the current dataset and split protocol. While this contributes to strong overall performance, it also highlights the importance of evaluating models across multiple tasks to ensure robustness.

6.7.4 Threshold Stability

Several configurations produced highly separable probability scores on the current dataset. In such cases, threshold calibration methods such as macro-F1 optimization or ROC-based Youden selection can become unstable, producing extreme threshold values. For this reason, the final main table reports fixed-threshold results at 0.5, while threshold-calibrated variants are treated as auxiliary experiments.

6.7.5 Robustness to Label Perturbation

Permutation testing confirms that the model does not exploit spurious correlations, as performance degrades to chance under label randomization.

6.8 Discussion, Limitations, and Chapter Summary

The results demonstrate that combining task-conditioned acoustic representations with structured ASR error modeling leads to strong performance in PD classification. The TC-PEFT branch effectively captures low-level acoustic characteristics, while the PAESL branch provides complementary information related to intelligibility and prompt adherence.

One notable observation is the strong performance of the PR task in several baseline settings. While this suggests that the word-list task contains highly discriminative speech information in the IPVS dataset, it also raises questions about dataset difficulty, task-specific separability, and possible cohort-related effects. Therefore, the results should be interpreted together with the multi-task fusion experiments, permutation testing, and future external validation.

Another key finding is the limited effectiveness of scalar ASR metrics such as WER and CER. These metrics fail to capture detailed error structure, whereas PAESL preserves alignment-aware information that contributes to improved classification.

Several configurations produced highly separable probability scores on the current dataset. While this supports the discriminative strength of the learned representations, it also makes threshold calibration sensitive to small score variations. Therefore, fixed-threshold results are used for the main comparison, and calibrated-threshold variants are treated only as auxiliary analyses.

Overall, the proposed framework provides a flexible and interpretable approach for integrating acoustic and linguistic information in speech-based PD assessment.

This chapter presented a two-branch framework for Parkinson's disease classification from speech, combining task-conditioned acoustic modeling (TC-PEFT) with prompt-aligned ASR error structure learning (PAESL). The proposed approach leverages self-supervised speech representations and structured transcription analysis to capture complementary aspects of speech impairment.

Experimental results demonstrate that the fusion of acoustic and ASR-derived features improves performance over single-branch models and baseline approaches. The framework achieves high accuracy and macro-F1 scores under subject-independent evaluation, with additional support from permutation testing and cohort-restricted analyses.

The findings highlight the importance of modeling both acoustic and linguistic aspects of speech, as well as the limitations of coarse ASR metrics. Future work will focus on extending the framework to longitudinal analysis and evaluating generalization across datasets, languages, and recording conditions.

Part IV

General Discussion and Conclusions

Chapter 7

Discussion, Limitations, and Future Work

7.1 Summary of Contributions

This thesis investigated AI methods for prediction and event detection from physiological time series, with a primary focus on continuous glucose monitoring and a complementary focus on speech-based neurological assessment. The contributions are organized according to the sub-objectives defined in Chapter 1 and the publication-aligned structure of Parts I–III.

Contribution 1 (SO1): CGM glucose prediction. Chapter 2 presented a personalized glucose forecasting pipeline based on Long Short-Term Memory modeling, evaluated using standard regression criteria. In addition to demonstrating strong predictive accuracy, the chapter highlighted the practical value of incorporating auxiliary physiological and behavioral features (e.g., heart rate and dietary logs) when available. The results provide evidence that multi-sensor context can improve forecasting performance relative to glucose-only history, while also reinforcing the importance of consistent preprocessing and evaluation under a clearly defined prediction horizon.

Contribution 2 (SO2): Systematic synthesis of CGM-based eating event detection and prediction. Chapter 3 provided a comprehensive review of AI-based methods for detecting and predicting eating events from CGM. The review contributed a structured taxonomy spanning model-based estimation, machine learning, deep learning, and hybrid approaches, and it summarized the datasets and labeling practices used across studies. A key conclusion is that cross-study comparability is still limited by heterogeneous labeling windows, differences in validation protocols, and inconsistent reporting of operational metrics such as false positives per day (FP/day) and detection delay. These findings motivate standardized evaluation practices and encourage future work that bridges algorithmic advances with clinically relevant operating constraints.

Contribution 3 (SO3): CGM meal episode detection with explicit decoding and operational evaluation. Chapter 4 introduced a CGM-only meal episode detection pipeline based on a BiLSTM

sliding-window model and constraint-based decoding. A central methodological contribution is the explicit separation between probability estimation and episode formation. By evaluating Logistic Regression and Random Forest baselines under the same feature set, decoding rules, and per-subject operating point selection, the chapter established a transparent comparison framework and demonstrated that compact sequence models can improve episode-level performance under controlled FP/day. The chapter also emphasized the importance of episode-level metrics for practical deployment, since clinically acceptable performance depends not only on sensitivity but also on alarm burden and timeliness.

Contribution 4 (SO4): Speech-based neurological assessment (MS and PD). Chapters 5 and 6 extended the thesis scope beyond CGM and demonstrated that rigorous time-series modeling protocols can be applied to neurological speech assessment. Chapter 5 investigated Multiple Sclerosis classification using LSTM/BiLSTM models with MFCC and acoustic feature representations, including strategies for dealing with data scarcity and class imbalance. Chapter 6 presented a two-branch Parkinson's disease framework combining task-conditioned parameter-efficient fine-tuning of a speech foundation model with prompt-aligned ASR error structure representations. The results show that integrating complementary representations acoustic embeddings and structured ASR-derived cues can yield strong subject-level screening performance under subject-independent cross-validation.

7.2 Limitations

Despite the breadth of investigated settings, several limitations constrain the generality of the presented results. These limitations also outline clear opportunities for future work.

7.2.1 Dataset scale, cohort diversity, and external validity

Across both CGM and speech domains, dataset scale and population diversity remain key limiting factors. The forecasting study in Chapter 2 is conducted on a prediabetes cohort with a limited number of participants, and the observed performance may not directly transfer to broader populations (e.g., insulin-treated T1D/T2D) without additional validation. In the CGM meal episode detection study (Chapter 4), evaluation was performed on OhioT1DM, which is a widely used benchmark, but still represents a specific setting with self-reported meal logs and fixed sampling characteristics. Similarly, the MS speech dataset in Chapter 5 contains an imbalanced healthy/control subset that required augmentation, and the PD study in Chapter 6 is evaluated on a specific prompted dataset with strong separability, which may not reflect the full variability of real clinical recordings.

7.2.2 Label noise and event definition ambiguity

Meal detection from CGM is fundamentally limited by label uncertainty. Many datasets provide meal start times only, and meal "episodes" must be constructed using fixed windows around start times. Such design choices directly affect both training supervision and evaluation metrics,

particularly detection delay and FP/day. In free-living conditions, under-reporting and timestamp misalignment further complicate interpretation of false positives: some detections may correspond to unlogged snacks or mis-timed annotations rather than genuine algorithmic error. Similarly, in speech assessment, clinical labels may not capture disease heterogeneity (stage, medication state, comorbidities), and prompt adherence may vary across participants, which impacts both acoustic and ASR-derived features.

7.2.3 Modeling scope and deployment constraints

This thesis emphasizes interpretable and reproducible pipelines, but practical deployment introduces additional requirements that are not fully addressed here. In CGM meal episode detection, even when FP/day is controlled, detection delay is bounded by physiological lag and the postprandial response time. For closed-loop applications, the cost of delayed detection may remain clinically significant unless combined with additional contextual signals or control strategies. In speech assessment, computational requirements for foundation models and ASR pipelines can be substantial, motivating efficiency considerations such as chunk-based processing, caching, and PEFT, but runtime and on-device feasibility were not evaluated in this thesis.

7.2.4 Comparability across tasks and metrics

While this thesis argues for episode-level evaluation in meal detection, the broader literature still lacks unified benchmarking standards. Differences in label definitions, tolerance windows, minimum-duration rules, and post-processing make cross-paper comparisons difficult. In speech tasks, reporting practices vary across datasets and studies, and there is no universal consensus on clinically actionable evaluation endpoints. Although the thesis adopts subject-independent validation where possible, broader standardization remains an open challenge.

7.3 Future Work

Future directions naturally follow from the limitations and from the emerging research trends identified in Chapters 3 and 4.

7.3.1 CGM: broader validation and stronger operational evaluation

A direct next step is to validate the CGM pipelines across additional cohorts and conditions, including different sensor brands, sampling rates, and patient populations. For meal episode detection, future studies should report complete operating curves that jointly describe sensitivity/recall, FP/day, and detection delay under consistent decoder definitions. Where possible, evaluation should also incorporate prospective or semi-prospective settings, to reduce dependence on retrospective labeling artifacts.

7.3.2 CGM: improved labeling and semi-automated ground truth refinement

Improving label quality is likely to yield substantial gains without necessarily changing model complexity. Future work may combine self-reports with auxiliary evidence (e.g., insulin events when available, contextual logs, or behavioral sensors) to refine meal timing and reduce ambiguity in ground truth. Even within CGM-only settings, data-driven identification of plausible meal windows can help flag potentially missing meal logs and enable more realistic error analysis.

7.3.3 Speech: robustness across tasks, conditions, and cohorts

For MS and PD speech assessment, future work should prioritize robustness across prompt tasks, recording conditions, and demographic cohorts. This includes evaluating cross-task generalization, investigating domain adaptation to different microphones/environments, and designing evaluation protocols that reflect realistic usage (e.g., home recordings and longitudinal monitoring). The fusion approach in Chapter 6 suggests that combining acoustic embeddings with structured linguistic/ASR cues is promising; future work can explore additional structured representations (e.g., phoneme-level alignments or prosodic error patterns) while maintaining interpretability.

7.3.4 Reproducibility: shared evaluation toolkits and reporting standards

A practical thesis-driven outcome is to standardize evaluation utilities across projects. For CGM meal episode detection, this includes a shared codebase for window construction, decoding, FP/day calibration, and event-level matching under transparent assumptions. For speech assessment, standardized subject-wise splits, reporting templates, and error analysis tools can improve reproducibility. Such toolkits would enable fair comparison of future methods and accelerate translation into clinical studies.

7.4 Concluding Remarks

This thesis presented a publication-aligned body of work on AI methods for physiological time series analysis, with an emphasis on clinically meaningful prediction and event detection. The CGM-focused contributions addressed glucose forecasting, synthesized the rapidly growing meal detection literature, and proposed a transparent CGM-only episode detection pipeline evaluated under operational constraints. The speech-focused contributions demonstrated that modern time-series modeling principles extend to neurological assessment, including MS classification and PD screening using foundation-model adaptation and structured ASR-derived representations.

Overall, the presented results support the broader conclusion that methodological rigor including careful preprocessing, leakage-aware evaluation, and task-aligned metrics is as critical as model complexity for building useful health sensing systems. The directions outlined above aim to strengthen external validity, improve label quality, and move toward deployable systems that can support real-world decision making in metabolic and neurological monitoring.

Bibliography

[1] D. F. de Carvalho, U. Kaymak, P. Van Gorp, and N. van Riel, "Data-driven meal events detection using blood glucose response patterns," *BMC Medical Informatics and Decision Making*, vol. 23, no. 1, p. 282, 2023.

[2] K. Kölle, A. L. Fougner, and Ø. Stavdahl, "Meal detection based on non-individualized moving horizon estimation and classification," in *2017 IEEE Conference on Control Technology and Applications (CCTA)*, pp. 529–535, IEEE, 2017.

[3] K. Kölle, T. Biester, S. Christiansen, A. L. Fougner, and Ø. Stavdahl, "Pattern recognition reveals characteristic postprandial glucose changes: Non-individualized meal detection in diabetes mellitus type 1," *IEEE journal of biomedical and health informatics*, vol. 24, no. 2, pp. 594–602, 2019.

[4] S. Faccioli, I. Sala-Mira, J.-L. Díez, A. Facchinetti, G. Sparacino, S. Del Favero, and J. Bondia, "Super–twisting-based meal detector for type 1 diabetes management: Improvement and assessment in a real-life scenario," *Computer Methods and Programs in Biomedicine*, vol. 219, p. 106736, 2022.

[5] E. M. Gamarra, F. Reiterer, P. Tkachenko, P. Schrangl, G. Freckmann, and W. Ipanaque, "Simple strategies for retrospective detection of meals in diabetes datasets," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 16380–16385, 2020.

[6] C. M. Ramkissoon, P. Herrero i Viñas, J. Bondia, and J. Vehí, "A novel meal detection algorithm for an artificial pancreas [pòster]," 2017.

[7] M. Ibrahim, A. Beneyto, I. Contreras, and J. Vehi, "An ensemble machine learning approach for the detection of unannounced meals to enhance postprandial glucose control," *Computers in Biology and Medicine*, vol. 171, p. 108154, 2024.

[8] L. de la Brosse, S. Bidet, N. Caleca, P. Calmels, T. Camalon, M. Rehn, P. Soulé, J. Place, and E. Renard, "Novel ai-based algorithm to detect and reconstruct meal real time using cgm data," in *Advanced Technologies & Treatments for Diabetes*, 2022.

[9] A. Tavasoli and H. Shakeri, "Online meal detection based on cgm data dynamics," *arXiv preprint arXiv:2507.00080*, 2025.

[10] J. Daniels, P. Herrero, and P. Georgiou, "A deep learning framework for automatic meal detection and estimation in artificial pancreas systems," *Sensors*, no. 2, p. 466, 2022.

[11] G. Freckmann, S. Schauer, A. Beltzer, D. Waldenmaier, S. Buck, A. Baumstark, N. Jendrike, M. Link, E. Zschornack, C. Haug, *et al.*, "Continuous glucose profiles in healthy people with fixed meal times and under everyday life conditions," *Journal of diabetes science and technology*, vol. 18, no. 2, pp. 407–413, 2024.

[12] C. Mosquera-Lopez, L. M. Wilson, J. El Youssef, W. Hilts, J. Leitschuh, D. Branigan, V. Gabo, J. H. Eom, J. R. Castle, and P. G. Jacobs, "Enabling fully automated insulin delivery through meal detection and size estimation using artificial intelligence," *NPJ digital medicine*, vol. 6, no. 1, p. 39, 2023.

[13] J. L. Godoy, J. E. Sereno, and P. S. Rivadeneira, "Meal detection and carbohydrate estimation based on a feedback scheme with application to the artificial pancreas," *Biomedical Signal Processing and Control*, vol. 68, p. 102715, 2021.

[14] M. Zheng, B. Ni, and S. Kleinberg, "Automated meal detection from continuous glucose monitor data through simulation and explanation," *Journal of the American Medical Informatics Association*, vol. 26, no. 12, pp. 1592–1599, 2019.

[15] S. Omidvar, A. R. Roghanizad, L. Chikwetu, G. Ash, J. Dunn, and B. J. Mortazavi, "Enhancing continuous glucose monitoring-based eating detection with wearable biomarkers," in *2022 IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI)*, pp. 1–4, IEEE, 2022.

[16] H. M. Ahmed, S. Maraoui, M. A. E. Sadek, B. Abdulrazak, C. Vandenberghe, S. C. Cunnane, and F. G. Blanchet, "Meal-time detection by means of long periods blood glucose level monitoring via iot technology," *arXiv preprint arXiv:2303.00223*, 2023.

[17] I. Dexcom, "Continuous glucose monitoring (cgm)." `https://www.dexcom.com/`, 2023. Accessed: 01 Nov 2025.

[18] Abbott, "Freestyle libre continuous glucose monitor." `https://www.abbott.com/freestyle-libre-2-continuous-glucose-monitor-cgm.html`, 2023. Accessed: 31 Oct 2025.

[19] Medtronic, "The guardian connect system." `https://www.medtronicdiabetes.com/products/guardian-connect-continuous-glucose-monitoring-system`, 2023.

[20] K. Turksoy, S. Samadi, J. Feng, E. Littlejohn, L. Quinn, and A. Cinar, "Meal detection in patients with type 1 diabetes: a new module for the multivariable adaptive artificial pancreas control system," *IEEE journal of biomedical and health informatics*, vol. 20, no. 1, pp. 47–54, 2015.

[21] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems*, 2020.

[22] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, *et al.*, "Wavlm: Large-scale self-supervised pre-training for full stack speech processing," *IEEE Journal of Selected Topics in Signal Processing*, 2022.

[23] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2021.

[24] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, vol. 202 of *Proceedings of Machine Learning Research*, pp. 28492–28518, 2023.

[25] C. Marling and R. Bunescu, "The ohiot1dm dataset for blood glucose level prediction: Update 2020," in *CEUR workshop proceedings*, vol. 2675, p. 71, 2020.

[26] T. Prioleau, A. Bartolome, R. Comi, and C. Stanger, "Diatrend: A dataset from advanced diabetes technology to enable development of novel analytic solutions," *Scientific Data*, vol. 10, no. 1, p. 556, 2023.

[27] G. Skoglund, B. B. Nilsson, C. F. Olsen, A. Bergland, and G. Hilde, "Facilitators and barriers for lifestyle change in people with prediabetes: a meta-synthesis of qualitative studies," *BMC public health*, vol. 22, no. 1, p. 553, 2022.

[28] M. Marik, "Machine learning based continuous glucose monitoring system," *Global Journal on Innovation, Opportunities and Challenges in AAI and Machine Learning*, vol. 6, no. 1, 2022.

[29] K. K. Patro, J. P. Allam, U. Sanapala, C. K. Marpu, N. A. Samee, M. Alabdulhafith, and P. Plawiak, "An effective correlation-based data modeling framework for automatic diabetes prediction using machine and deep learning techniques," *BMC bioinformatics*, vol. 24, no. 1, p. 372, 2023.

[30] B. Bent, P. J. Cho, M. Henriquez, A. Wittmann, C. Thacker, M. Feinglos, M. J. Crowley, and J. P. Dunn, "Engineering digital biomarkers of interstitial glucose from noninvasive smartwatches," *NPJ Digital Medicine*, vol. 4, no. 1, p. 89, 2021.

[31] A. D. Zahedani, T. McLaughlin, A. Veluvali, N. Aghaeepour, A. Hosseinian, S. Agarwal, J. Ruan, S. Tripathi, M. Woodward, N. Hashemi, *et al.*, "Digital health application integrating wearable data and behavioral patterns improves metabolic health," *NPJ digital medicine*, vol. 6, no. 1, p. 216, 2023.

[32] D. Adams and E. Nsugbe, "A pilot on the endocrine effects of hormonal replacement therapy on menopausal t1 diabetics using wearable sensors," *Engineering Proceedings*, vol. 27, no. 1, p. 19, 2022.

[33] P. Cho, J. Kim, B. Bent, and J. Dunn, "BIG IDEAs Lab Glycemic Variability and Wearable Device Data," *PhysioNet*, Sept. 2023. Version 1.1.2.

[34] D. Kalita and K. B. Mirza, "Glucose prediction using wide-deep lstm network for accurate insulin dosing in artificial pancreas," in *2022 44th Annual international conference of the IEEE engineering in medicine & biology society (EMBC)*, pp. 4426–4429, IEEE, 2022.

[35] L. Fayaz, R. Joseph, B. Ankayarkanni, S. Princemary, and P. Asha, "Multi-scale and context aware optimized glucose prediction using neural networks," in *2023 7th International Conference on Trends in Electronics and Informatics (ICOEI)*, pp. 820–825, IEEE, 2023.

[36] F. Iacono, L. Magni, and C. Toffanin, "Personalized lstm models for glucose prediction in type 1 diabetes subjects," in *2022 30th Mediterranean conference on control and automation (MED)*, pp. 324–329, IEEE, 2022.

[37] H. Butt, I. Khosa, and M. A. Iftikhar, "Feature transformation for efficient blood glucose prediction in type 1 diabetes mellitus patients," *Diagnostics*, vol. 13, no. 3, p. 340, 2023.

[38] Z. Wang, X. Xiao, Y. Pang, W. Su, and T. Kikkawa, "Noninvasive, intelligent blood glucose monitoring on fingertip using dual-band fusion and lstm-r network," *IEEE Sensors Journal*, vol. 24, no. 3, pp. 3465–3476, 2023.

[39] Y. Sun, H. Cano-Garcia, E. Kallos, F. O'Brien, A. Akintonde, D.-E. Motei, O. Ancu, R. W. A. Mackenzie, and P. Kosmas, "Random forest analysis of combined millimeter-wave and near-infrared sensing for noninvasive glucose detection," *IEEE Sensors Journal*, vol. 23, no. 17, pp. 20294–20309, 2023.

[40] D. N. Katsarou, E. I. Georga, M. Christou, S. Tigas, C. Papaloukas, and D. I. Fotiadis, "Short term glucose prediction in patients with type 1 diabetes mellitus," in *2022 44th Annual International Conference Of The IEEE Engineering In Medicine & Biology Society (EMBC)*, pp. 329–332, IEEE, 2022.

[41] A. Ramachandran, S. Ananth, *et al.*, "Deep learning based time series modelling for glucose level prediction of type-1 diabetes," in *2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pp. 1–6, IEEE, 2023.

[42] D. Kalita and K. B. Mirza, "Ls-grunet: glucose forecasting using deep learning for closed-loop diabetes management," in *2022 IEEE 7th International conference for Convergence in Technology (I2CT)*, pp. 1–6, IEEE, 2022.

[43] M. Jaloli, W. Lipscomb, and M. Cescon, "Incorporating the effect of behavioral states in multi-step ahead deep learning based multivariate predictors for blood glucose forecasting in type 1 diabetes," *BioMedInformatics*, vol. 2, no. 4, pp. 715–726, 2022.

[44] K. M. Bermingham, H. A. Smith, J. T. Gonzalez, E. L. Duncan, A. M. Valdes, P. W. Franks, L. Delahanty, H. S. Dashti, R. Davies, G. Hadjigeorgiou, *et al.*, "Glycaemic variability, assessed with continuous glucose monitors, is associated with diet, lifestyle and health in people without diabetes," *Research square*, pp. rs–3, 2023.

[45] J. Brummer, C. Glasbrenner, S. Hechenbichler Figueroa, K. Koehler, and C. Höchsmann, "Continuous glucose monitoring for automatic real-time assessment of eating events and nutrition: a scoping review," *Frontiers in Nutrition*, vol. 10, p. 1308348, 2024.

[46] M. R. Askari, M. Rashid, X. Sun, M. Sevil, A. Shahidehpour, K. Kawaji, and A. Cinar, "Detection of meals and physical activity events from free-living data of people with diabetes," *Journal of Diabetes Science and Technology*, vol. 17, no. 6, pp. 1482–1492, 2023.

[47] Y. Dong, J. Scisco, M. Wilson, E. Muth, and A. Hoover, "Detecting periods of eating during free-living by tracking wrist motion," *IEEE journal of biomedical and health informatics*, vol. 18, no. 4, pp. 1253–1260, 2013.

[48] C. Wang, T. S. Kumar, W. De Raedt, G. Camps, H. Hallez, and B. Vanrumste, "Eating speed measurement using wrist-worn imu sensors towards free-living environments," *IEEE journal of biomedical and health informatics*, vol. 28, no. 10, pp. 5816–5828, 2024.

[49] Y. Dong, A. Hoover, J. Scisco, and E. Muth, "A new method for measuring meal intake in humans via automated wrist motion tracking," *Applied psychophysiology and biofeedback*, vol. 37, no. 3, pp. 205–215, 2012.

[50] S. Sharma and A. Hoover, "Top-down detection of eating episodes by analyzing large windows of wrist motion using a convolutional neural network," *Bioengineering*, vol. 9, no. 2, p. 70, 2022.

[51] M. Farooq, J. M. Fontana, and E. Sazonov, "A novel approach for food intake detection using electroglottography," *Physiological measurement*, vol. 35, no. 5, p. 739, 2014.

[52] A. Bedri, R. Li, M. Haynes, R. P. Kosaraju, I. Grover, T. Prioleau, M. Y. Beh, M. Goel, T. Starner, and G. Abowd, "Earbit: using wearable sensors to detect eating episodes in unconstrained environments," *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies*, vol. 1, no. 3, pp. 1–20, 2017.

[53] Y. Bi, M. Lv, C. Song, W. Xu, N. Guan, and W. Yi, "Autodietary: A wearable acoustic sensor system for food intake recognition in daily life," *IEEE Sensors Journal*, vol. 16, no. 3, pp. 806–816, 2015.

[54] C. Wang, T. S. Kumar, W. De Raedt, G. Camps, H. Hallez, and B. Vanrumste, "Eat-radar: Continuous fine-grained intake gesture detection using fmcw radar and 3d temporal convolutional network with attention," *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 2, pp. 1000–1011, 2023.

[55] J. Qiu, F. P.-W. Lo, S. Jiang, Y.-Y. Tsai, Y. Sun, and B. Lo, "Counting bites and recognizing consumed food from videos for passive dietary monitoring," *IEEE Journal of Biomedical and Health Informatics*, vol. 25, no. 5, pp. 1471–1482, 2020.

[56] E. K. Presseller, M. N. Parker, F. Zhang, S. Manasse, and A. S. Juarascio, "Continuous glucose monitoring as an objective measure of meal consumption in individuals with binge-spectrum eating disorders: A proof-of-concept study," *European eating disorders review*, vol. 32, no. 4, pp. 828–837, 2024.

[57] D. F. de Carvalho, U. Kaymak, P. Van Gorp, and N. van Riel, "Population and individual level meal response patterns in continuous glucose data," in *International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems*, pp. 235–247, Springer, 2022.

[58] B. Lu, Y. Cui, P. Belsare, C. Stanger, X. Zhou, and T. Prioleau, "Mealtime prediction using wearable insulin pump data to support diabetes management," *Scientific Reports*, vol. 14, no. 1, p. 21013, 2024.

[59] C. J. Popp, C. Wang, A. Hoover, L. A. Gomez, M. Curran, D. E. St-Jules, S. Barua, M. A. Sevick, and S. Kleinberg, "Objective determination of eating occasion timing: combining self-report, wrist motion, and continuous glucose monitoring to detect eating occasions in adults with prediabetes and obesity," *Journal of Diabetes Science and Technology*, vol. 18, no. 2, pp. 266–272, 2024.

[60] A. Roy, B. Grosman, A. Benedetti, B. Engheta, D. Miller, M. Laron-Hirsh, Y. Cohen, R. Ré, S. N. Edd, R. Vigersky, *et al.*, "An automated insulin delivery system with automatic meal bolus based on a hand-gesturing algorithm," *Diabetes Technology & Therapeutics*, vol. 26, no. 9, pp. 633–643, 2024.

[61] M. Kashif, S. Flesca, and P. Veltri, "Glstm: On using lstm for glucose level prediction," in *pHealth 2024*, pp. 103–107, IOS Press, 2024.

[62] N. Fera, A. Procopio, P. Zaffino, A. Cutruzzolà, C. Irace, and C. Cosentino, "Modeling strategies for cgm data: A scoping review of mechanistic, machine learning, and hybrid approaches in diabetes management," *Computers in Biology and Medicine*, vol. 205, p. 111601, 2026.

[63] S. H. Park, R. Oh, S. Kim, S. H. Yang, H. Lee, J. Y. Kim, S. H. Cho, S. Park, G. Kim, J. H. Kim, *et al.*, "Identification of clinically meaningful automatically detected postprandial glucose excursions in individuals with type 1 diabetes using personal continuous glucose monitoring," *Diabetes Research and Clinical Practice*, p. 112951, 2025.

[64] M. Athanasiou, K. Zarkogianni, K. Karytsas, and K. S. Nikita, "An lstm-based approach towards automated meal detection from continuous glucose monitoring in type 1 diabetes mellitus," in *2021 IEEE 21st International Conference on Bioinformatics and Bioengineering (BIBE)*, pp. 1–5, IEEE, 2021.

[65] H. Chen, N. Paoletti, S. A. Smolka, and S. Lin, "Committed moving horizon estimation for meal detection and estimation in type 1 diabetes," in *2019 american control conference (acc)*, pp. 4765–4772, IEEE, 2019.

[66] O. M. Staal, S. Sælid, A. L. Fougner, and Ø. Stavdahl, "Meal estimation from continuous glucose monitor data using kalman filtering and hypothesis testing," in *2019 IEEE 58th Conference on Decision and Control (CDC)*, pp. 5654–5661, IEEE, 2019.

[67] V. Palacios, D. M.-K. Woodbridge, and J. L. Fry, "Machine learning-based meal detection using continuous glucose monitoring on healthy participants: an objective measure of participant compliance to protocol," in *2021 43rd annual international conference of the IEEE engineering in Medicine & Biology Society (EMBC)*, pp. 7032–7035, IEEE, 2021.

[68] R. Pai, S. Barua, B. S. Kim, M. McDonald, R. A. Wierzchowska-McNew, A. Pai, N. E. Deutz, D. Kerr, and A. Sabharwal, "Estimating breakfast characteristics using continuous glucose monitoring and machine learning in adults with or at risk of type 2 diabetes," *Journal of Diabetes Science and Technology*, p. 19322968241274800, 2024.

[69] R. Edward-Alfonso, V. Rodolfo, and C. José-Jorge, "Meal detection and carbohydrates counting for in-silico type 1 diabetic patients based on supervised learning," *IFAC-PapersOnLine*, vol. 59, no. 2, pp. 79–84, 2025.

[70] J. Pavan, M. F. Villa-Tamayo, and M. Breton, "An algorithm for retrospectively detecting missing meal records in type 1 diabetes datasets," *IFAC-PapersOnLine*, vol. 59, no. 2, pp. 85–90, 2025.

[71] S. Samadi, K. Turksoy, I. Hajizadeh, J. Feng, M. Sevil, and A. Cinar, "Meal detection and carbohydrate estimation using continuous glucose sensor data," *IEEE journal of biomedical and health informatics*, vol. 21, no. 3, pp. 619–627, 2017.

[72] F. D'Antoni, M. Bertazzoni, L. Vollero, and M. Merone, "Identification of the optimal meal detection strategy for adults, adolescents, and children with type 1 diabetes: an in silico validation," in *2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC)*, pp. 1725–1730, IEEE, 2023.

[73] D. Borghäll, "Automatic event detection oncontinuous glucose datausing neural networks," 2023.

[74] M. R. Askari, M. Rashid, X. Sun, M. Sevil, A. Shahidehpour, K. Kawaji, and A. Cinar, "Meal and physical activity detection from free-living data for discovering disturbance patterns of glucose levels in people with diabetes," *BioMedInformatics*, vol. 2, no. 2, pp. 297–317, 2022.

[75] L. Bertrand, N. Cleyet-Marrel, and Z. Liang, "The role of continuous glucose monitoring in automatic detection of eating activities," in *2021 IEEE 3rd global conference on life sciences and technologies (LifeTech)*, pp. 313–314, IEEE, 2021.

[76] M. Sontakke, F. Cameron, and B. W. Bequette, "Probabilistic meal detection and estimation in type 1 diabetes with extreme shape variability," *Journal of Diabetes Science and Technology*, vol. 19, no. 6, pp. 1471–1480, 2025.

[77] A. Bartolome, S. Shah, and T. Prioleau, "Glucomine: A case for improving the use of wearable device data in diabetes management," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 3, pp. 1–24, 2021.

[78] C. Stanger, T. Kowatsch, H. Xie, I. Nahum-Shani, F. Lim-Liberty, M. Anderson, P. Santhanam, S. Kaden, B. Rosenberg, *et al.*, "A digital health intervention (sweetgoals) for young adults with type 1 diabetes: Protocol for a factorial randomized trial," *JMIR Research Protocols*, vol. 10, no. 2, p. e27109, 2021.

[79] S. G. James, M. E. G. Armstrong, A. A. O'Kane, H. Emerson, and Z. S. Abdallah, "Brist1d dataset: Young adults with type 1 diabetes in the uk using smartwatches," *arXiv preprint arXiv:2507.17757*, 2025.

[80] R. Gutierrez-Osuna, D. Kerr, B. Mortazavi, and A. Das, "CGMacros: a scientific dataset for personalized nutrition and diet monitoring," *PhysioNet*, Jan. 2025. Version 1.0.0.

[81] M. K. Wolff, P. Calhoun, E. M. Aiello, Y. Qin, and S. F. Royston, "Metabonet: The largest publicly available consolidated data set for type 1 diabetes management," *Journal of Diabetes Science and Technology*, p. 19322968261441637, 2026.

[82] J. P. Corbett, L. Hsu, S. A. Brown, L. Kollar, K. Vleugels, B. Buckingham, M. D. Breton, and R. A. Lal, "Smartwatch gesture-based meal reminders improve glycaemic control," *Diabetes, Obesity and Metabolism*, vol. 24, no. 8, pp. 1667–1670, 2022.

[83] T. Koutny, M. Kukrál, J. Romová, and J. Vašátko, "Meta-analysis of technologies for diabetes treatment: Glycemic control, prediction, meal and physical activity detection," *Progress in Biomedical Engineering*, 2026.

[84] G. Annuzzi, A. Apicella, P. Arpaia, L. Bozzetto, S. Criscuolo, E. De Benedetto, M. Pesola, and R. Prevete, "Exploring nutritional influence on blood glucose forecasting for type 1 diabetes using explainable ai," *IEEE journal of biomedical and health informatics*, vol. 28, no. 5, pp. 3123–3133, 2023.

[85] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[86] D. F. de Carvalho, U. Kaymak, P. Van Gorp, and N. van Riel, "Data-driven meal events detection using blood glucose response patterns," *BMC Medical Informatics and Decision Making*, vol. 23, no. 1, p. 282, 2023.

[87] S. Samadi, M. Rashid, K. Turksoy, J. Feng, I. Hajizadeh, N. Hobbs, C. Lazaro, M. Sevil, E. Littlejohn, and A. Cinar, "Automatic detection and estimation of unannounced meals for

multivariable artificial pancreas system," *Diabetes technology & therapeutics*, vol. 20, no. 3, pp. 235–246, 2018.

[88] A. J. Thompson, S. E. Baranzini, J. Geurts, B. Hemmer, and O. Ciccarelli, "Multiple sclerosis," *The Lancet*, vol. 391, no. 10130, pp. 1622–1636, 2018.

[89] C. Walton, R. King, L. Rechtman, W. E. Kaye, E. Leray, R. A. Marrie, N. P. Robertson, N. L. Rocca, B. M. J. Uitdehaag, I. van der Mei, M. T. Wallin, A. Helme, C. A. Napier, N. Rijke, and P. Baneke, "Rising prevalence of multiple sclerosis worldwide: Insights from the atlas of ms, third edition," *Multiple Sclerosis (Houndmills, Basingstoke, England)*, vol. 26, pp. 1816 – 1821, 2020.

[90] J. R. Duffy, "Motor speech disorders: Substrates, differential diagnosis, and management," *Elsevier Health Sciences.*, 2020.

[91] J. Rusz, T. Tykalová, G. Salerno, S. Bancone, J. Scarpelli, and M. T. Pellecchia, "Distinctive speech signature in cerebellar and parkinsonian subtypes of multiple system atrophy," *Journal of Neurology*, vol. 266, pp. 1394–1404, 2019.

[92] P. Vizza, D. Mirarchi, G. Tradigo, M. Redavide, R. B. Bossio, and P. Veltri, "Vocal signal analysis in patients affected by multiple sclerosis," *Procedia Computer Science*, vol. 108, pp. 1205–1214, 2017. International Conference on Computational Science, ICCS 2017, 12-14 June 2017, Zurich, Switzerland.

[93] S. Hawi, J. Alhozami, R. AlQahtani, D. AlSafran, M. Alqarni, and L. El Sahmarany, "Automatic parkinson's disease detection based on the combination of long-term acoustic features and mel frequency cepstral coefficients (mfcc)," *Biomedical Signal Processing and Control*, vol. 78, p. 104013, 2022.

[94] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, pp. 1735–1780, 1997.

[95] S. Belabbas, D. Addou, and S. A. Selouani, "Pathological voice classification system based on cnn-bilstm network using speech enhancement and multi-stream approach," *International Journal of Speech Technology*, vol. 27, no. 2, pp. 483–502, 2024.

[96] M. Gonzalez-Machorro, P. Hecker, U. Reichel, H. N. Hammer, R. Hoepner, L. Pedrotti, A. Zmutt, H. Sagha, J. van Beek, F. Eyben, *et al.*, "Towards supporting an early diagnosis of multiple sclerosis using vocal features," *Age (Omaha)*, vol. 36, pp. 10–15, 2023.

[97] P. Vizza, G. Tradigo, D. Mirarchi, R. B. Bossio, N. Lombardo, G. Arabia, A. Quattrone, and P. Veltri, "Methodologies of speech analysis for neurodegenerative diseases evaluation," *International Journal of Medical Informatics*, vol. 122, pp. 45–54, 2019.

[98] K. Dashtipour, A. Tafreshi, J. Lee, and B. Crawley, "Speech disorders in parkinson's disease: pathophysiology, medical management and surgical approaches," *Neurodegenerative Disease Management*, 2018.

[99] A. Favaro *et al.*, "Interpretable speech features vs. dnn embeddings: What to use in the automatic assessment of parkinson's disease in multi-lingual scenarios," *Computers in Biology and Medicine*, 2023.

[100] C. D. Rios-Urrego *et al.*, "Automatic speech-based assessment to discriminate parkinson's disease from essential tremor with a cross-language approach," *npj Digital Medicine*, 2024.

[101] M. A. Little, P. E. McSharry, E. J. Hunter, J. Spielman, and L. O. Ramig, "Suitability of dysphonia measurements for telemonitoring of parkinson's disease," *IEEE Transactions on Biomedical Engineering*, vol. 56, no. 4, pp. 1015–1022, 2009.

[102] A. Tsanas, M. A. Little, P. E. McSharry, and L. O. Ramig, "Novel speech signal processing algorithms for high-accuracy classification of parkinson's disease," *IEEE Transactions on Biomedical Engineering*, vol. 59, no. 5, pp. 1264–1271, 2012.

Curriculum

Muhammad Kashif was born on April 01, 1991 in Swabi City, Khyber Pakhtunkhwa Province, Pakistan.

Education

2024–2025 **KU Leuven, Belgium**
Visiting PhD Student / International Scholar
e-Media Research Lab (ESAT–STADIUS), Campus GroupT Leuven

2023–2026 **University of Calabria, Italy**
PhD Student, Information and Communications Technology (ICT)
Department of Informatics, Modeling, Electronics, and Systems (DIMES)

2016–2019 **IIUI Islamabad, Pakistan**
MS in Computer Science
Faculty of Basic and Applied Sciences

2009–2013 **Peshawar University, Pakistan**
BS in Computer Science
Faculty of Numerical and Physical Sciences

List of Publications

Journal Articles

M. Kashif, C. Wang, S. Flesca, and P. Veltri, "AI-Based Detection and Prediction of Eating Events Using Continuous Glucose Monitoring: A Comprehensive Review," *Biomed. Signal Process. Control*, vol. 127, article no. 111207, 2026.

Conference Proceedings

M. Kashif, P. Vizza, G. Tradigo, and S. Flesca, "Task-Conditioned PEFT with Prompt-Aligned ASR Error Structure Fusion for Parkinson's Speech Assessment," *in Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*, 2026, pp. 1-2.

M. Kashif, C. Wang, S. Flesca, and P. Veltri, "Meal Episode Detection from Continuous Glucose Monitoring Using a BiLSTM Sliding-Window and Constraint-Based Decoding," *34th European Signal Processing Conference (EUSIPCO)*, 2026, pp. 856-860.

M. Kashif, P. Vizza, A. Delfino, R. B. Bossio, G. Tradigo, S. Flesca, and P. Veltri, "Using LSTM-Based Model on Vocal Signal Analysis for the Classification of Multiple Sclerosis," in *Lecture Notes in Computer Science*, Springer Nature Switzerland, 2025, pp. 192–196.

M. Kashif, S. Flesca, and P. Veltri, "GLSTM: On using LSTM for glucose level prediction," *Stud. Health Technol. Inform.*, vol. 314, pp. 103–107, 2024.